Received

JUL 6 2018

ND Oil & Gas Division

WELL COMPLETION OR RECOMPLETION REPORT - FORM 6

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 2468 (04-2010)

Well File No. **28978**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Designate Type of Completion
☑ Oil Well ☐ EOR Well ☐ Recompletion ☐ Deepened Well ☐ Added Horizontal Leg ☐ Extended Horizontal Leg
☐ Gas Well ☐ SWD Well ☐ Water Supply Well ☐ Other:

Well Name and Number: **Wade Federal 5300 21-30 13B**			Spacing Unit Description: **Sec. 29/30 T153N R100W**
Operator: **Oasis Petroleum North America**	Telephone Number: **(281) 404-9591**		Field: **Baker**
Address: **1001 Fannin, Suite 1500**			Pool: **Bakken**
City: **Houston**	State: **TX**	Zip Code: **77002**	Permit Type: ☐ Wildcat ☑ Development ☐ Extension

LOCATION OF WELL

At Surface	Qtr-Qtr	Section	Township	Range	County
1574 F N L **270** F W L	**LOT2**	**30**	**153 N**	**100 W**	**McKenzie**

Spud Date	Date TD Reached	Drilling Contractor and Rig Number	KB Elevation (Ft)	Graded Elevation (Ft)
September 25, 2014	**November 11, 2014**	**Nabors B25**	**1999**	**2024**

Type of Electric and Other Logs Run (See Instructions)
MWD/GR from KOP to TD; CBL from int. td to surface

CASING & TUBULARS RECORD (Report all strings set in well)

Well Bore	String Type	String Size (Inch)	Top Set (MD Ft)	Depth Set (MD Ft)	Hole Size (Inch)	Weight (Lbs/Ft)	Anchor Set (MD Ft)	Packer Set (MD Ft)	Sacks Cement	Top of Cement
Surface Hole	Surface	9 5/8	0	6030	13 1/2	36			953	0
Vertical Hole	Intermediate	7	0	11000	8 3/4	29/32			852	6114
Lateral1	Liner	4 1/2	10059	20535	6	11.6			515	

PERFORATION & OPEN HOLE INTERVALS

Well Bore	Well Bore TD Drillers Depth (MD Ft)	Completion Type	Open Hole/Perforated Interval (MD,Ft) Top	Bottom	Kick-off Point (MD Ft)	Top of Casing Window (MD Ft)	Date Perf'd or Drilled	Date Isolated	Isolation Method	Sacks Cement
Lateral1	20560	Perforations	11000	20535	10118		03/16/2015			

PRODUCTION

Current Producing Open Hole or Perforated Interval(s), This Completion, Top and Bottom, (MD Ft)	Name of Zone (If Different from Pool Name)
Lateral 1- 11000' to 20535'	

Date Well Completed (SEE INSTRUCTIONS)	Producing Method	Pumping-Size & Type of Pump	Well Status (Producing or Shut-In)
April 22, 2015	**Flowing**		**Producing**

Date of Test	Hours Tested	Choke Size	Production for Test	Oil (Bbls)	Gas (MCF)	Water (Bbls)	Oil Gravity-API (Corr.)	Disposition of Gas
05/01/2015	**24**	**22 /64**		**874**	**787**	**1412**	**42.0 °**	**Sold**

Flowing Tubing Pressure (PSI)	Flowing Casing Pressure (PSI)	Calculated 24-Hour Rate	Oil (Bbls)	Gas (MCF)	Water (Bbls)	Gas-Oil Ratio
1100	**2350**		**874**	**787**	**1412**	**900**

GEOLOGICAL MARKERS

Formation	MD (Ft)	TVD (Ft)
Dakota	**5350**	**5350**
Kibbey Lime	**8312**	**8311**
First Charles Salt	**8472**	**8471**
UB	**9035**	**9034**
Base Last Salt	**9114**	**9113**
Ratcliffe	**9162**	**9161**
Mission Canyon	**9332**	**9331**
Lodgepole	**9888**	**9887**
False Bakken	**10771**	**10772**
Upper Bakken	**10802**	**10624**
Middle Bakken	**10861**	**10642**
Target	**10937**	**10657**

PLUG BACK INFORMATION

Well Bore	Type of Plug	Top (Ft)	Bottom (Ft)	Sacks Cement

CORES CUT

Top (Ft)	Bottom (Ft)	Formation	Top (Ft)	Bottom (Ft)	Formation

Drill Stem Test

Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								

Well Specific Stimulations

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
03/16/2015	Bakken	11000	20535	50	187400	Barrels

Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)	Maximum Treatment Rate (BBLS/Min)
Sand Frac		6583070	9040	40.0

Details
40/70 White: 999510 lbs
20/40 White: 4600820 lbs
20/40 CRC: 982740 lbs

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units

Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)	Maximum Treatment Rate (BBLS/Min)

Details

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units

Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)	Maximum Treatment Rate (BBLS/Min)

Details

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units

Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)	Maximum Treatment Rate (BBLS/Min)

Details

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units

Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)	Maximum Treatment Rate (BBLS/Min)

Details

ADDITIONAL INFORMATION AND/OR LIST OF ATTACHMENTS

This supplemental report provides frac data.

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.	Email Address	Date
	dbusch@oasispetroleum.com	07/05/2018
Signature	Printed Name	Title
	Daniel Busch	Regulatory Specialist

SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Received
JUL 06 2016
ND Oil & Gas Division

Well File No. **28978**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☐ Notice of Intent	Approximate Start Date	☐ Drilling Prognosis	☐ Spill Report
☑ Report of Work Done	Date Work Completed **June 20, 2016**	☐ Redrilling or Repair	☐ Shooting
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date	☐ Casing or Liner	☐ Acidizing
		☐ Plug Well	☐ Fracture Treatment
		☐ Supplemental History	☑ Change Production Method
		☐ Temporarily Abandon	☐ Reclamation
		☐ Other **Well is now on pump**	

Well Name and Number					
Wade Federal 5300 21-30 13B					
Footages	Qtr-Qtr	Section	Township	Range	
1574 F N L 270 F W L	**LOT 2**	**30**	**153 N**	**100 W**	
Field	Pool	County			
Baker	**Bakken**	**McKenzie**			

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Effective 06/20/2016 the above referenced well was converted to rod pump.

End of Tubing: 2-7/8" L-80 tubing @ 9533.19'

Pump @ 9717.86'

Company **Oasis Petroleum North America LLC**		Telephone Number **281-404-9436**
Address **1001 Fannin, Suite 1500**		
City **Houston**	State **TX**	Zip Code **77002**
Signature	Printed Name **Jennifer Swenson**	
Title **Regulatory Specialist**	Date **July 5, 2016**	
Email Address **jswenson@oasispetroleum.com**		

FOR STATE USE ONLY

☑ Received	☐ Approved
Date 7-28-2016	
By	
Title JARED THUNE Engineering Technician	

SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

RECEIVED OCT 2015 ND OIL & GAS DIVISION

Well File No. **28978**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☐ Notice of Intent	Approximate Start Date	☐ Drilling Prognosis	☐ Spill Report
		☐ Redrilling or Repair	☐ Shooting
☑ Report of Work Done	Date Work Completed **August 10, 2015**	☐ Casing or Liner	☐ Acidizing
		☐ Plug Well	☐ Fracture Treatment
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.		☐ Supplemental History	☑ Change Production Method
	Approximate Start Date	☐ Temporarily Abandon	☐ Reclamation
		☐ Other **Well is now on pump**	

Well Name and Number					
Wade Federal 5300 21-30 13B					
Footages	Qtr-Qtr	Section	Township	Range	
1574 F N L 270 F W L	**LOT 2**	**30**	**153 N**	**100 W**	
Field	Pool	County			
Baker	**Bakken**	**McKenzie**			

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Effective 08/10/2015 the above referenced well is on pump.

End of Tubing: 2-7/8" L-80 tubing @ 10015.78'

Pump: ESP @ 9891.55'

Company **Oasis Petroleum North America LLC**		Telephone Number **281-404-9436**
Address **1001 Fannin, Suite 1500**		
City **Houston**	State **TX**	Zip Code **77002**
Signature	Printed Name **Jennifer Swenson**	
Title **Regulatory Specialist**	Date **October 14, 2015**	
Email Address **jswenson@oasispetroleum.com**		

FOR STATE USE ONLY

☑ Received	☐ Approved
Date 10/26/2015	
By Taylor Roth	
Title TAYLOR ROTH Engineering Technician	

AUTHORIZATION TO PURCHASE AND TRANSPORT OIL FROM LEASE -- Form 8

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5698 (03-2000)

Well File No.
28978
NDIC CTB No.
To be assigned

228303

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND FOUR COPIES.

JUN 2015 RECEIVED ND OIL & GAS DIVISION

Well Name and Number	Qtr-Qtr	Section	Township	Range	County
WADE FEDERAL 5300 21-30 13B	LOT2	30	153	100	Williams

Operator	Telephone Number	Field
Oasis Petroleum North America LLC	(281) 404-9573	BAKER

Address	City	State	Zip Code
1001 Fannin, Suite 1500	Houston	TX	77002

Name of First Purchaser	Telephone Number	% Purchased	Date Effective
Oasis Petroleum Marketing LLC	(281) 404-9627	100%	April 1, 2015

Principal Place of Business	City	State	Zip Code
1001 Fannin, Suite 1500	Houston	TX	77002

Field Address	City	State	Zip Code

Transporter	Telephone Number	% Transported	Date Effective
Hiland Crude, LLC	(580) 616-2058	75%	April 1, 2015 ~~April 15, 2015~~

Address	City	State	Zip Code
P.O. Box 3886	Enid	OK	73702

The above named producer authorizes the above named purchaser to purchase the percentage of oil stated above which is produced from the lease designated above until further notice. The oil will be transported by the above named transporter.

Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective

Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective

Other Transporters Transporting From This Lease	% Transported	Date Effective
Hofmann Trucking	28%	April 15, 2015

Other Transporters Transporting From This Lease	% Transported	Date Effective

Comments

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.		Date
Signature	Printed Name	Title
	Dina Barron	Mktg. Contracts Administrator

Above Signature Witnessed By:

Signature	Printed Name	Title
	Alexa Cardona	Marketing Assistant

JUL 2015 RECEIVED ND OIL & GAS DIVISION

FOR STATE USE ONLY

Date Approved	JUL 13 2015
By	Eric Robertson
Title	Oil & Gas Production Analyst

Industrial Commission of North Dakota
Oil and Gas Division
Verbal Approval To Purchase and Transport Oil

Well or Facility No **28978**

Tight Hole **Yes**

OPERATOR

Operator	Representative	Rep Phone
OASIS PETROLEUM NORTH AMERICA LL	**Todd Hanson**	**(701) 577-1632**

WELL INFORMATION

Well Name
WADE FEDERAL 5300 21-30 13B

Inspector
Richard Dunn

Well Location	QQ	Sec	Twp		Rng	
	LOT2	**30**	**153**	**N**	**100**	**W**

County
MCKENZIE

Field
BAKER

Footages
1574 Feet From the **N** Line
270 Feet From the **W** Line

Pool
BAKKEN

Date of First Production Through Permanent Wellhead **4/22/2015** **This Is The First Sales**

PURCHASER / TRANSPORTER

Purchaser	Transporter
OASIS PETROLEUM MARKETING LLC	**HOFMANN TRUCKING, LLC**

TANK BATTERY

Single Well Tank Battery Number :

SALES INFORMATION **This Is The First Sales**

ESTIMATED BARRELS TO BE SOLD		ACTUAL BARRELS SOLD		DATE
15000	BBLS	**237**	BBLS	**4/22/2015**
	BBLS		BBLS	
	BBLS		BBLS	
	BBLS		BBLS	
	BBLS		BBLS	
	BBLS		BBLS	
	BBLS		BBLS	
	BBLS		BBLS	
	BBLS		BBLS	

DETAILS

Must also forward Forms 6 & 8 to State prior to reaching 15000 Bbl estimate or no later than required time frame for submitting those forms.

Start Date	**4/22/2015**
Date Approved	**5/18/2015**
Approved By	**Richard Dunn**

SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

APR 2015 RECEIVED ND OIL & GAS DIVISION

Well File No. **28978**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

- [x] Notice of Intent — Approximate Start Date: **March 31, 2015**
- [ ] Report of Work Done — Date Work Completed:
- [ ] Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. — Approximate Start Date:

- [ ] Drilling Prognosis
- [ ] Spill Report
- [ ] Redrilling or Repair
- [ ] Shooting
- [ ] Casing or Liner
- [ ] Acidizing
- [ ] Plug Well
- [ ] Fracture Treatment
- [ ] Supplemental History
- [ ] Change Production Method
- [ ] Temporarily Abandon
- [ ] Reclamation
- [x] Other **Change well status to CONFIDENTIAL**

Well Name and Number					
Wade Federal 5300 21-30 13B					
Footages	Qtr-Qtr	Section	Township	Range	
1574 F N L 270 F W L	**LOT2**	**30**	**153 N**	**100 W**	
Field	Pool	County			
Baker	**BAKKEN**	**McKenzie**			

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Effective immediately, we request CONFIDENTIAL STATUS for the above referenced well.

This well has not been completed.

OFF CONFIDENTIAL 10/01/15.

Company	Telephone Number
Oasis Petroleum North America LLC	**281-404-9436**
Address: **1001 Fannin, Suite 1500**	
City: **Houston**	State: **TX** / Zip Code: **77002**
Signature:	Printed Name: **Jennifer Swenson**
Title: **Regulatory Specialist**	Date: **March 31, 2015**
Email Address: **jswenson@oasispetroleum.com**	

FOR STATE USE ONLY

- [ ] Received
- [x] Approved

Date: 4/08/15

By: Alice D. Webber

Title: **Engineering Technician**

SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Well File No. **28976**

28978

AUG 20 2014 RECEIVED ND OIL & GAS DIVISION

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☑ Notice of Intent	Approximate Start Date **September 14, 2014**	☐ Drilling Prognosis	☐ Spill Report
☐ Report of Work Done	Date Work Completed	☐ Redrilling or Repair	☐ Shooting
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date	☐ Casing or Liner	☐ Acidizing
		☐ Plug Well	☐ Fracture Treatment
		☐ Supplemental History	☐ Change Production Method
		☐ Temporarily Abandon	☑ Reclamation
		☐ Other **offsite pit**	

Well Name and Number					
Wade Federal 5300 21-30 12T					
Footages		Qtr-Qtr	Section	Township	Range
1640 F N L	270 F W L	SWNW	30	153 N	100 W
Field	Pool	County			
Baker	**Bakken**	**McKenzie**			

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum North America LLC respectfully requests to use an offsite pit for this well. The following wells will also use this pit:

Wade Federal 5300 31-30 2B – 28554
Wade Federal 5300 41-30 3T2 – 28555
Wade Federal 5300 41-30 4T – 28394
Wade Federal 5300 41-30 5T2 – 28556
Wade Federal 5300 41-30 6B – 28425 – TH
Wade Federal 5300 41-30 7T – 28557
Wade Federal 5300 41-30 8T2 – 28558 – TH
Wade Federal 5300 41-30 9B – 28744
Wade Federal 5300 21-30 13B – 28978
Wade Federal 5300 21-30 14T2 – 28977

Attached are the plats for the offsite pit location.

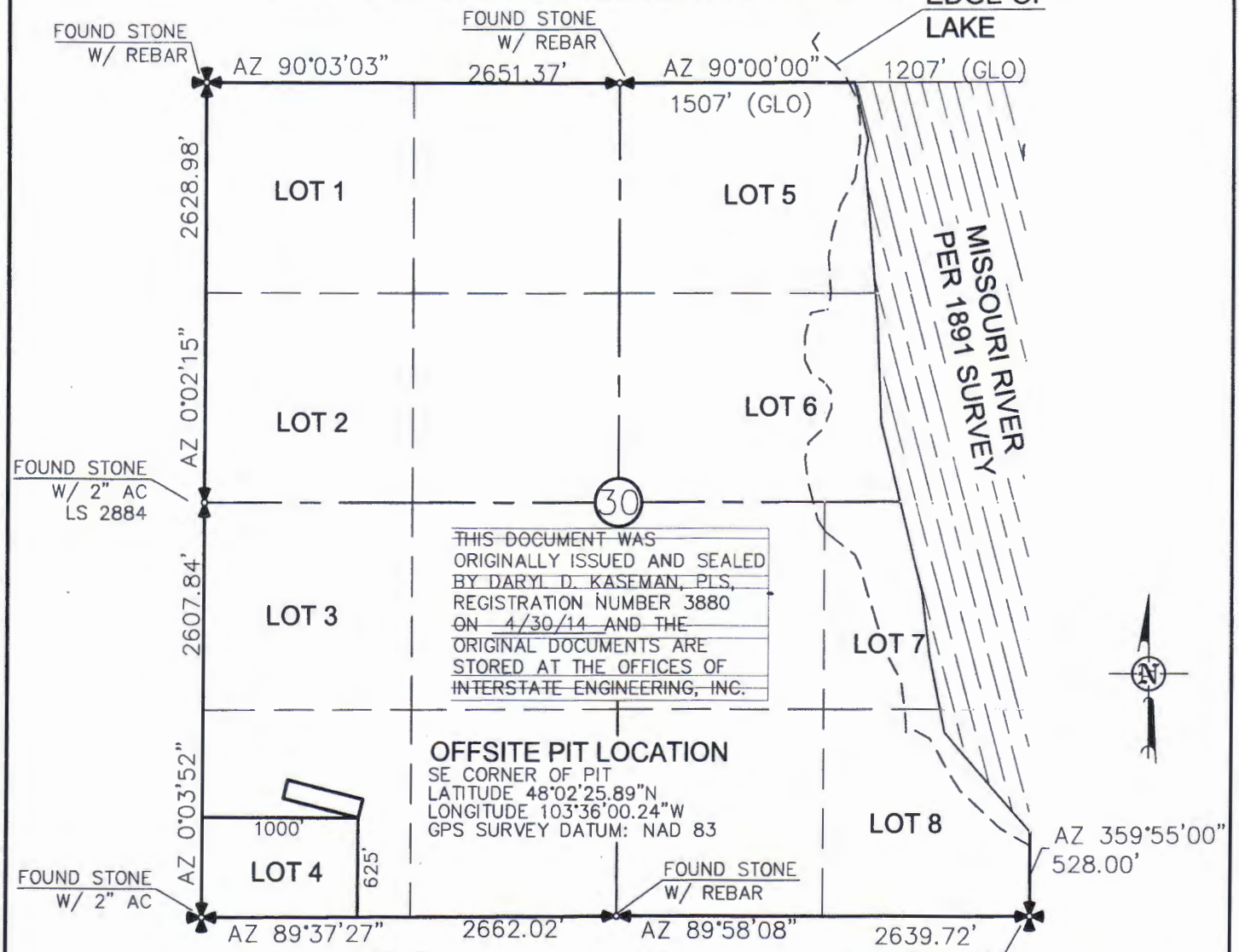
Company **Oasis Petroleum North America LLC**		Telephone Number **281-404-9589**
Address **1001 Fannin, Suite 1500**		
City **Houston**	State **TX**	Zip Code **77002**
Signature Sonja Rolfes	Printed Name **Sonja Rolfes**	
Title **Regulatory Analyst**	Date **August 20, 2014**	
Email Address **srolfes@oasispetroleum.com**		

FOR STATE USE ONLY

☐ Received	☒ Approved
Date 2-9-15	
By	
Title	

OFFSITE PIT LOCATION PLAT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"AS-STAKED OFFSITE PIT FOR WADE FEDERAL 5300 31-30 2B, WADE FEDERAL 5300 41-30 3T2, WADE FEDERAL 5300 41-30 4T, WADE FEDERAL 5300 41-30 5T2, WADE FEDERAL 5300 41-30 6B, WADE FEDERAL 5300 41-30 7T, WADE FEDERAL 5300 41-30 8T2, & WADE FEDERAL 5300 41-30 9B"
625 FEET FROM SOUTH LINE AND 1000 FEET FROM WEST LINE
SECTION 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



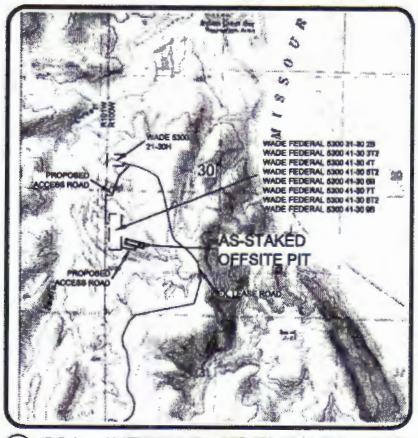
THIS DOCUMENT WAS ORIGINALLY ISSUED AND SEALED BY DARYL D. KASEMAN, PLS, REGISTRATION NUMBER 3880 ON 4/30/14 AND THE ORIGINAL DOCUMENTS ARE STORED AT THE OFFICES OF INTERSTATE ENGINEERING, INC.

– MONUMENT – RECOVERED

– MONUMENT – NOT RECOVERED

0 1000
1" = 1000'

VICINITY MAP



STAKED ON 4/30/14
VERTICAL CONTROL DATUM WAS BASED UPON CONTROL POINT 705 WITH AN ELEVATION OF 2158.3'

THIS SURVEY AND PLAT IS BEING PROVIDED AT THE REQUEST OF ERIC BAYES OF OASIS PETROLEUM. I CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS WORK PERFORMED BY ME OR UNDER MY SUPERVISION AND IS TRUE AND CORRECT TO THE BEST OF MY KNOWLEDGE AND BELIEF.

DARYL D. KASEMAN LS-3880

REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE: 4/30/14
NORTH DAKOTA

© 2014, INTERSTATE ENGINEERING, INC.

1/3
SHEET NO.

INTERSTATE ENGINEERING
Professionals you need, people you trust

Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com
Other offices in Minnesota, North Dakota and South Dakota

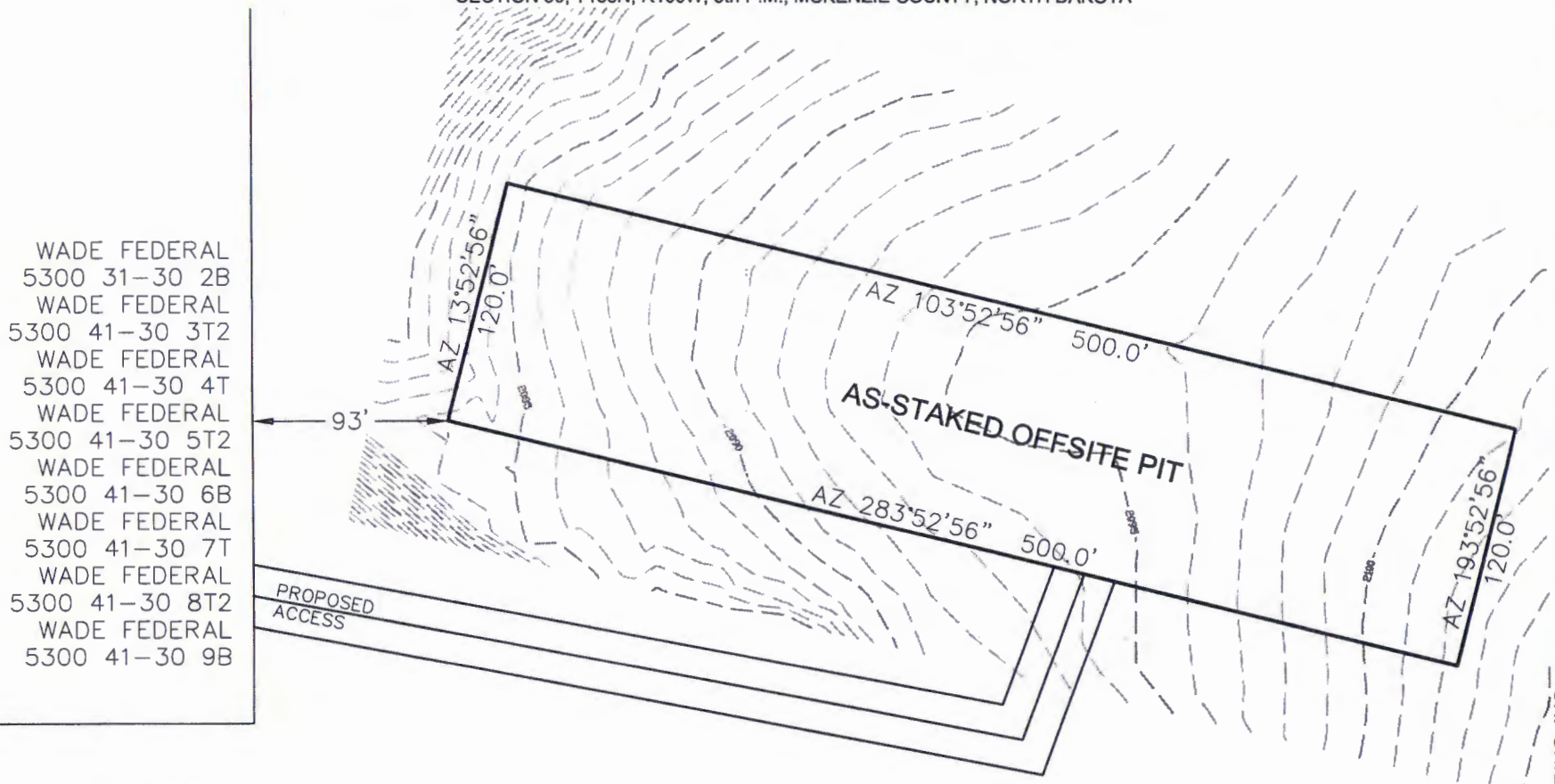
OASIS PETROLEUM NORTH AMERICA, LLC
OFFSITE PIT LOCATION PLAT
SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H.		Project No.: S13-09-381.09	
Checked By: D.D.K.		Date: APRIL 2014	

Revision No.	Date	By	Description

PAD LAYOUT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"AS-STAKED OFFSITE PIT FOR WADE FEDERAL 5300 31-30 2B, WADE FEDERAL 5300 41-30 3T2, WADE FEDERAL 5300 41-30 4T, WADE FEDERAL 5300 41-30 5T2, WADE FEDERAL 5300 41-30 6B, WADE FEDERAL 5300 41-30 7T, WADE FEDERAL 5300 41-30 8T2, & WADE FEDERAL 5300 41-30 9B"
625 FEET FROM SOUTH LINE AND 1000 FEET FROM WEST LINE
SECTION 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE: 4/30/14
NORTH DAKOTA

THIS DOCUMENT WAS ORIGINALLY ISSUED AND SEALED BY DARYL D. KASEMAN, PLS, REGISTRATION NUMBER 3880 ON 4/30/14 AND THE ORIGINAL DOCUMENTS ARE STORED AT THE OFFICES OF INTERSTATE ENGINEERING, INC.

NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

0 80
1" = 80'

Revision No.	Date	By	Description

OASIS PETROLEUM NORTH AMERICA, LLC
PAD LAYOUT
SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S13-09-381.09
Checked By: D.D.K. Date: APRIL 2014

Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com
Other offices in Montana, North Dakota and South Dakota

INTERSTATE ENGINEERING
Professionals you need, people you trust

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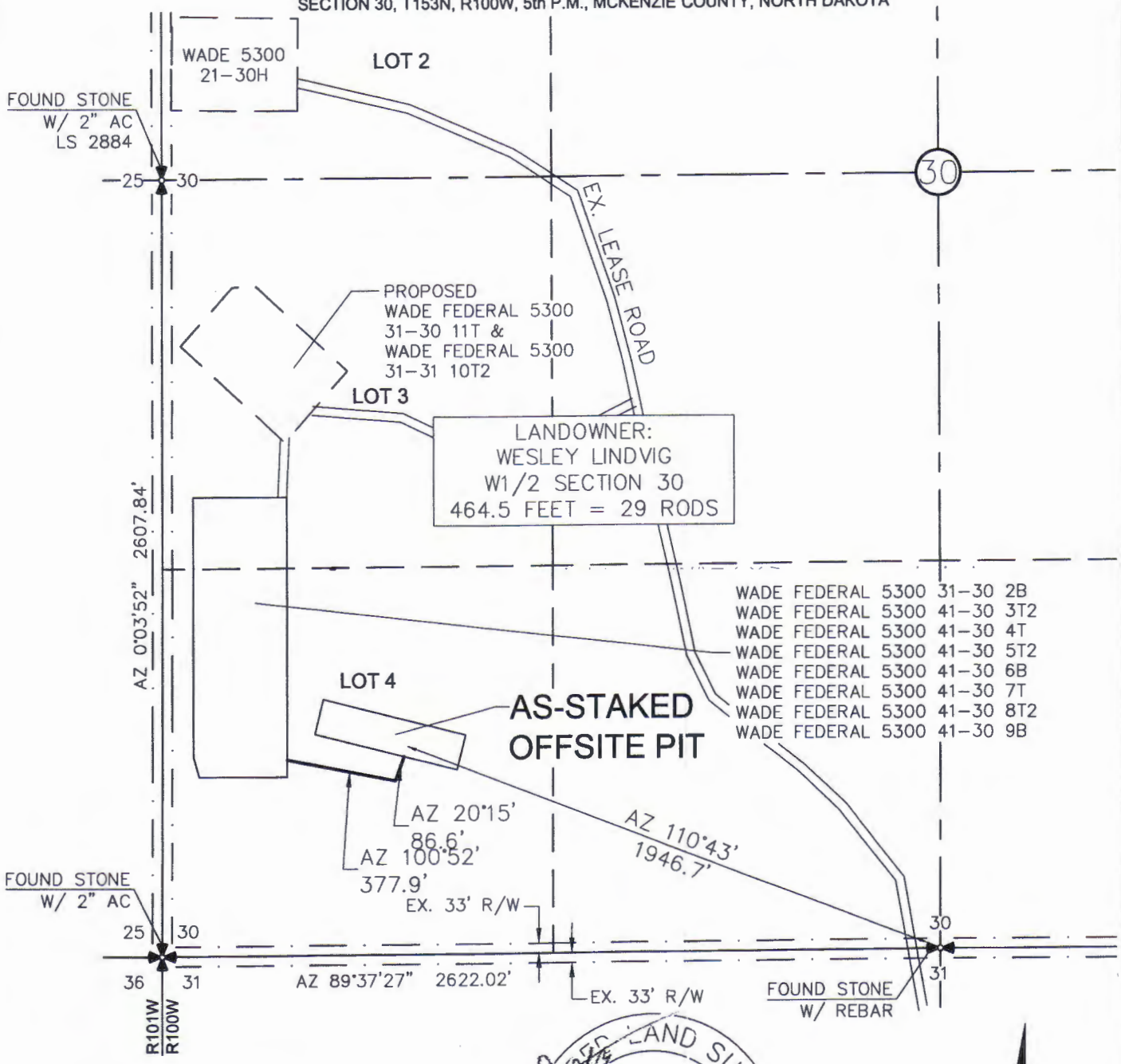
ACCESS APPROACH

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"AS-STAKED OFFSITE PIT FOR WADE FEDERAL 5300 31-30 2B, WADE FEDERAL 5300 41-30 3T2, WADE FEDERAL 5300 41-30 4T, WADE FEDERAL 5300 41-30 5T2, WADE FEDERAL 5300 41-30 6B, WADE FEDERAL 5300 41-30 7T, WADE FEDERAL 5300 41-30 8T2, & WADE FEDERAL 5300 41-30 9B"
625 FEET FROM SOUTH LINE AND 1000 FEET FROM WEST LINE
SECTION 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



THIS DOCUMENT WAS ORIGINALLY ISSUED AND SEALED BY DARYL D. KASEMAN, PLS, REGISTRATION NUMBER 3880 ON 4/30/14 AND THE ORIGINAL DOCUMENTS ARE STORED AT THE OFFICES OF INTERSTATE ENGINEERING, INC.

REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE: 4/30/14
NORTH DAKOTA

0 500
1" = 500'

NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

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3/3
SHEET NO.

INTERSTATE ENGINEERING
Professionals you need, people you trust

Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
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Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
ACCESS APPROACH
SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H.	Project No.: S13-09-381.09		
Checked By: D.D.K.	Date: APRIL 2014		

Revision No.	Date	By	Description

OFF-SITE PIT AGREEMENT

In consideration of the sum of [redacted] paid by Oasis Petroleum North America LLC ("Oasis") the undersigned surface owners, Wesley Lindvig and Barbara Lindvig, for themselves and their heirs, successors, administrators and assigns, hereby acknowledge the receipt and sufficiency of said payment in full and complete settlement for and as a release of all claim for loss, damage or injury to the hereafter described surface property arising out of the off-site cuttings pit, in which the cuttings from the **Wade Federal 5300 21-30 13B, Wade Federal 5300 21-30 14T2** wells will be buried, located on the approximately two (2.0) acre tract of land identified on the plat attached hereto as Exhibit "A" and which is situated on the following described real property located in McKenzie County, State of North Dakota, towit:

Township 153 North, Range 100 West, 5th P.M.
Section 30: Lots 3 & 4 a/k/a W½SW¼

The undersigned knows that Oasis Petroleum North America LLC is the operator and will be drilling the **Wade Federal 5300 21-30 13B, Wade Federal 5300 21-30 14T2** wells. The undersigned further states that they are fully aware that the cuttings generated from the drilling of the **Wade Federal 5300 21-30 13B, Wade Federal 5300 21-30 14T2** wells will be buried in the pit on the above described location.

Dated this 19 day of May, 2014.

SURFACE OWNER(S)

Wesley Lindvig
Wesley Lindvig

Barbara J. Lindvig
Barbara Lindvig

BJL W.G.L.

* Location will be fenced after Construction.
* Pit will be reclaimed to Owners Satisfaction

BJL W.G.L.

ACKNOWLEDGMENT INDIVIDUAL

State of North Dakota)
)
County of McKenzie)

BE IT REMEMBERED, That on this 19 day of May, 2014 before me, a Notary Public, in and for said County and State, personally appeared Wesley Lindvig and Barbara Lindvig, to me known to be the identical persons described in and who executed the within and foregoing instrument and acknowledged to me to that they executed the same as their free and voluntary act and deed for the uses and purposes therein set forth.

IN WITNESS WHEREOF, I have hereunto set my official signature and affixed my notarial seal, the day and year last above written.

My Commission expires:

JAMES DEMORRETT
Notary Public
State of North Dakota
My Commission Expires Aug. 16, 2018

Notary Public

SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

DEC 2014 RECEIVED ND OIL & GAS DIVISION

Well File No. **28978**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☑ Notice of Intent	Approximate Start Date **December 12, 2014**	☐ Drilling Prognosis	☐ Spill Report
☐ Report of Work Done	Date Work Completed	☐ Redrilling or Repair	☐ Shooting
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date	☐ Casing or Liner	☐ Acidizing
		☐ Plug Well	☐ Fracture Treatment
		☐ Supplemental History	☐ Change Production Method
		☐ Temporarily Abandon	☐ Reclamation
		☐ Other	**Waiver from tubing/packer requirement**

Well Name and Number					
Wade Federal 5300 21-30 13B					
Footages	Qtr-Qtr	Section	Township	Range	
1574 F N L 270 F W L	LOT2	30	153 N	100 W	
Field	Pool	County			
BAKKEN	Bakken	MCKENZIE			

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum North America LLC requests a variance to NDAC 43-02-03-21 for the tubing/packer requirement: Casing, tubing, and cementing requirements during the completion period immediately following the upcoming fracture stimulation.

The following assurances apply:

1. the well is equipped with new 29# and 32# casing at surface with an API burst rating of 11,220 psi;
2. The Frac design will ise a safety factor of 0.85 API burst rating to determine the maximum pressure;
3. Damage to the casing during the frac would be detected immediately by montioring equipement;
4. The casing is exposed to significantly lower rates and pressures during flowback than during the frac job;
5. The frac fluid and formation fluids have very low corrosion and erosion rates;
6. Production equipment will be installed as soon as possible after the well ceases flowing;
7. A 300# gauge will be installed on the surface casing during the flowback period

Company **Oasis Petroleum North America LLC**		Telephone Number **281-404-9436**
Address **1001 Fannin, Suite 1500**		
City **Houston**	State **TX**	Zip Code **77002**
Signature	Printed Name **Jennifer Swenson**	
Title **Regulatory Assistant**	Date **December 12, 2014**	
Email Address **jswenson@oasispetroleum.com**		

FOR STATE USE ONLY

☐ Received	☒ Approved
Date	December 22, 2014
By	
Title	PETROLEUM ENGINEER

RYAN DIRECTIONAL SERVICES, INC.
A NABORS COMPANY

19510 Oil Center Blvd
Houston, TX 77073
Bus 281.443.1414
Fax 281.443.1676

Monday, December 15, 2014

State of North Dakota

Subject: **Surveys**

Re: **Oasis**
Wade Federal 5300 21-30 13B
McKenzie, ND

Enclosed, please find the original and one copy of the survey performed on the above-referenced well by Ryan Directional Services, Inc.. Other information required by your office is as follows:

Surveyor Name	*Surveyor Title*	*Borehole Number*	*Start Depth*	*End Depth*	*Start Date*	*End Date*	*Type of*	*TD Straight Line Projection*
Daniel Ogden	MWD Operator	O.H.	0'	2018'	09/24/14	09/26/14	MWD	2018'
Kyle Gilliam	MWD Operator	O.H.	2018'	20498'	10/27/14	11/11/14	MWD	20560'

If any other information is required please contact the undersigned at the letterhead address or phone number.

Douglas Hudson
Well Planner

RYAN DIRECTIONAL SERVICES, INC.
A NABORS COMPANY

Ryan Directional Services, Inc.
19510 Oil Center Blvd.
Houston, Texas 77073
Bus: 281.443.1414
Fax: 281.443.1676

Friday, September 26, 2014

State of North Dakota
County of McKenzie

Subject: **Survey Certification Letter**

Survey Company: **Ryan Directional Services, Inc.**
Job Number: **8112**
Survey Job Type: **Ryan MWD**
Customer: **Oasis Petroleum N.A. LLC**
Well Name: **Wade Federal 5300 21-30 13B**
Rig Name: **Nabors B-25**

Surface: **48° 2' 55.790N, 103° 36' 10.970W**
A.P.I. No: **33-053-06131**
Location: **McKenzie, ND**
RKB Height: **25'**
Distance to Bit: **52'**

Surveyor Name	*Surveyor Title*	*Borehole Number*	*Start Depth*	*End Depth*	*Start Date*	*End Date*	*Type of*	*TD Straight Line Projection*
Daniel Ogden	MWD Supervisor	OH	0'	2018'	09/24/14	09/26/14	MWD	2070'

The data and calculations for this survey have been checked by me and conform to the calibration standards and operational procedures set forth by Ryan Directional Services, Inc. I am authorized and qualified to review the data, calculations and these reports; the reports represents true and correct Directional Surveys of this well based on the original data, the minimum curvature method, corrected to True North and obtained at the well site.

Daniel Ogden
MWD Supervisor
Ryan Directional Services, Inc.

RYAN DIRECTIONAL SERVICES, INC.
A NABORS COMPANY

Ryan Directional Services, Inc.
19510 Oil Center Blvd.
Houston, Texas 77073
Bus: 281.443.1414
Fax: 281.443.1676

Tuesday, November 11, 2014

State of North Dakota
County of McKenzie

Subject: **Survey Certification Letter**

Survey Company: **Ryan Directional Services, Inc.**
Job Number: **8138**
Survey Job Type: **Ryan MWD**
Customer: **Oasis Petroleum N.A. LLC**
Well Name: **Wade Federal 5300 21-30 13B**
Rig Name: **Nabors B-25**

Surface: **48 02' 55.79"N 103 36' 10.97" W**
A.P.I. No: **33-053-06130**
Location: **McKenzie, ND**
RKB Height: **25'**
Distance to Bit: **62'**

Surveyor Name	*Surveyor Title*	*Borehole Number*	*Start Depth*	*End Depth*	*Start Date*	*End Date*	*Type of*	*TD Straight Line Projection*
Kyle Gilliam	MWD Supervisor	OH	2018'	20498'	10/27/14	11/11/14	MWD	20560'

The data and calculations for this survey have been checked by me and conform to the calibration standards and operational procedures set forth by Ryan Directional Services, Inc. I am authorized and qualified to review the data, calculations and these reports; the reports represents true and correct Directional Surveys of this well based on the original data, the minimum curvature method, corrected to True North and obtained at the well site.

Kyle Gilliam
MWD Supervisor
Ryan Directional Services, Inc.

Report #: 1

Date: 25-Sep-14

RYAN DIRECTIONAL SERVICES, INC.

A NABORS COMPANY

Ryan Job # 8112

SURVEY REPORT

Customer:	Oasis Petroleum N.A. LLC
Well Name:	Wade Federal 5300 21-30 13B
Rig #:	Nabors B-25
API #:	33-053-06131
Calculation Method:	Minimum Curvature Calculation

MWD Operator:	Daniel Ogden
Directional Drillers:	RPM
Survey Corrected To:	True North
Vertical Section Direction:	85.39
Total Correction:	8.15
Temperature Forecasting Model (Chart Only):	Logarithmic

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
Tie in to Zero Depth									
Tie In	0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1	149	0.80	334.90	60.00	149.00	-0.36	0.94	-0.44	0.54
2	244	0.90	333.80	64.00	243.98	-0.87	2.21	-1.05	0.11
3	337	1.20	346.40	64.00	336.97	-1.29	3.81	-1.60	0.41
4	399	1.20	344.80	71.00	398.96	-1.51	5.07	-1.93	0.05
5	461	1.30	346.30	73.00	460.94	-1.74	6.38	-2.26	0.17
6	524	1.20	356.60	75.00	523.93	-1.84	7.73	-2.47	0.39
7	615	1.10	342.80	77.00	614.91	-2.01	9.52	-2.79	0.32
8	707	1.10	354.90	77.00	706.89	-2.21	11.24	-3.13	0.25
9	799	0.80	29.20	77.00	798.88	-1.86	12.68	-2.89	0.68
10	891	1.30	32.80	80.00	890.86	-0.87	14.12	-2.01	0.55
11	983	1.10	27.60	84.00	982.84	0.23	15.78	-1.04	0.25
12	1075	0.90	36.30	86.00	1074.83	1.18	17.15	-0.20	0.27
13	1167	1.10	2.00	87.00	1166.82	1.75	18.61	0.26	0.67
14	1260	1.10	341.20	87.00	1259.80	1.64	20.35	0.00	0.43
15	1352	1.10	334.10	89.00	1351.78	1.10	21.98	-0.67	0.15
16	1444	1.10	322.40	89.00	1443.77	0.30	23.47	-1.59	0.24
17	1537	0.90	330.40	91.00	1536.75	-0.50	24.81	-2.50	0.26
18	1630	0.80	308.40	91.00	1629.74	-1.28	25.85	-3.37	0.36
19	1723	0.80	302.10	93.00	1722.73	-2.28	26.60	-4.43	0.09
20	1816	1.10	320.30	95.00	1815.72	-3.31	27.63	-5.55	0.45
21	1909	0.90	334.80	95.00	1908.71	-4.08	28.98	-6.43	0.35
22	2002	0.70	333.30	96.00	2001.70	-4.55	30.15	-7.00	0.22
23	2018	1.20	335.30	100.00	2017.69	-4.64	30.39	-7.11	3.13
24	2118	1.10	341.30	73.00	2117.67	-5.24	32.25	-7.85	0.16
25	2183	1.30	334.60	75.00	2182.66	-5.65	33.51	-8.37	0.38
26	2276	1.30	324.20	82.00	2275.64	-6.57	35.31	-9.44	0.25
27	2369	1.40	325.60	86.00	2368.61	-7.68	37.11	-10.70	0.11
28	2462	1.10	3.70	89.00	2461.59	-8.12	38.94	-11.28	0.93
29	2556	1.10	3.60	91.00	2555.57	-7.86	40.74	-11.17	0.00
30	2649	1.10	327.00	95.00	2648.56	-8.16	42.38	-11.60	0.74
31	2742	0.90	339.90	96.00	2741.54	-8.78	43.81	-12.34	0.32
32	2835	0.60	125.30	100.00	2834.54	-8.60	44.22	-12.19	1.54
33	2929	1.10	150.60	104.00	2928.53	-7.84	43.15	-11.35	0.65
34	3022	0.90	154.80	105.00	3021.51	-7.21	41.71	-10.60	0.23
35	3115	0.90	170.80	107.00	3114.50	-6.89	40.32	-10.17	0.27
36	3208	1.10	154.70	111.00	3207.49	-6.52	38.80	-9.67	0.37
37	3302	1.50	168.10	113.00	3301.46	-6.05	36.78	-9.03	0.53
38	3395	0.50	72.60	116.00	3394.45	-5.50	35.71	-8.39	1.75
39	3488	0.80	43.70	118.00	3487.45	-4.61	36.30	-7.56	0.47
40	3581	0.80	45.30	120.00	3580.44	-3.63	37.22	-6.65	0.02
41	3674	0.70	37.30	122.00	3673.43	-2.76	38.13	-5.84	0.16
42	3768	0.80	28.60	123.00	3767.42	-2.01	39.17	-5.18	0.16
43	3861	0.70	31.90	125.00	3860.41	-1.32	40.22	-4.57	0.12
44	3954	0.50	23.40	127.00	3953.41	-0.79	41.07	-4.11	0.23
45	4048	0.70	11.50	129.00	4047.40	-0.44	42.01	-3.83	0.25
46	4141	0.70	25.30	131.00	4140.40	0.00	43.08	-3.47	0.18
47	4234	0.20	64.80	131.00	4233.39	0.44	43.66	-3.08	0.60
48	4327	0.10	61.10	132.00	4326.39	0.66	43.77	-2.87	0.11
49	4420	0.10	313.70	134.00	4419.39	0.68	43.87	-2.85	0.17
50	4513	0.40	250.20	136.00	4512.39	0.31	43.81	-3.22	0.39
51	4607	0.40	239.10	140.00	4606.39	-0.30	43.53	-3.81	0.08
52	4700	0.60	219.40	140.00	4699.39	-0.93	42.99	-4.40	0.28
53	4793	0.70	224.40	141.00	4792.38	-1.69	42.21	-5.10	0.12
54	4886	0.50	217.10	141.00	4885.38	-2.39	41.48	-5.74	0.23
55	4979	0.70	218.40	143.00	4978.37	-3.05	40.71	-6.34	0.22
56	5072	1.20	193.80	145.00	5071.36	-3.74	39.32	-6.93	0.68
57	5165	1.40	192.90	147.00	5164.33	-4.39	37.27	-7.41	0.22
58	5258	0.60	182.40	147.00	5257.32	-4.79	35.67	-7.69	0.88
59	5351	0.50	158.90	150.00	5350.31	-4.74	34.81	-7.56	0.26
60	5444	0.70	160.60	152.00	5443.31	-4.48	33.89	-7.23	0.22

Report #: 1

Date: 25-Sep-14

RYAN DIRECTIONAL SERVICES, INC.

A NABORS COMPANY

Ryan Job # 8112

SURVEY REPORT

Customer:	Oasis Petroleum N.A. LLC
Well Name:	Wade Federal 5300 21-30 13B
Rig #:	Nabors B-25
API #:	33-053-06131
Calculation Method:	Minimum Curvature Calculation

MWD Operator:	Daniel Ogden
Directional Drillers:	RPM
Survey Corrected To:	True North
Vertical Section Direction:	85.39
Total Correction:	8.15
Temperature Forecasting Model (Chart Only):	Logarithmic

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
61	5537	1.10	161.20	154.00	5536.30	-4.11	32.51	-6.75	0.43
62	5630	1.80	161.80	158.00	5629.27	-3.55	30.28	-6.01	0.75
63	5723	1.20	154.00	154.00	5722.24	-2.85	28.02	-5.12	0.68
64	5815	1.20	150.40	158.00	5814.22	-2.10	26.31	-4.22	0.08
65	5909	1.20	144.80	159.00	5908.19	-1.18	24.65	-3.17	0.12
66	6002	1.50	162.80	161.00	6001.17	-0.42	22.70	-2.25	0.55
67	6090	0.70	216.90	122.00	6089.15	-0.52	21.16	-2.23	1.40
68	6181	0.60	241.20	123.00	6180.15	-1.33	20.49	-2.98	0.32
69	6274	0.90	60.40	134.00	6273.15	-1.11	20.62	-2.77	1.61
70	6368	1.00	53.90	140.00	6367.13	0.26	21.47	-1.47	0.16
71	6461	0.80	46.10	143.00	6460.12	1.45	22.39	-0.35	0.25
72	6554	0.70	49.60	149.00	6553.11	2.42	23.21	0.55	0.12
73	6647	0.50	32.90	152.00	6646.11	3.13	23.92	1.21	0.28
74	6740	0.60	37.90	156.00	6739.10	3.70	24.65	1.73	0.12
75	6833	0.20	4.80	159.00	6832.10	4.06	25.19	2.04	0.48
76	6926	0.00	288.30	163.00	6925.10	4.08	25.35	2.05	0.22
77	7019	1.40	3.50	165.00	7018.09	4.24	26.49	2.12	1.51
78	7112	1.80	12.50	170.00	7111.06	4.83	29.05	2.51	0.51
79	7205	1.70	7.40	172.00	7204.01	5.55	31.84	3.00	0.20
80	7298	1.60	3.30	172.00	7296.97	6.02	34.51	3.25	0.17
81	7391	1.10	11.60	174.00	7389.95	6.44	36.68	3.51	0.58
82	7484	1.00	12.30	176.00	7482.93	6.93	38.34	3.86	0.11
83	7578	0.80	10.30	177.00	7576.92	7.34	39.79	4.15	0.22
84	7671	0.50	357.10	179.00	7669.91	7.52	40.83	4.25	0.36
85	7764	0.40	32.50	181.00	7762.91	7.72	41.51	4.40	0.31
86	7857	0.30	284.10	183.00	7855.91	7.69	41.85	4.34	0.61
87	7951	0.30	243.80	185.00	7949.91	7.23	41.80	3.88	0.22
88	8044	0.50	221.90	185.00	8042.91	6.71	41.39	3.39	0.27
89	8137	0.40	172.20	188.00	8135.90	6.43	40.76	3.16	0.42
90	8230	0.20	225.20	188.00	8228.90	6.32	40.33	3.09	0.35
91	8324	0.30	217.40	192.00	8322.90	6.03	40.02	2.83	0.11
92	8417	0.20	223.30	195.00	8415.90	5.75	39.71	2.57	0.11
93	8510	0.10	290.00	197.00	8508.90	5.56	39.62	2.38	0.20
94	8603	0.10	343.50	199.00	8601.90	5.47	39.72	2.28	0.10
95	8696	0.40	217.90	201.00	8694.90	5.23	39.54	2.06	0.50
96	8790	0.30	257.80	203.00	8788.90	4.76	39.23	1.62	0.27
97	8883	0.50	260.70	203.00	8881.90	4.12	39.12	0.98	0.22
98	8976	0.30	280.40	203.00	8974.89	3.48	39.09	0.34	0.26
99	9069	0.40	280.90	201.00	9067.89	2.93	39.20	-0.22	0.11
100	9163	0.20	302.20	204.00	9161.89	2.48	39.35	-0.68	0.24
101	9256	0.50	305.30	203.00	9254.89	2.04	39.67	-1.15	0.32
102	9349	0.40	290.50	203.00	9347.89	1.44	40.02	-1.78	0.16
103	9442	0.40	278.40	204.00	9440.88	0.83	40.18	-2.41	0.09
104	9535	0.40	267.30	204.00	9533.88	0.19	40.21	-3.06	0.08
105	9629	0.40	312.90	203.00	9627.88	-0.36	40.42	-3.62	0.33
106	9722	0.50	278.70	203.00	9720.88	-0.98	40.70	-4.26	0.30
107	9815	0.40	263.50	204.00	9813.87	-1.70	40.73	-4.99	0.17
108	9908	0.40	265.70	204.00	9906.87	-2.35	40.66	-5.63	0.02
109	10001	0.40	272.00	206.00	9999.87	-2.99	40.65	-6.28	0.05
110	10063	0.40	301.50	208.00	10061.87	-3.38	40.77	-6.68	0.33
111	10095	0.40	312.70	188.00	10093.87	-3.55	40.91	-6.86	0.24
112	10126	2.20	44.40	192.00	10124.86	-3.17	41.40	-6.52	7.25
113	10157	6.10	53.60	195.00	10155.77	-1.32	42.81	-4.78	12.72
114	10188	10.20	56.10	197.00	10186.45	2.47	45.32	-1.17	13.27
115	10220	13.90	56.30	199.00	10217.74	8.30	49.03	4.38	11.56
116	10250	14.90	55.30	201.00	10246.80	14.79	53.23	10.55	3.43
117	10281	16.20	54.60	201.00	10276.66	21.95	58.00	17.35	4.24
118	10312	19.90	54.60	204.00	10306.13	30.20	63.56	25.18	11.94
119	10343	23.50	55.60	203.00	10334.93	40.10	70.11	34.58	11.67
120	10374	27.60	56.40	203.00	10362.89	51.75	77.58	45.67	13.27

Report #: 1

Date: 25-Sep-14

RYAN DIRECTIONAL SERVICES, INC.

A NABORS COMPANY

Ryan Job # 8112

SURVEY REPORT

Customer:	Oasis Petroleum N.A. LLC
Well Name:	Wade Federal 5300 21-30 13B
Rig #:	Nabors B-25
API #:	33-053-06131
Calculation Method:	Minimum Curvature Calculation

MWD Operator:	Daniel Ogden
Directional Drillers:	RPM
Survey Corrected To:	True North
Vertical Section Direction:	85.39
Total Correction:	8.15
Temperature Forecasting Model (Chart Only):	Logarithmic

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
121	10405	31.70	57.70	203.00	10389.83	65.25	85.91	58.54	13.39
122	10436	35.30	58.20	203.00	10415.68	80.44	94.99	73.04	11.65
123	10468	39.00	56.80	206.00	10441.18	97.51	105.38	89.33	11.86
124	10499	43.00	57.20	206.00	10464.57	115.40	116.45	106.38	12.93
125	10530	46.50	56.40	206.00	10486.58	134.56	128.40	124.64	11.44
126	10561	47.80	52.40	204.00	10507.67	154.03	141.63	143.11	10.35
127	10592	51.00	51.60	204.00	10527.84	173.67	156.13	161.65	10.51
128	10623	54.70	52.50	204.00	10546.56	194.32	171.31	181.13	12.16
129	10654	58.10	55.30	204.00	10563.71	216.33	186.51	202.00	13.30
130	10685	62.20	57.50	204.00	10579.14	239.85	201.38	224.39	14.59
131	10716	66.50	58.20	206.00	10592.56	264.62	216.24	248.05	14.02
132	10747	69.10	58.20	206.00	10604.27	290.15	231.37	272.44	8.39
133	10778	69.20	58.60	206.00	10615.30	315.97	246.55	297.11	1.25
134	10809	70.10	58.50	206.00	10626.08	341.90	261.71	321.91	2.92
135	10840	73.50	57.90	206.00	10635.76	368.09	277.23	346.93	11.12
136	10871	76.00	57.10	208.00	10643.92	394.52	293.30	372.15	8.44
137	10903	78.20	56.90	208.00	10651.06	421.96	310.29	398.31	6.90
138	10934	82.30	57.00	208.00	10656.31	448.82	326.94	423.91	13.23
139	10965	86.90	57.40	210.00	10659.23	476.01	343.66	449.85	14.89
140	10990	89.50	57.20	212.00	10660.01	498.06	357.16	470.87	10.43
141	11051	89.90	56.00	221.00	10660.33	551.52	390.73	521.80	2.07
142	11082	90.30	56.00	217.00	10660.28	578.53	408.07	547.50	1.29
143	11112	90.20	56.30	219.00	10660.15	604.70	424.78	572.41	1.05
144	11143	89.70	57.50	221.00	10660.17	631.95	441.71	598.38	4.19
145	11174	89.50	57.30	217.00	10660.39	659.32	458.41	624.50	0.91
146	11205	89.40	57.90	221.00	10660.69	686.74	475.02	650.67	1.96
147	11236	88.50	59.60	221.00	10661.26	714.45	491.10	677.17	6.20
148	11267	88.70	59.70	219.00	10662.01	742.36	506.76	703.91	0.72
149	11298	89.30	60.50	221.00	10662.55	770.39	522.21	730.78	3.23
150	11329	90.00	62.40	222.00	10662.74	798.72	537.02	758.01	6.53
151	11360	90.00	63.10	219.00	10662.74	827.33	551.22	785.57	2.26
152	11391	90.20	63.60	222.00	10662.69	856.06	565.12	813.27	1.74
153	11421	89.60	65.30	224.00	10662.74	884.08	578.06	840.34	6.01
154	11452	89.80	65.80	221.00	10662.90	913.24	590.89	868.56	1.74
155	11483	89.80	65.90	224.00	10663.01	942.45	603.57	896.85	0.32
156	11514	88.60	68.00	226.00	10663.44	971.86	615.71	925.37	7.80
157	11545	88.70	67.90	222.00	10664.18	1001.42	627.34	954.09	0.46
158	11576	88.80	68.50	224.00	10664.85	1031.03	638.85	982.87	1.96
159	11607	88.60	70.60	226.00	10665.55	1060.85	649.68	1011.90	6.80
160	11638	88.70	70.60	224.00	10666.29	1090.81	659.97	1041.14	0.32
161	11669	88.40	71.00	226.00	10667.07	1120.80	670.17	1070.40	1.61
162	11701	87.80	72.90	230.00	10668.13	1151.91	680.07	1100.81	6.22
163	11732	88.00	72.70	226.00	10669.27	1182.14	689.24	1130.40	0.91
164	11762	88.40	73.50	228.00	10670.21	1211.44	697.95	1159.09	2.98
165	11793	88.70	75.10	230.00	10670.99	1241.85	706.34	1188.92	5.25
166	11824	88.80	75.70	226.00	10671.67	1272.37	714.15	1218.92	1.96
167	11855	89.30	76.40	230.00	10672.18	1302.96	721.62	1249.00	2.77
168	11886	89.20	78.00	231.00	10672.59	1333.64	728.49	1279.22	5.17
169	11916	89.10	77.90	230.00	10673.03	1363.38	734.75	1308.56	0.47
170	11947	89.00	78.80	231.00	10673.55	1394.14	741.01	1338.92	2.92
171	11978	88.80	79.60	233.00	10674.14	1424.96	746.82	1369.36	2.66
172	12009	89.10	79.90	235.00	10674.71	1455.80	752.33	1399.86	1.37
173	12040	89.50	79.90	235.00	10675.09	1486.66	757.77	1430.38	1.29
174	12071	89.70	79.50	237.00	10675.31	1517.50	763.31	1460.88	1.44
175	12101	89.60	79.60	233.00	10675.49	1547.35	768.75	1490.38	0.47
176	12133	89.30	80.20	235.00	10675.80	1579.20	774.37	1521.88	2.10
177	12164	88.40	81.10	237.00	10676.42	1610.08	779.40	1552.46	4.11
178	12194	88.90	80.50	233.00	10677.13	1639.98	784.20	1582.07	2.60
179	12225	89.10	81.60	235.00	10677.67	1670.89	789.02	1612.69	3.61
180	12256	87.90	82.90	237.00	10678.48	1701.83	793.20	1643.39	5.71

Report #: 1

Date: 25-Sep-14

RYAN DIRECTIONAL SERVICES, INC.

A NABORS COMPANY

Ryan Job # 8112

SURVEY REPORT

Customer:	Oasis Petroleum N.A. LLC
Well Name:	Wade Federal 5300 21-30 13B
Rig #:	Nabors B-25
API #:	33-053-06131
Calculation Method:	Minimum Curvature Calculation

MWD Operator:	Daniel Ogden
Directional Drillers:	RPM
Survey Corrected To:	True North
Vertical Section Direction:	85.39
Total Correction:	8.15
Temperature Forecasting Model (Chart Only):	Logarithmic

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
181	12286	87.80	82.80	233.00	10679.60	1731.78	796.93	1673.14	0.47
182	12318	87.90	83.50	235.00	10680.80	1763.73	800.74	1704.89	2.21
183	12349	87.40	85.70	237.00	10682.08	1794.70	803.66	1735.72	7.27
184	12380	87.10	86.10	235.00	10683.56	1825.66	805.87	1766.61	1.61
185	12411	87.40	86.40	237.00	10685.05	1856.62	807.90	1797.50	1.37
186	12441	88.80	87.30	239.00	10686.05	1886.59	809.54	1827.44	5.55
187	12473	88.80	86.90	237.00	10686.72	1918.57	811.16	1859.39	1.25
188	12504	88.70	88.10	237.00	10687.39	1949.54	812.51	1890.36	3.88
189	12536	89.00	89.70	239.00	10688.03	1981.48	813.13	1922.34	5.09
190	12631	88.80	89.40	242.00	10689.86	2076.21	813.87	2017.32	0.38
191	12725	90.90	89.90	244.00	10690.10	2169.94	814.45	2111.31	2.30
192	12819	89.70	89.10	244.00	10689.61	2263.70	815.27	2205.31	1.53
193	12913	90.10	89.40	244.00	10689.78	2357.48	816.50	2299.30	0.53
194	13007	90.80	90.50	242.00	10689.04	2451.18	816.58	2393.29	1.39
195	13101	89.70	90.10	248.00	10688.63	2544.83	816.09	2487.29	1.25
196	13195	87.20	91.30	249.00	10691.17	2638.39	814.94	2581.24	2.95
197	13288	87.20	91.90	249.00	10695.71	2730.73	812.35	2674.09	0.64
198	13382	86.80	92.00	248.00	10700.63	2823.99	809.15	2767.91	0.44
199	13476	88.30	91.00	251.00	10704.65	2917.36	806.70	2861.79	1.92
200	13570	88.40	90.50	251.00	10707.36	3010.91	805.47	2955.74	0.54
201	13663	89.00	89.90	255.00	10709.47	3103.56	805.14	3048.71	0.91
202	13757	89.80	89.10	255.00	10710.45	3197.31	805.96	3142.70	1.20
203	13851	90.30	88.50	257.00	10710.37	3291.15	807.93	3236.68	0.83
204	13945	90.10	90.00	255.00	10710.04	3384.93	809.16	3330.67	1.61
205	14039	90.20	90.00	255.00	10709.80	3478.63	809.16	3424.67	0.11
206	14134	89.30	89.30	257.00	10710.21	3573.36	809.74	3519.67	1.20
207	14229	89.50	89.30	258.00	10711.20	3668.13	810.90	3614.65	0.21
208	14323	90.60	89.00	258.00	10711.12	3761.93	812.30	3708.64	1.21
209	14418	90.20	88.90	260.00	10710.46	3856.74	814.04	3803.62	0.43
210	14513	86.20	88.80	257.00	10713.44	3951.51	815.94	3898.54	4.21
211	14608	88.00	89.70	258.00	10718.25	4046.16	817.18	3993.40	2.12
212	14703	89.00	90.90	258.00	10720.74	4140.78	816.69	4088.37	1.64
213	14797	89.70	90.00	260.00	10721.80	4234.41	815.95	4182.36	1.21
214	14892	91.40	90.20	258.00	10720.89	4329.08	815.78	4277.35	1.80
215	14986	89.60	89.90	262.00	10720.07	4422.76	815.70	4371.34	1.94
216	15081	86.50	89.40	258.00	10723.30	4517.43	816.28	4466.27	3.31
217	15176	88.60	89.90	262.00	10727.36	4612.07	816.86	4561.18	2.27
218	15271	88.50	89.30	260.00	10729.77	4706.79	817.52	4656.14	0.64
219	15365	91.00	89.60	260.00	10730.18	4800.54	818.42	4750.13	2.68
220	15460	90.50	90.00	264.00	10728.93	4895.25	818.76	4845.12	0.67
221	15555	89.50	90.00	262.00	10728.93	4989.94	818.76	4940.12	1.05
222	15649	89.30	90.80	264.00	10729.92	5083.58	818.10	5034.11	0.88
223	15744	90.70	90.40	266.00	10729.92	5178.18	817.10	5129.11	1.53
224	15839	91.30	89.70	264.00	10728.26	5272.86	817.02	5224.09	0.97
225	15934	90.80	90.90	266.00	10726.52	5367.49	816.52	5319.07	1.37
226	16028	91.00	90.00	264.00	10725.04	5461.11	815.79	5413.05	0.98
227	16123	88.40	89.30	266.00	10725.54	5555.84	816.37	5508.04	2.83
228	16218	89.50	89.60	264.00	10727.28	5650.58	817.28	5603.02	1.20
229	16313	89.10	90.00	267.00	10728.44	5745.29	817.61	5698.01	0.60
230	16407	90.20	90.10	266.00	10729.02	5838.98	817.53	5792.01	1.18
231	16502	89.60	90.90	266.00	10729.18	5933.60	816.70	5887.00	1.05
232	16596	88.40	90.70	269.00	10730.82	6027.17	815.39	5980.98	1.29
233	16690	88.40	90.70	269.00	10733.45	6120.73	814.24	6074.94	0.00
234	16785	88.70	89.40	271.00	10735.85	6215.38	814.16	6169.90	1.40
235	16880	88.90	89.10	271.00	10737.84	6310.14	815.40	6264.87	0.38
236	16975	89.10	90.10	269.00	10739.50	6404.87	816.06	6359.86	1.07
237	17069	89.30	89.70	267.00	10740.81	6498.57	816.23	6453.85	0.48
238	17164	89.20	91.30	271.00	10742.05	6593.18	815.40	6548.83	1.69
239	17258	89.60	90.70	271.00	10743.04	6686.73	813.76	6642.81	0.77
240	17353	90.20	89.60	267.00	10743.20	6781.40	813.51	6737.81	1.32

Report #: 1

Date: 25-Sep-14

RYAN DIRECTIONAL SERVICES, INC.

A NABORS COMPANY

Ryan Job # 8112

SURVEY REPORT

Customer:	Oasis Petroleum N.A. LLC
Well Name:	Wade Federal 5300 21-30 13B
Rig #:	Nabors B-25
API #:	33-053-06131
Calculation Method:	Minimum Curvature Calculation

MWD Operator:	Daniel Ogden
Directional Drillers:	RPM
Survey Corrected To:	True North
Vertical Section Direction:	85.39
Total Correction:	8.15
Temperature Forecasting Model (Chart Only):	Logarithmic

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
241	17448	89.60	89.40	271.00	10743.37	6876.15	814.34	6832.80	0.67
242	17543	90.10	88.80	271.00	10743.62	6970.95	815.83	6927.79	0.82
243	17637	89.60	89.40	271.00	10743.87	7064.75	817.31	7021.78	0.83
244	17732	90.00	89.40	271.00	10744.20	7159.52	818.30	7116.77	0.42
245	17826	90.70	90.20	271.00	10743.62	7253.24	818.63	7210.77	1.13
246	17921	90.60	90.90	273.00	10742.55	7347.85	817.72	7305.76	0.74
247	18015	91.00	90.40	271.00	10741.23	7441.44	816.65	7399.74	0.68
248	18110	90.30	91.00	273.00	10740.15	7536.03	815.49	7494.73	0.97
249	18205	91.20	90.50	271.00	10738.91	7630.60	814.25	7589.71	1.08
250	18300	89.60	90.70	275.00	10738.25	7725.20	813.25	7684.70	1.70
251	18394	90.50	89.40	275.00	10738.17	7818.89	813.17	7778.70	1.68
252	18489	89.10	89.90	273.00	10738.50	7913.62	813.75	7873.69	1.56
253	18584	89.10	90.20	273.00	10739.99	8008.30	813.67	7968.68	0.32
254	18679	90.10	91.50	275.00	10740.65	8102.86	812.26	8063.66	1.73
255	18773	90.20	91.50	273.00	10740.41	8196.33	809.80	8157.63	0.11
256	18868	90.20	91.10	275.00	10740.08	8290.82	807.64	8252.61	0.42
257	18963	89.50	91.40	273.00	10740.32	8385.32	805.57	8347.58	0.80
258	19058	90.30	91.20	275.00	10740.49	8479.82	803.41	8442.56	0.87
259	19152	90.00	90.70	273.00	10740.24	8573.37	801.86	8536.54	0.62
260	19247	89.00	91.30	273.00	10741.07	8667.91	800.20	8631.52	1.23
261	19342	89.60	90.50	273.00	10742.23	8762.47	798.71	8726.50	1.05
262	19437	90.50	90.00	275.00	10742.15	8857.12	798.29	8821.50	1.08
263	19532	88.30	88.00	275.00	10743.15	8951.92	799.95	8916.47	3.13
264	19626	88.50	87.40	275.00	10745.77	9045.81	803.72	9010.36	0.67
265	19721	89.10	87.70	276.00	10747.76	9140.72	807.78	9105.25	0.71
266	19816	89.80	87.00	276.00	10748.67	9235.66	812.17	9200.14	1.04
267	19911	90.40	87.50	273.00	10748.51	9330.61	816.73	9295.03	0.82
268	20005	87.60	90.00	275.00	10750.15	9424.41	818.78	9388.98	3.99
269	20100	88.20	89.80	276.00	10753.63	9519.05	818.95	9483.92	0.67
270	20195	89.60	89.40	276.00	10755.45	9613.78	819.61	9578.89	1.53
271	20290	90.50	89.50	278.00	10755.37	9708.54	820.52	9673.89	0.95
272	20384	89.30	87.40	278.00	10755.53	9802.40	823.06	9767.85	2.57
273	20479	89.00	85.90	276.00	10756.94	9897.36	828.61	9862.67	1.61
274	20498	88.70	85.50	276.00	10757.32	9916.36	830.04	9881.61	2.63
275	20650	88.70	85.50	276.00	10760.77	10068.32	841.96	10033.11	0.00

Oasis Petroleum N. A., LLC

Wade Federal 5300 21-30 13B

1,574' FNL & 270' FWL
Lot 2 Section 30, T153N, R100W
Baker Field / Middle Bakken
McKenzie County, North Dakota

BOTTOM HOLE LOCATION:

818.22' north & 9,943.24' east of surface location or approx.
755.75' FSL & 300.13' FEL, NE NE Section 29, T153N, R100W

Prepared for:
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WELL EVALUATION

Wade Federal 5300 21-30 13B

Figure 1: Nabors B25 drilling the Oasis Petroleum, Wade Federal 5300 21-30 13B well during November of 2014, south of Williston in McKenzie County, North Dakota. (Photos by Hannah Thatcher, well-site geologist)

INTRODUCTION

The Oasis Petroleum North America, LLC - Wade Federal 5300 21-30 13B well [Lot 2 Section 30, T153N, R100W] is located approximately 31 miles south of Williston, North Dakota **(Figure 1)**. The well was spud on October 26, 2014 by Nabors rig B25, and represents a continuation of Oasis Petroleum's development of the Middle Bakken and Three Forks within Baker Field. The well was planned to drill an approximately 9,623' long lateral, along a proposed azimuth of 85.39°. Upon completion of drilling operations, the well will be enhanced for production by a multistage fracture stimulation.

OFFSET CONTROL INFORMATION

Three offsets were used to establish a prognosis of formation tops and thicknesses to be used as a control for the Wade Federal 5300 21-30 13B **(Figure 2).** Offset formation thicknesses were compared and an appropriate curve landing target was selected based on the average of these thicknesses. Structural data was also used during the lateral. The three offsets include the Wade Federal 5300 21-30 12T which was spud on September 30th, 2014, is located on the same pad as the subject well and was considered the primary offset. The Chalmers 5301 44-24 2TR was spud on the June 11th, 2014 and is located 0.50 miles north of the subject well. The Wade Federal 5300 21-30 14T2 was spud on September 28th, 2014 and is located on the same pad as the subject well.

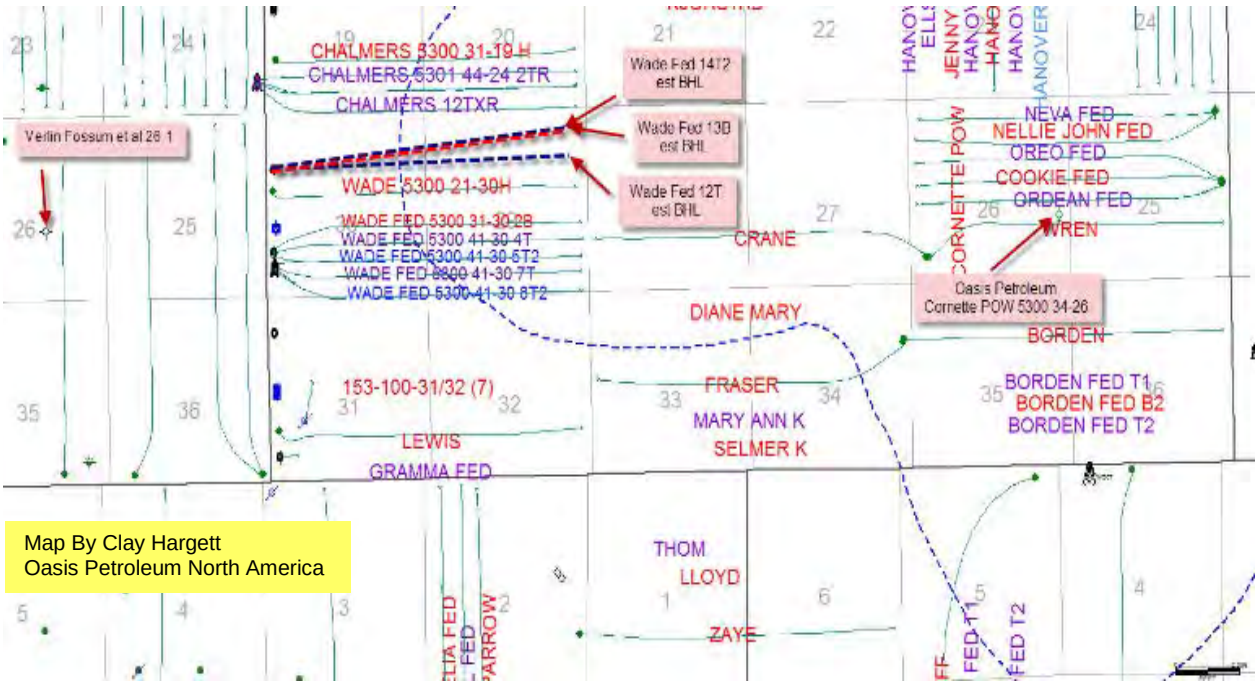


Figure 2: Offsetting control wells in relation to the Wade Federal 5300 21-30 13B well.

GEOLOGIC EVALUATION

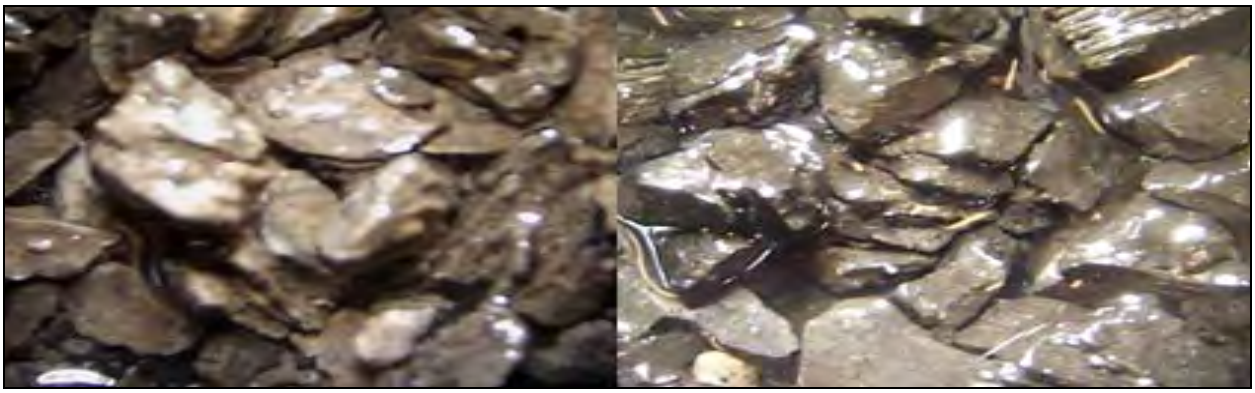
Methods:

Geologic supervision was provided by Sunburst Consulting, Inc. with two pairs of well-site geologists. A digital gas detector and chromatograph were interfaced with a Pason electronic data recorder system (EDR). The unit of measure for gas on this well was units, which was defined as 100 units equaling 1% methane equivalent in air. The EDR provided drill rate, on-off bottom and pump strokes to the gas detection computer and received total gas information from Sunburst for viewing around location. Lagged samples were caught by the rig crew in 30' intervals from 8,200' to 10,810' MD, 10' intervals from 10,810' to 11,050' MD, and 30' intervals from 11,050' to 20,560' MD. Wet and dry cuttings were examined under a bi-ocular microscope and complete lithologic descriptions and sampling intervals are provided in the lithology document within this evaluation. The set of dry cuttings collected during the duration of the well were sent to the State of North Dakota.

Zones of Interest:

The **Mission Canyon Formation** was drilled at 9,331' TVD (-7,307') and is comprised largely of light gray, *lime* mudstone followed by gray to tan, earthy textured, argillaceous *lime* mudstone. Gas shows within the Mission Canyon ranged from 30 to 80 units against 10.9 pound/gallon mud weight.

The top of the **Lodgepole Formation** was logged at 9,887' TVD (-7,863'). In general the Lodgepole can be described as a medium to dark gray-brown, argillaceous *lime* mudstone with crystalline texture and trace amounts of disseminated pyrite **(Figure 3)**. The False Bakken, drilled at 10,613' TVD (-8,589'), is comprised of very dark brown to black, calcareous, slightly pyritic shale **(Figure 4)**. Strong hydrocarbon shows in the lower 100' Lodgepole were as high as 297 units, which suggest that some of the oil and gas from the Bakken shale may be exploiting fractures.



Figures 3 & 4: Wet cutting of the Lodgepole Formation (left) and False Bakken sub-interval (right) at 10x.

The **Bakken Formation** has four formal members, an upper and lower black, fissile, organic-rich pyritic shale, and a clastic and carbonate middle and lower member. The Upper Bakken Shale Member was drilled at 10,624' TVD (-8,600') with sample returns typically described as black, carbonaceous, *petroliferous* shale with trace amounts of disseminated pyrite **(Figure 5)**. The Middle Bakken Member, penetrated at 10,642' TVD (-8,618'), consists of a varying thickness and sequence of siltstone, limestone and silty sandstone **(Figure 6)**. *Trace, spotty light brown oil staining* was present along with gas shows as high as 1,545 units.

Figures 5 & 6: Wet cuttings of the upper Bakken (left) & Middle Bakken silty sandstone (right) at 10x.

Geo-steering:

Kick-off point was established from the isopach of the "base last salt" marker to the Three Forks "target" from the offset wells. While drilling the curve, measured gamma signatures were compared to those of the three offsets and aided in the landing of the curve. The landing target was confirmed by the depth of the False Bakken, which was consistent with the offset wells. The curve successfully landed within the Middle Bakken Member at a depth of 11,050' MD (10,660' TVD). Directional tools were then pulled out of the hole and a string of 7" casing was set to 11,006' MD and then cemented by Schlumberger.

Samples from the target zone varied in porosity, oil staining, color, cementation, and pyrite content. Stratigraphic location in the target zone was based on these sample observations along with gas shows, drill rates and gamma ray values.

The Oasis Petroleum North America, LLC prospect geologist defined the target zone as a 10' zone that began 15' below the upper shale and ended 15 feet above the lower shale. The target zone consisted of an upper silty sandstone interval, reading gamma from 115-120 counts (A marker). The center of the target interval was comprised of a lighter colored silty sandstone, with a higher gamma signature ranging between 120 and 130 counts (B marker). The base of the target zone was characterized as a clean, silty sandstone with gamma ranging from 70 to 110 counts (C marker). A-C gamma markers were used for determining depth within the target interval **(Figure 7)**.

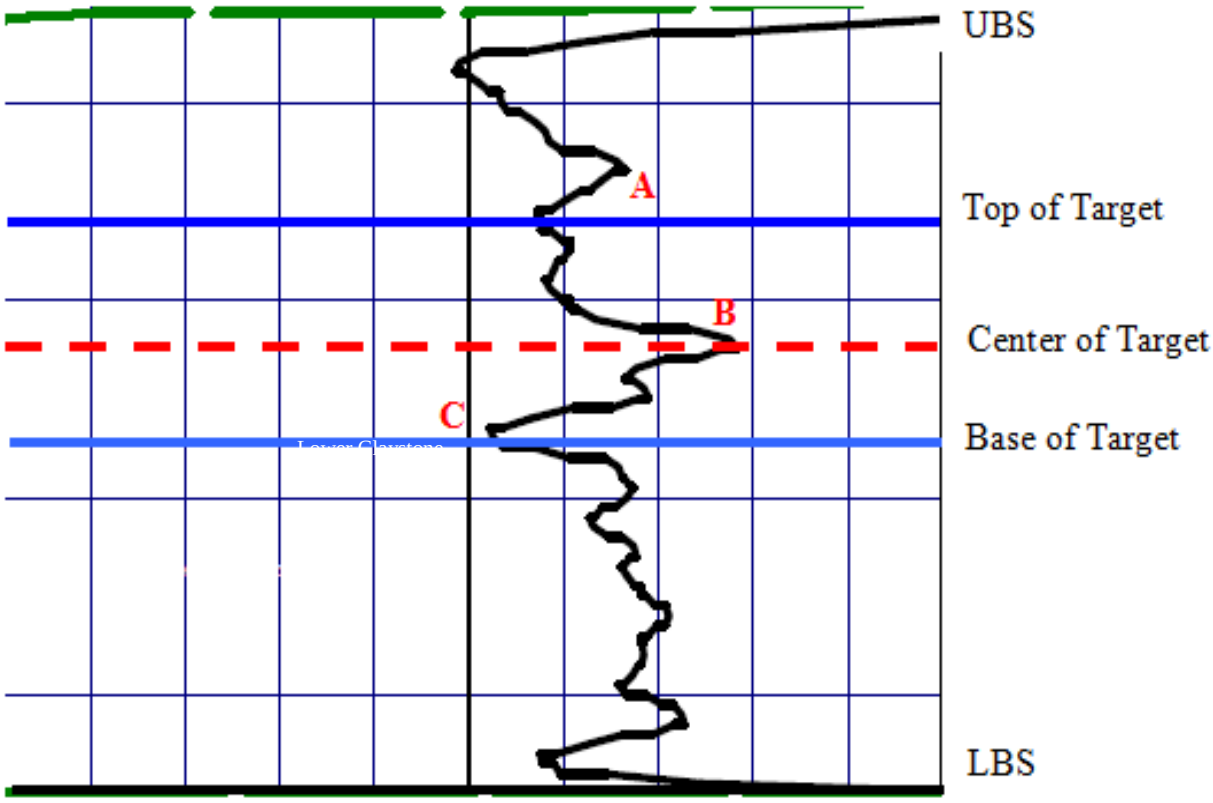


Figure 7: Offset, TVD gamma profile of the Middle Bakken Member (0-200 api. scale).

Using the aforementioned information gathered during drilling, offset log information and the structural data provided from Oasis Petroleum North America, LLC., well-site geologists were able to anticipate, and interpret the local, apparent dip during drilling of the Wade Federal 5300 21-30 13B well. A total depth of 20,560' MD was reached on November 10th, 2014 at 18:15 CT. The target resultant was 100% within the Middle Bakken Member. The resulting structure of the Middle Bakken was a drop in TVD of 85.70' over 9,699', resulting in an overall down-dip of -0.51° as portrayed on the dip profile **(Figure 8)**.

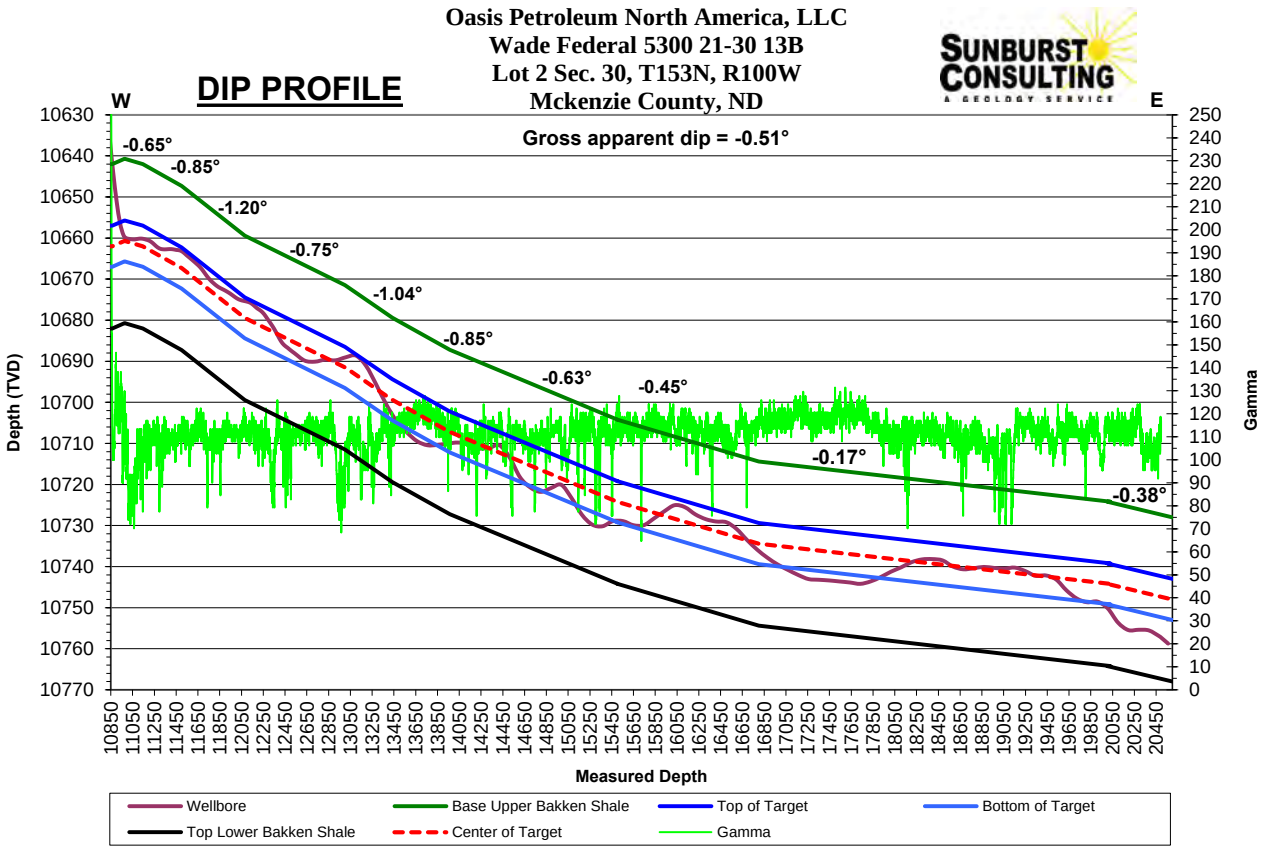


Figure 8: Well profile representing estimated dip value & gamma for the Wade Federal 5300 21-30 13B.

Hydrocarbons:

Gas monitoring and fluid gains were monitored to evaluate the viability of this reservoir during the drilling of the well. In the invert mud system, hydrostatic conditions were maintained near balance. This allowed for gas and fluid gains from the well to be monitored.

Background gas observed during the lateral ranged from 50 to 450 units. The lateral was drilled with saltwater drilling fluid, with a mud weight of 9.5-10.50 pounds/gallon. Gas shows typically ranged from 200 to 300 units. In the latter half of the lateral, gas shows ranged up to 1,618 units and connection gases were observed up to 1,850 units **(Figure 9)**. C1-C4 gas components were observed throughout the course of the lateral. Trip gases were observed as high as 200 units. Oil shows were very light throughout the lateral ranging from 0-3% in sample. When present it presented as a *light brown spotty oil stain that yielded a slow to moderate streaming to diffuse light green cut fluorescence.*

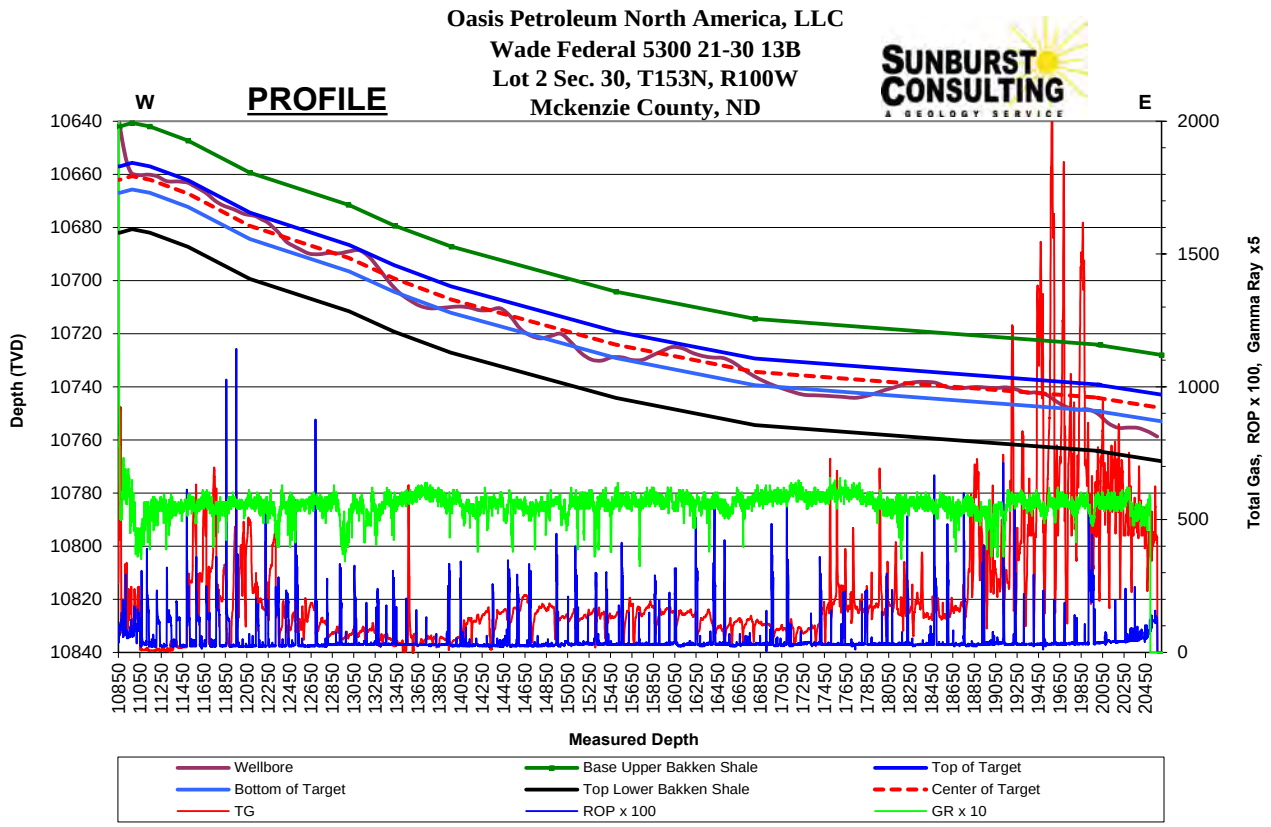


Figure 9: Well profile illustrating total gas values for the Wade Federal 5300 21-30 13B.

SUMMARY

Nabors rig B25 successfully drilled a two-section, horizontal well within the Middle Bakken Member at the Wade Federal 5300 21-30 13B. A net of 9,699' was drilled within the Middle Bakken. A mud program consisting of diesel invert (9.6–10.5 ppg), during the vertical and curve build sections, and saline based mud (9.5-10.5 ppg), during the lateral maintained stable hole conditions and permitted adequate analysis of gas concentrations.

Projections of dip were successfully used to maintain the well-bore in the Middle Bakken target for 77% of the lateral. Samples from the target consisted of a light to medium gray, silty sandstone with trace to occasional amounts of disseminated pyrite. Intergranular porosity was generally seen throughout the entire lateral. Hydrocarbon shows in the target zone were low to moderate. Samples from the ideal zone contained spotty light brown oil staining.

The Wade Federal 5300 21-30 13B will be fitted with a 4 ½" production liner and swell packers in preparation for a fracture stimulation that will determine the true commercial value of the well. The well currently awaits fracture stimulation.

Respectfully submitted,
Hannah Thatcher
Daniel Haynes
Sunburst Consulting, Inc.
November 11, 2014

WELL DATA SUMMARY

OPERATOR:	**Oasis Petroleum N. A., LLC**
ADDRESS:	1001 Fannin Street, Suite 1500 Houston, Texas 77002
WELL NAME:	**Wade Federal 5300 21-30 13B**
API #:	33-053-06131-00-00
WELL FILE #:	28978
SURFACE LOCATION:	1,574' FNL & 270' FWL Lot 2 Section 30, T153N, R100W
FIELD/ OBJECTIVE:	Baker Field / Middle Bakken
COUNTY, STATE	McKenzie County, North Dakota
BASIN:	Williston
WELL TYPE:	Bakken Horizontal
ELEVATION:	GL: 1,999' KB: 2,024'
SPUD/ RE-ENTRY DATE:	October 26, 2014
BOTTOM HOLE LOCATION:	818.22' north & 9,943.24' east of surface location or approx. 755.75' FSL & 300.13' FEL, NE NE Section 29, T153N, R100W
CLOSURE COORDINATES:	Closure Azimuth: 85.30° Closure Distance: 9,976.85'
TOTAL DEPTH / DATE:	20,560' on November 10, 2014 77% within target interval
TOTAL DRILLING DAYS:	15.5 days
CONTRACTOR:	Nabors B25
PUMPS:	H & H Triplex (stroke length - 12")

TOOLPUSHERS:	Casey Pippenger, Bruce Walter
FIELD SUPERVISORS:	Mike Crow, Travis Handran
CHEMICAL COMPANY:	Fluid Control
MUD ENGINEER:	Keith McCarty, Warren Carlson
MUD TYPE:	Fresh water in surface hole Diesel invert in curve; Salt water in lateral
MUD LOSSES:	Invert Mud: 864 bbls., Salt Water: Not tracked
PROSPECT GEOLOGIST:	Kellie Rassmusen
WELLSITE GEOLOGISTS:	Hannah Thatcher, Daniel Haynes Derry Callender, David Mueller
GEOSTEERING SYSTEM:	Sunburst Digital Wellsite Geological System
ROCK SAMPLING:	30' from 8,200' - 10,810' 10' from 10,810' -11,080' 30' from 11,080' - 20,560' (TD)
SAMPLE EXAMINATION:	trinocular microscope & fluoroscope
SAMPLE CUTS:	Trichloroethylene
GAS DETECTION:	MSI (Mudlogging Systems, Inc.) TGC - total gas with chromatograph Serial Number(s): ML-382
ELECTRIC LOGS:	none attempted
DRILL STEM TESTS:	none attempted
DIRECTIONAL DRILLERS:	RPM Mike Crow, Jordon Jensen
MWD:	Ryan Directional Service Inc. Daniel Ogden

CASING:

Surface: 13.375" 36# J-55 set to 2,070'

Intermediate Surface: 9 5/8" 40# HCL-180 set to 6,050'

Intermediate: 7" 32# P-110 set to 11,006'

SAFETY/ H_2S MONITORING:

Oilind Safety

KEY OFFSET WELLS:

Oasis Petroleum North America

Wade Federal 5300 21-30 12T

SW NW Section 30, T153N, R100W

McKenzie Co., ND

KB: 2,024'

Oasis Petroleum North America

Wade Federal 5300 21-30 14T2

SW NW Section 30, T153N, R100W

McKenzie Co., ND

KB: 2,024'

Oasis Petroleum North America

Chalmers 5301 44-24 4T2R

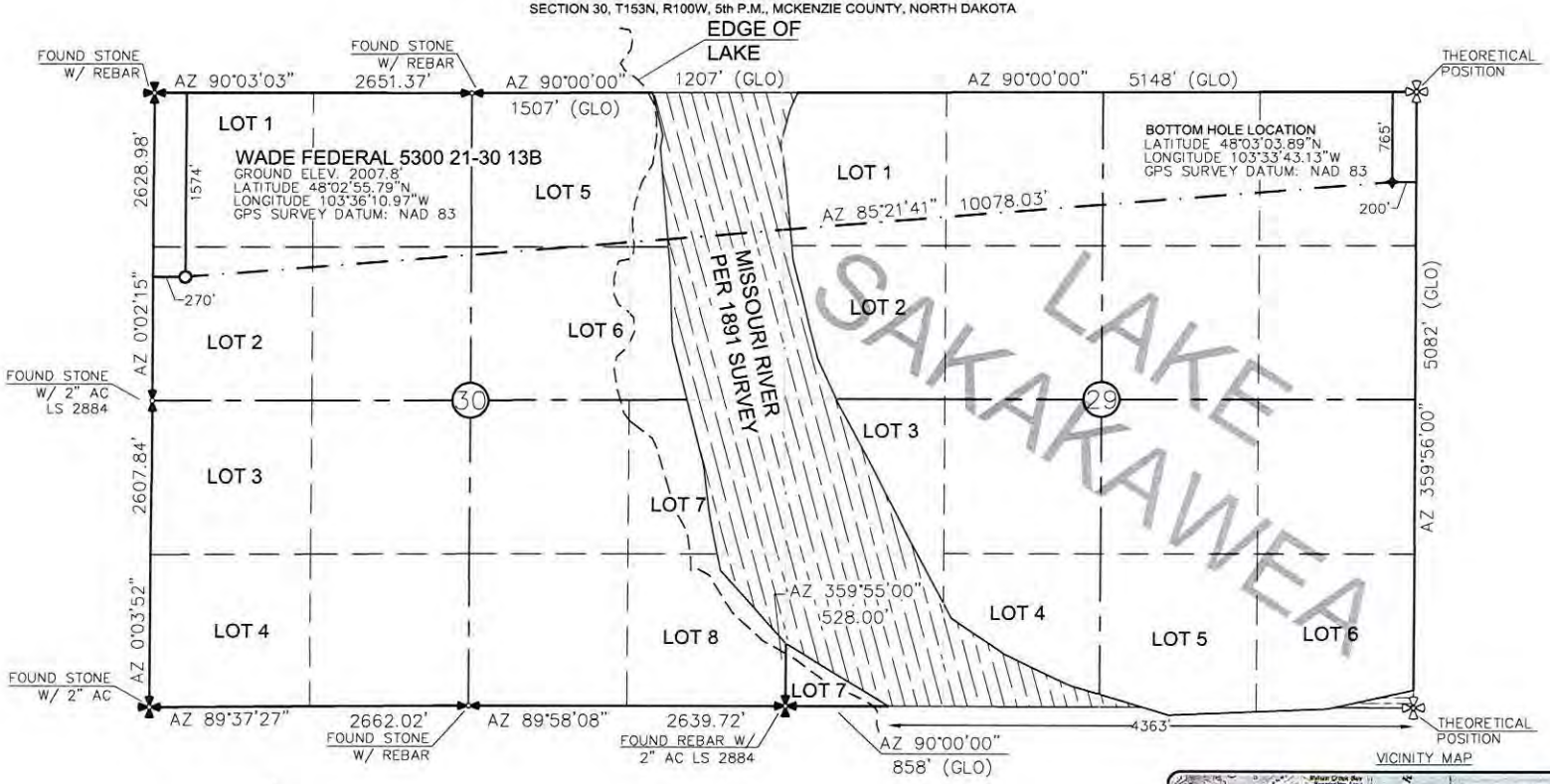
SE SE Section 24, T153N, R101W

McKenzie Co., ND

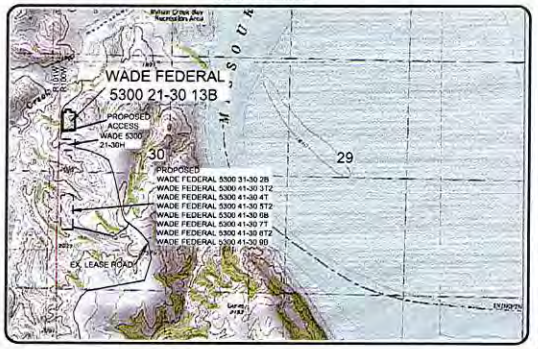
KB: 1,968'

WELL LOCATION PLAT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"WADE FEDERAL 5300 21-30 13B"
1574 FEET FROM NORTH LINE AND 270 FEET FROM WEST LINE
SECTION 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



VICINITY MAP



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DARYL D. KASEMAN LS-3880

REGISTERED PROFESSIONAL LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE: 3/19/14
NORTH DAKOTA

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OASIS PETROLEUM NORTH AMERICA, LLC
WELL LOCATION PLAT
SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.K. Project No.: S14-09-019.01
Checked By: D.D.K. Date: MAR. 2014

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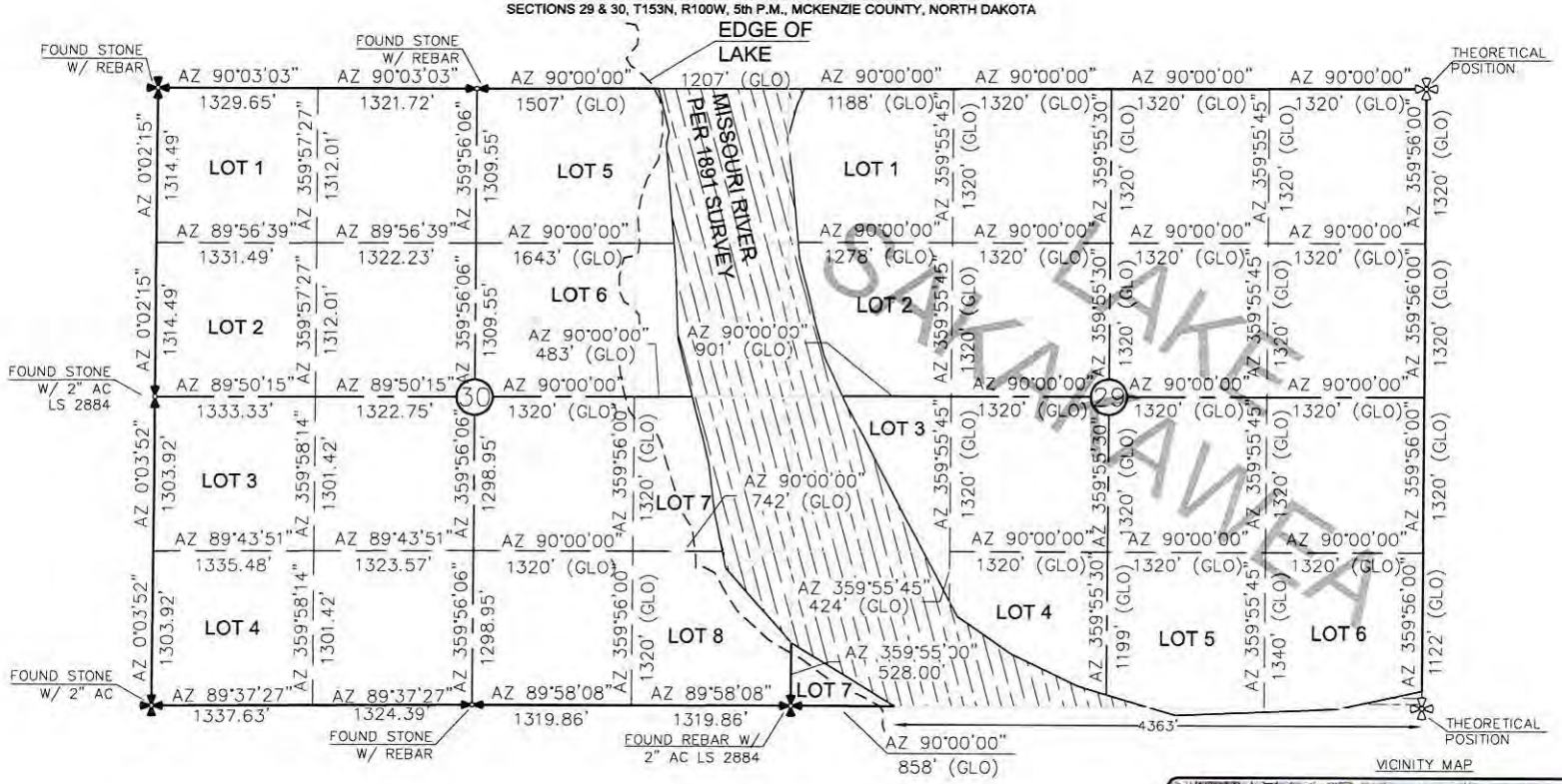
1574 FEET FROM NORTH LINE AND 270 FEET FROM WEST LINE
SECTIONS 29 & 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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LS-3880
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NORTH DAKOTA

VICINITY MAP

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Revision No.	Date	By	Description

OASIS PETROLEUM NORTH AMERICA, LLC
SECTION BREAKDOWN
SECTIONS 29 & 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S14-09-019.01
Checked By: D.D.K. Date: MAR. 2014

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PAD LAYOUT

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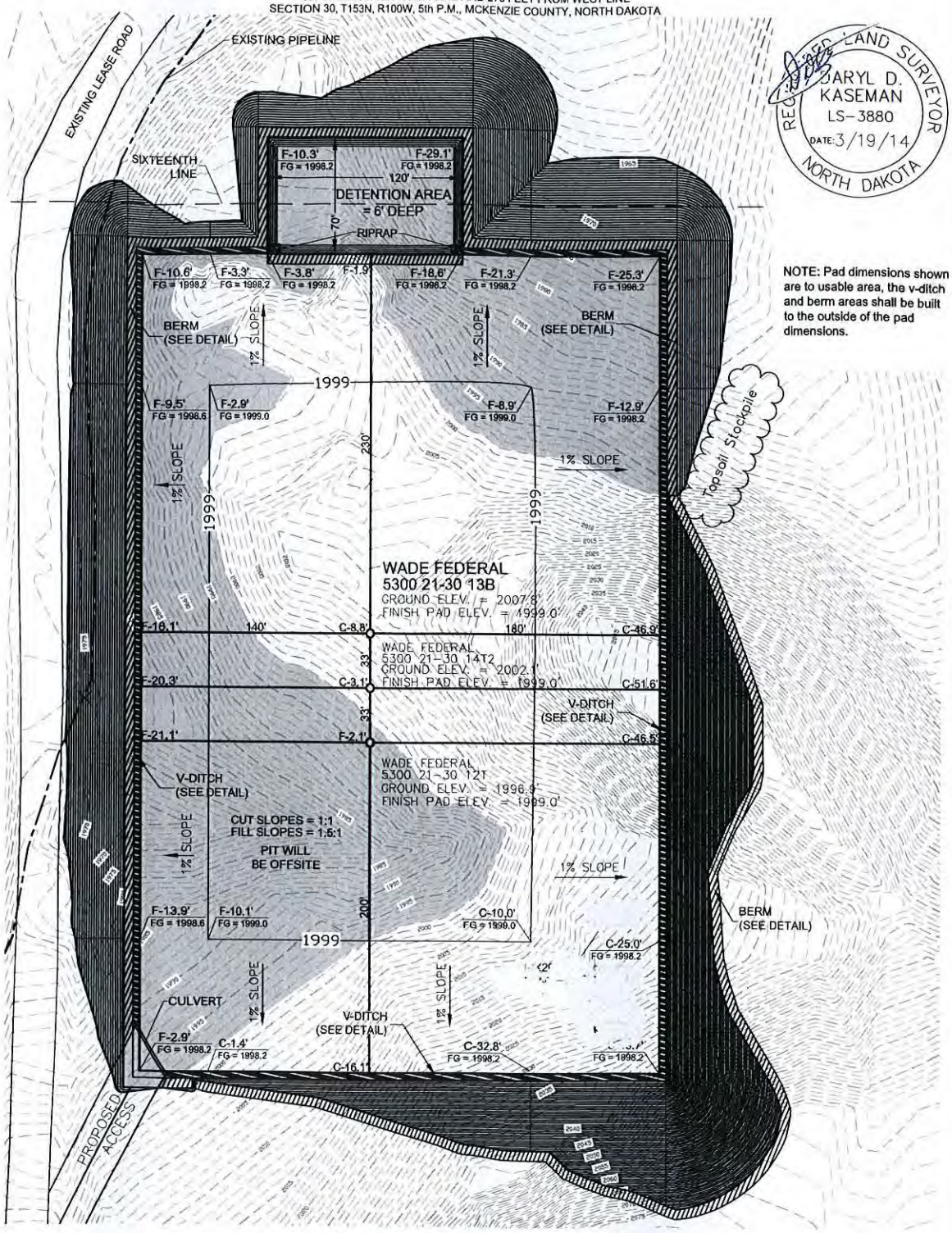
"WADE FEDERAL 5300 21-30 13B"
1574 FEET FROM NORTH LINE AND 270 FEET FROM WEST LINE
SECTION 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



NOTE: Pad dimensions shown are to usable area, the v-ditch and berm areas shall be built to the outside of the pad dimensions.

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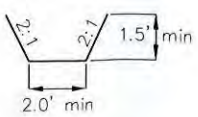
V-DITCH DETAIL



Proposed Contours
Original Contours
BERM
DITCH

0 60
1" = 60'

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NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

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OASIS PETROLEUM NORTH AMERICA, LLC
PAD LAYOUT
SECTION 30, T153N, R100W
MKENZIE COUNTY, NORTH DAKOTA

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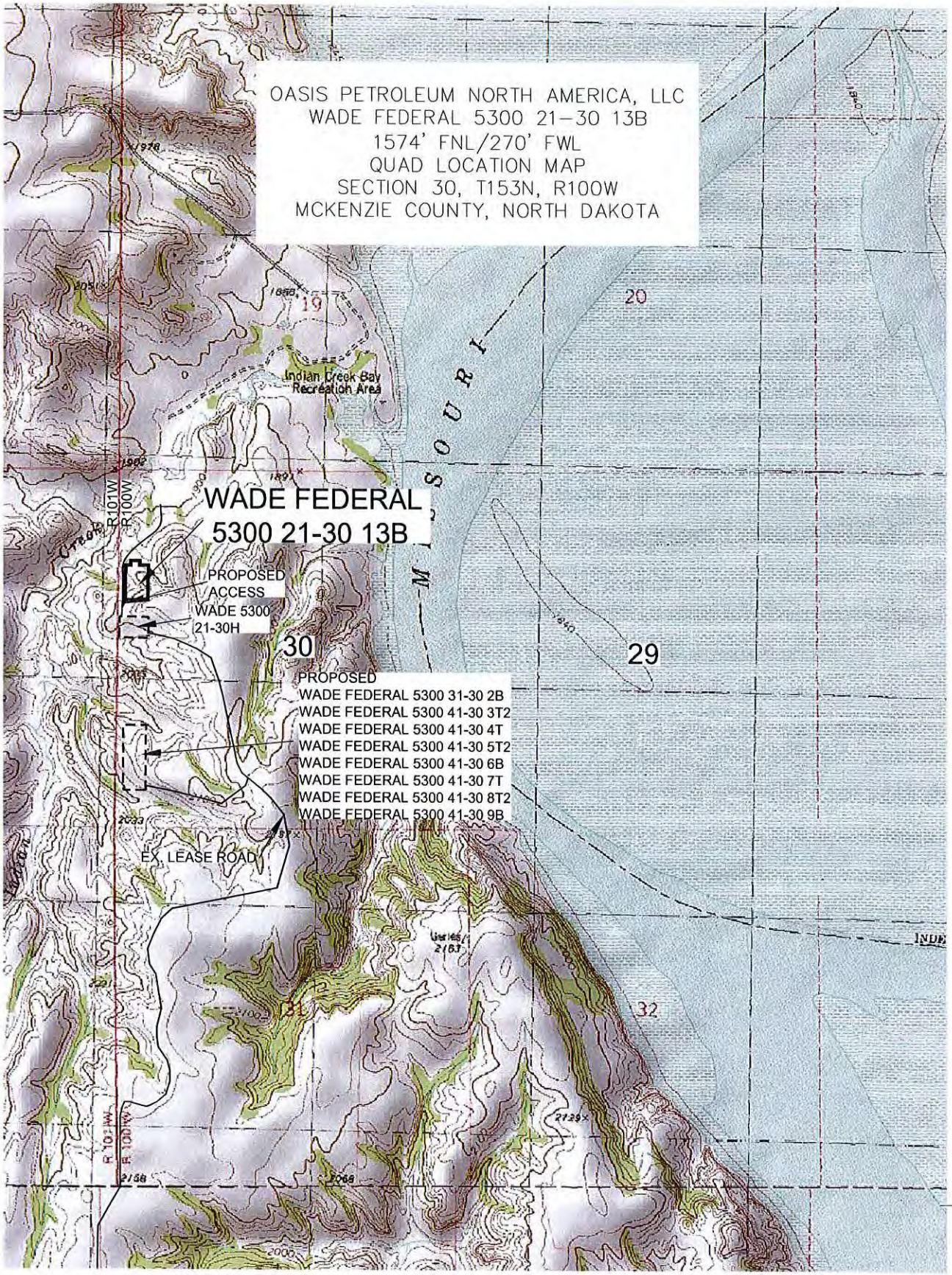

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OASIS PETROLEUM NORTH AMERICA, LLC
QUAD LOCATION MAP
SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

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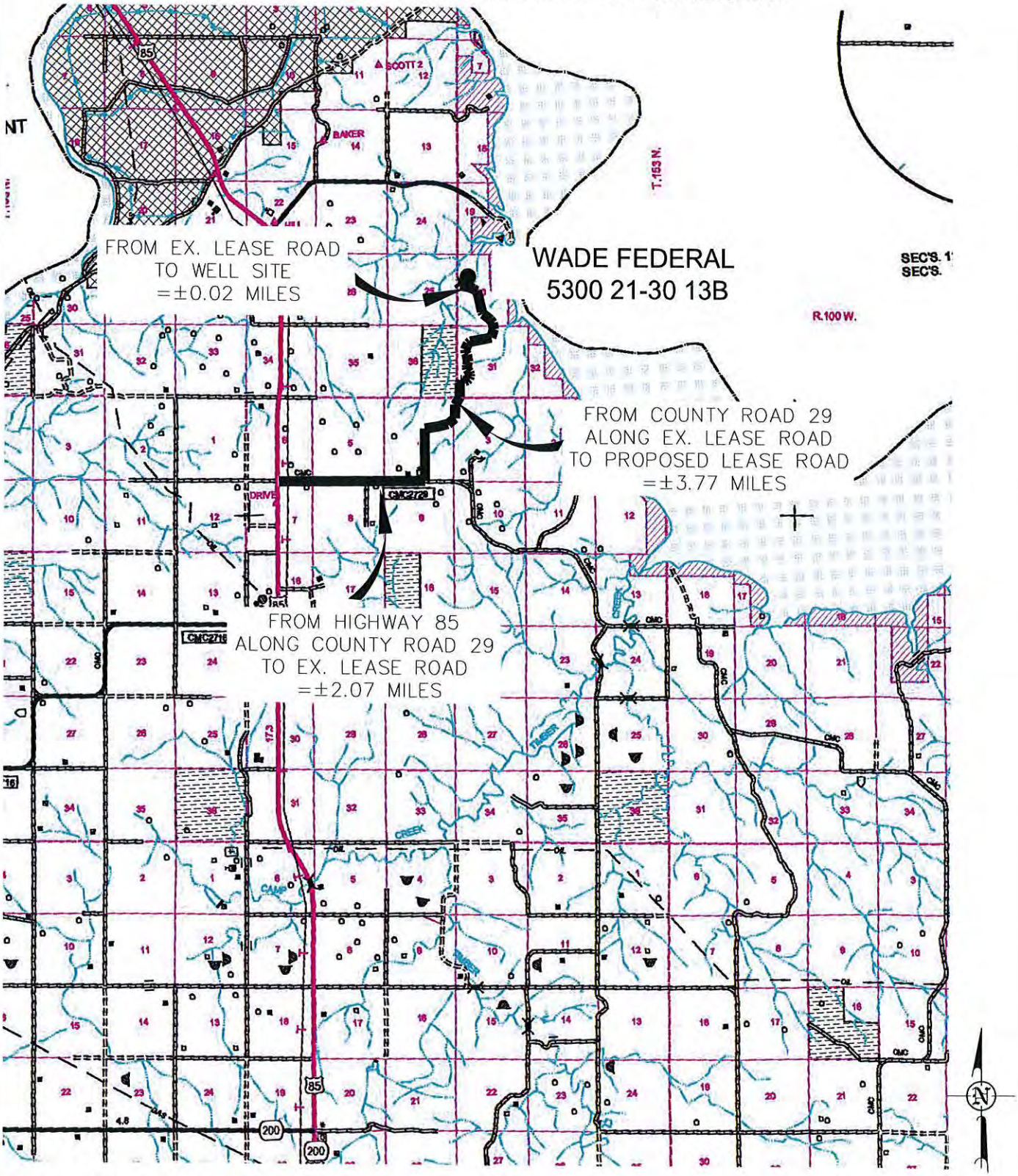
Checked By:	D.D.K.	Date:	MAR. 2014

Revision No.	Date	By	Description

COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"WADE FEDERAL 5300 21-30 13B"
1574 FEET FROM NORTH LINE AND 270 FEET FROM WEST LINE
SECTION 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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COUNTY ROAD MAP
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MCKENZIE COUNTY, NORTH DAKOTA

Drawn By:	B.H.H.	Project No.:	S14-09-019.01
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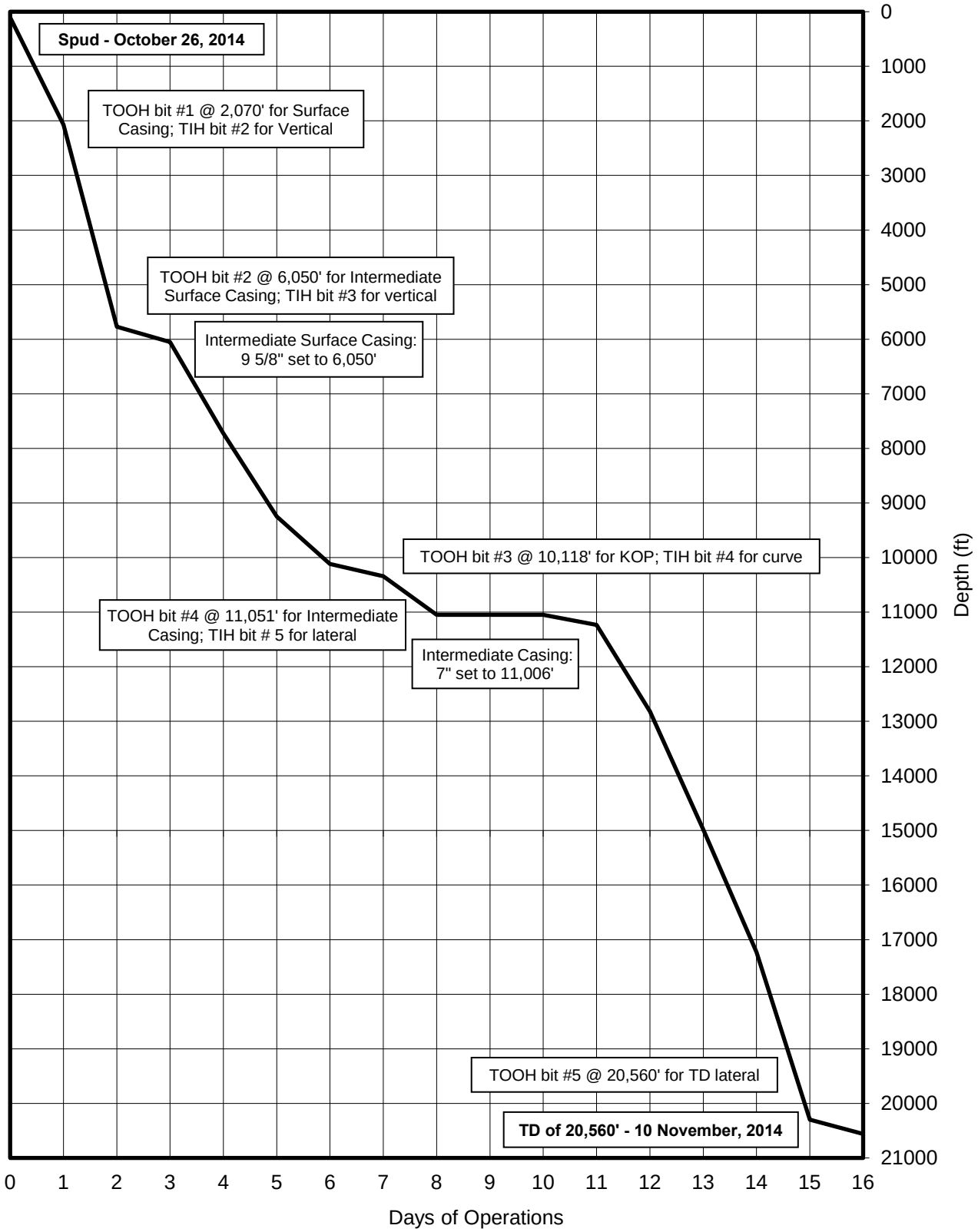
TIME VS. DEPTH

Oasis Petroleum N. A., LLC

Wade Federal 5300 21-30 13B



MORNING REPORT SUMMARY

Day	Date 2014	Depth (0600 Hrs)	24 Hr Footage	Bit #	WOB (Klbs) RT	RPM (RT)	WOB (Klbs) MM	RPM (MM)	PP	SPM 1	SPM 2	GPM	24 Hr Activity Summary	Formation
0	10/26	105'	0'	-	-	-	-	-	-	-	-	-	Rig up, Nipple Up, Choke Line, Test BOP, Pick up BHA.	-
1	10/27	2,070'	4896	1	20	70	-	275	1800	80	80	562	Drill 105' to 2,070', circulate and condition, TOOH, R/U casers, run casing, R/D casers, circulate and condition, R/U cementers, circulate/cement/displace, R/D cementers, install well heads	Pierre
2	10/28	5,772'	3702	2	25	70	10	327	3800	95	95	668	P/U BHA, TIH, FIT test, drill out float & shoe, drill F/2070 to 2160, waiting on third party tools trouble shoot MWD, drill F/2160 to 3630, well control BOP drill, drill F/3630 to 5772	-
3	10/29	6,050'	4897	2	20	30	15	275	3500	80	80	562	Drill F/5772 to 6050, circulate and condition, TOOH, install/remove wear bushing, pull rotating head, install trip nipple, TOOH, L/D BHA, install/remove wear bushing, R/U casers, run casing, weatherford verify landing jt, landed, circulate and condition, R/U cementers, primary cementing, L/D landing joint, R/D floor, dail, ext., elevators, run pack off, install/remove wear bushing, install pack off	Swift
4	10/30	7,725'	1675	3	22	50	25	275	3550	80	80	562	Install/remove wear bushing, weatherford pack off & test, P/U BHA, TIH, install/remove wear bushing, install trip nipple, circulate and condition, drilling cement, float and shoe assembly, float @6032 & shoe @6050, pressure test, drill F/6050 to 6235, rig service, drill F/6235 to 7725	Amsden
5	10/31	9,248'	1523	3	33	55	25	275	3750	80	80	562	Drill F/7725 to 8050, rig service, trouble shoot HMI screens & fix, drill F/8050 to 8626, BOP drill, drill F/8626 to 9248	Ratcliffe
6	11/1	10,118'	870	3	35	40	-	310	3600	75	75	632	Drill from 9,248' to 9,837', service top drive, removed cable wrapped in shives, drill from 9,837' to 10,118, trip out of hole'.	Lodgepole
7	11/2	10,340'	222	4	10	15	40	258	3600	75	75	527	Trip out of hole, lay down BHA, cased hole logs, CBL, RU loggers & held PJSM, cased hole logs, RD loggers, pick up BHA, trip in hole, install/remove wear bushing, remove trip nipple, install Rot head, circulate and condition test MWD & fill hole, trip in hole, circulate and condition fill and test, cut drilling line, trip in hole from 6,049' to 10,118', slide drilling troughing, service rig trouble shoot HMI screens, slide drilling curve from 10,118' to 10340	Lodgepole
8	11/3	11,051'	711	4	25	30	40	258	3800	75	75	527	Run casing, R/D casers, circulate cement, displace (circulate)	Middle Bakken
9	11/4	11,051'	0	3	35	40	45	310	3400	75	75	527	Drill F/9481-9944, Service Rig, Drill 9944-10210 circulate and condition, TOOH	Middle Bakken
10	11/5	11,051'	0	4	25	30	40	258	3600	75	75	527	Run casing, R/D casers, circulate cement, displace (circulate), nipple down BOPS/set casing slips and cut casing, nipple down BOPS and lift, stack/cleaning tanks, run casing/set casing slips, run casing/cut off casing, nipple up BOPS, function test BOPS/blinds/rams/annular/HCR, pre job safety/pressure testing/rig up pressure tester, test BOPS/test well head to BOP (250 low/ 5000 high), pressure test CSG/SHOE/held 2500 psi for 30 min.	Middle Bakken

MORNING REPORT SUMMARY

Day	Date 2014	Depth (0600 Hrs)	24 Hr Footage	Bit #	WOB (Klbs) RT	RPM (RT)	WOB (Klbs) MM	RPM (MM)	PP	SPM 1	SPM 2	GPM	24 Hr Activity Summary	Formation
11	11/6	11,236'	185	5	20	50	21	138	2450	0	80	281	Service top drive, change out lower well control, changing out transducer for MWD, out quill/back up wrench die/stabbing guide, pick up drill pipe from BHA to 1,120', downtime -powered catwalk change skate cable, pick up drill pipe from 1,120' to 8,200', pick up drill pipe from 8,200' to 10,200', change rotating head/rubber/remove trip nipple/install rotating head, pick up drill pipe from 10,200' to 10,331', drilling cement from 10,330' to 10,351, drilling cement/shoe track from 10,910' to 11,051', formation integrity test (FIT)/held 1800 psi for 30 min, drill from 11,051' to 11,236'.	Middle Bakken
12	11/7	12,819'	1583	5	15	50	12	138	3800	0	80	281	Drill F/11,236' to 11,948', service top drive, greased block crown, lower well and wash pipe, drill from 11,948' to 12,819'	Middle Bakken
13	11/8	14,985'	2166	5	21	50	19	138	3900	75	0	281	Rotary drilling F/12802 to 13632, Service rig finctioned HCR and tested casing cage, Rotary drilling F/13632 T/ 14985	Middle Bakken
14	11/9	17,226'	2241	5	24	50	25	138	330	75	0	281	Rotary drilling F/14954 to 16052, Service top drive, Rotary drilling F/16052 to 17226	Middle Bakken
15	11/10	20,297'	3071	5	18	50	17	138	2525	64	0	281	Rotary drilling F/17194 T/18267, Service rig, Rotary drilling F/18267 T/20297	Middle Bakken
16	11/11	20,560'	263	5	18	50	17	138	2525	64	0	281	Rotary drilling F/20297 T/20560	Middle Bakken

DAILY MUD SUMMARY

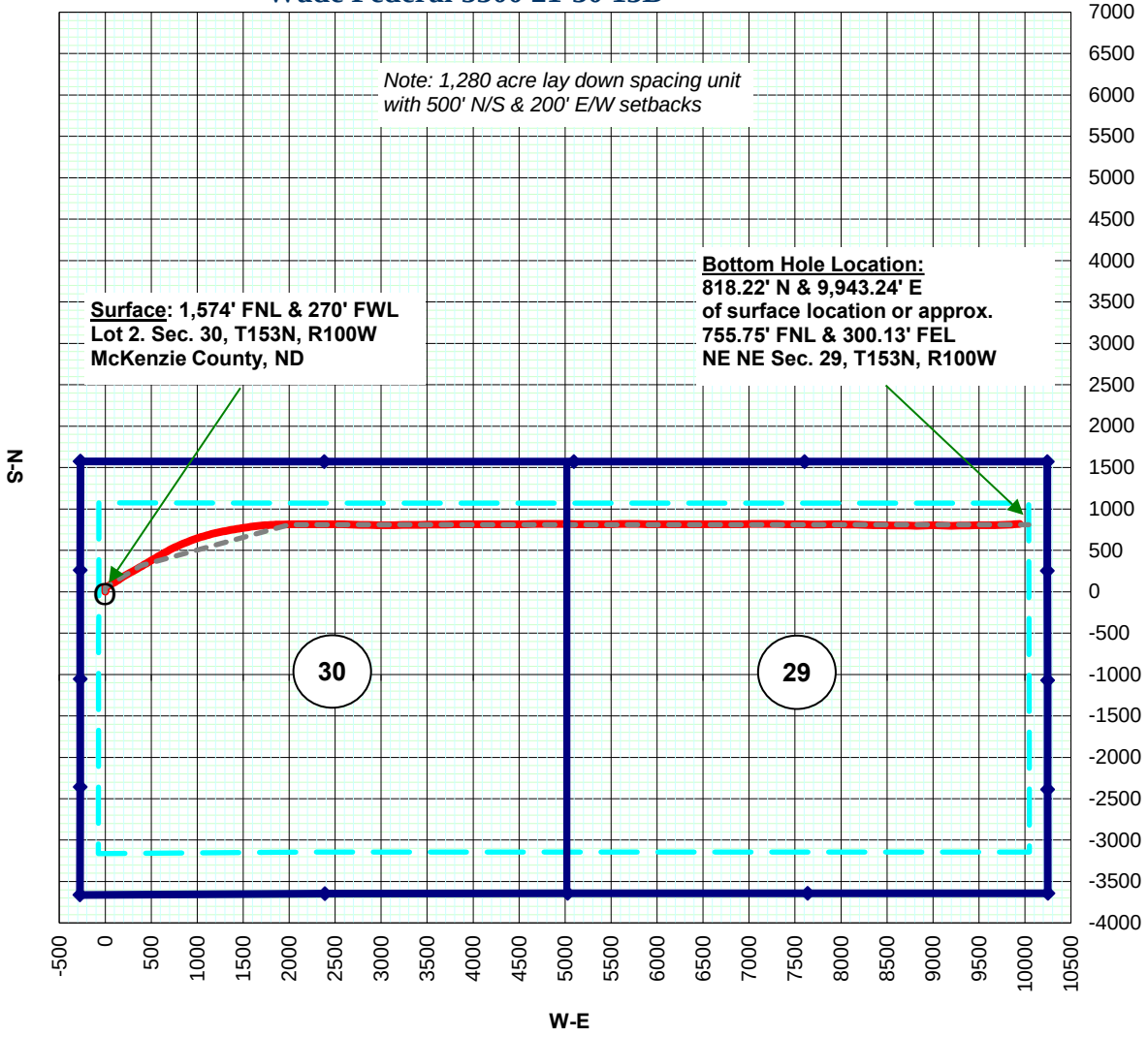
Date 2014	Mud Depth	Mud WT (ppg)	VIS (sec/ qt)	PV (cP)	YP (lbs/ 100 ft^2)	Gels (lbs/ 100 ft^2)	600/ 300	NAP/ H_20 (ratio)	NAP/ H_20 (% by vol)	Cake (API/ HTHP)	Cor. Solids (%)	Oil/ H_20 (%)	Alk	pH	Excess Lime (lb/bbl)	Cl^- (mg/L)	LGS/ HGS (%)	Salinity (ppm)	Electrical Stability	Gain/ Loss (bbls)
10/26	2,070'	11.3	78	212	12	14/19/-	56/34	78/22	64/18	3	18	64/18	2/0	-	2.6	38k	3.6/12.3	258,111	560	-/221
10/27	2,326'	10.7	87	22	13	14/19/-	57/35	76.5/23.5	65/20	3	15	65/20	2/0	-	2.6	39k	2.5/10.3	243,204	635	-/10
10/28	5,734'	11.5	54	23	12	13/18/-	58/35	77.8/22.2	63/18	3	19	63/18	2/0	-	2.6	39k	4.2/12.3	263,117	550	-/169
10/29	6,050'	9.9	50	14	10	10/15/-	38/24	79.3/20.7	69/18	3	13	69/18	2/0	-	2.6	39k	4.2/6.2	263,117	635	-/331
10/30	8,045'	9.9	47	17	10	10/15/-	44/27	79.2/20.8	68.5/18	3	11.1	68.5/18	2.2/0	-	2.8	39k	4.1/6.8	263,117	625	-/65
10/31	9,571'	10	49	18	12	12/17/-	48/30	77.0/23.0	67/20	3	10.4	67/20	2.2	-	2.8	42k	3.5/6.8	257,102	755	-/175
11/01	10,118'	10.15	53	18	12	12/17/-	48/30	77.9/22.1	67/19	3	14	67/19	2.2	-	2.8	42k	4.3/6.9	264,320	745	-/69
11/02	10,449'	10.1	48	18	12	13/18/-	48/30	80.2/19.8	69/17	3	14	69/17	2.3	-	3	48k	5.0/6.7	264,320	777	-/44
11/03	11,051'	10.4	62	17	14	12/17/-	48/31	81.2/18.8	69/16	3	15	69/16	2.3	-	3	42k	4.4/8.2	264,320	875	-/10
11/04	11,051'	9.6	28	1	1	-	3/2	-	-	-	8.7	-	-	8	-	140k	0.1/0.4	-	-	-/1413
11/05	11,051'	Change mud from diesel invert to salt water																		
11/06	11,236'	9.6	28	1	1	-	3/1	-	0/91.3	-	-	-	-	8	-	137k	0.1/1	-	-	-
11/07	12,819'	9.5	28	1	1	-	3/0	-	0/91.4	-	-	-	-	9	-	137k	0.1/0	-	-	-
11/08	14,985'	9.5	28	1	1	-	3/1	-	0/91.5	-	-	-	-	8	-	137k	0.1/1	-	-	-
11/09	17,226'	9.7	28	1	1	-	3/2	-	0/91.6	-	-	-	-	9	-	137k	0.1/0	-	-	-
11/10	19,435'	9.4	29	1	1	-	-3/2	-	0/91.7	-	-	-	-	8.5	-	147k	0.1/0	-	-	-

BOTTOM HOLE ASSEMBLY RECORD

Bit Data											Motor Data				Reason For Removal
Bit #	Size (in.)	Type	Make	Model	Depth In	Depth Out	Footage	Hours	Σ hrs	Vert. Dev.	Make	Model	Bend	Rev/Gal	
1	17 1/2	PDC	Hughes	HC605S	0'	2,070'	2,070'	15	15	Surface	-	-	-	-	TD surface
2	12 1/4	PDC	NOV	DS616D	2,070'	6,050'	3,980'	18.2	33.2	Vertical	-		-	-	Vertical
3	8 3/4	PDC	Varrel	-	6,050'	10,118'	4,068'	56.5	89.7	Vertical	Hunting	-	-	0.29	Vertical
4	8 3/4	PDC	Varel	6V513PEDU	10,118'	11,051'	933'	36	125.7	Curve	Baker	-	2.50°	0.29	TD curve
5	6	PDC	Varel	6V513PEDU	11,051'	20,560'	9,509'	94.75	220.45	Lateral	Baker	-	2.50°	0.29	TD curve

PLAN VIEW

Oasis Petroleum North America, LLC
Wade Federal 5300 21-30 13B



Oasis Petroleum North America, LLC
Wade Federal 5300 21-30 13B
Lot 2 Sec. 30, T153N, R100W
McKenzie County, ND

SUNBURST CONSULTING
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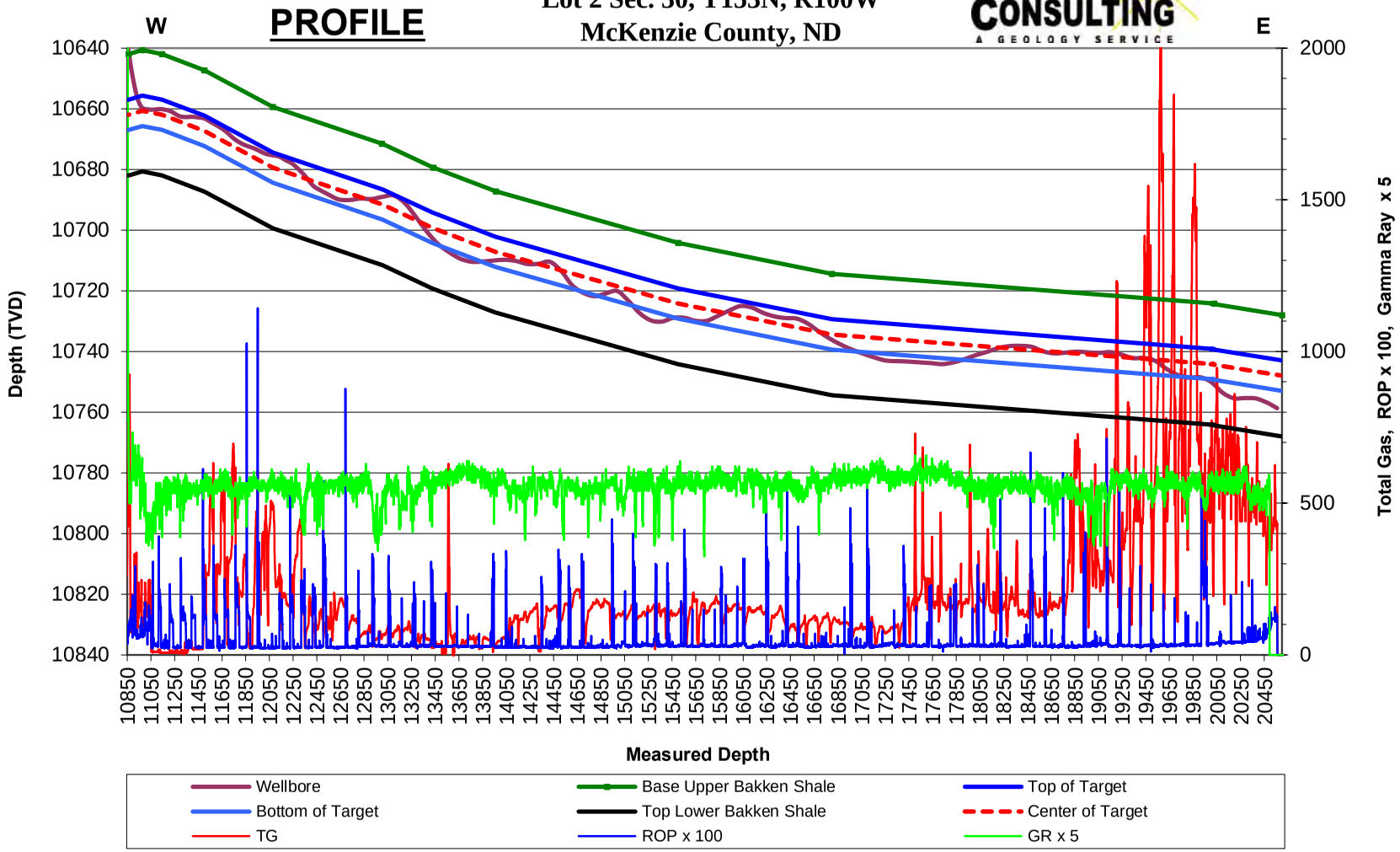

FORMATION MARKERS & DIP ESTIMATES

Oasis Petroleum North America, LLC - Wade Federal 5300 21-30 13B

Dip Change Points	MD	TVD	TVD diff.	MD diff.	Dip	Dipping up/down	Type of Marker
Marker proj. to M.B. top							
M. Bakken entry	10,861'	10,642.00					Gamma
Center of target	10,976'	10,640.70	-1.30	115.00	***0.65***	Up	Gamma
Low gamma above center	11,143'	10,642.00	1.30	167.00	***-0.45***	Down	Gamma
Top of target	11,500'	10,647.30	5.30	357.00	***-0.85***	Down	Gamma
Low gamma at top of target	12,079'	10,659.43	12.13	579.00	***-1.20***	Down	Gamma
Low gamma above center	13,000'	10,671.50	12.07	921.00	***-0.75***	Down	Gamma
High gamma at base	13,435'	10,679.40	7.90	435.00	***-1.04***	Down	Gamma
High gamma at base	13,963'	10,687.23	7.83	528.00	***-0.85***	Down	Gamma
High gamma at base	15,502'	10,704.20	16.97	1539.00	***-0.63***	Down	Gamma
Low gamma at center	16,800'	10,714.40	10.20	1298.00	***-0.45***	Down	Gamma
High gamma at base	20,030'	10,724.20	9.80	3230.00	***-0.17***	Down	Gamma
	20,560'	10,727.70	3.50	530.00	***-0.38***	Down	
Gross Dip							
Initial M. Bakken Contact	10,861'	10,642.00					
Projected Final M.B. Contact	20,560'	10,727.70	85.70	9699.00	***-0.51***	Down	Projection

Oasis Petroleum North America, LLC
Wade Federal 5300 21-30 13B
Lot 2 Sec. 30, T153N, R100W
McKenzie County, ND

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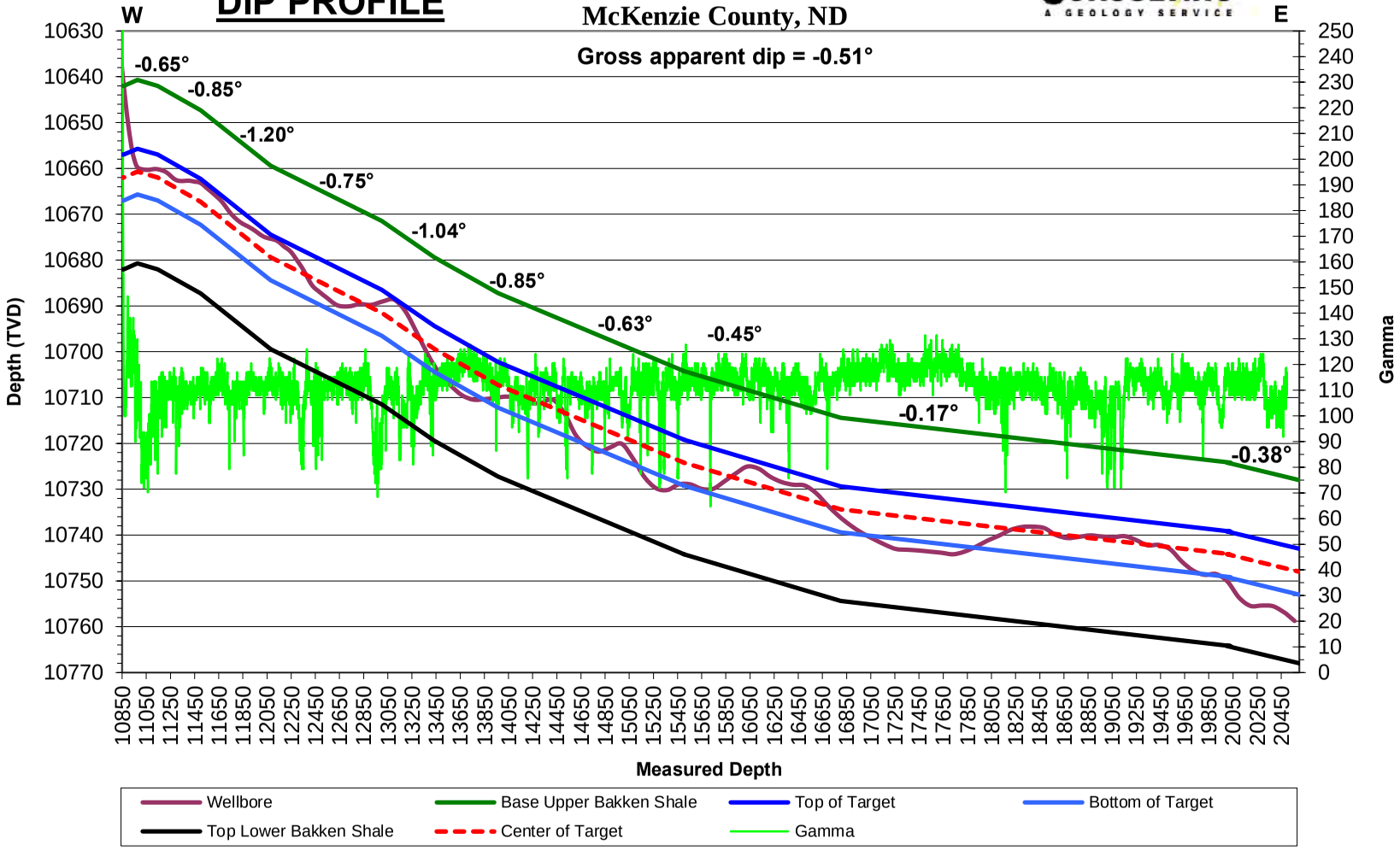
DIP PROFILE



SUNBURST CONSULTING, INC.

Operator:	Oasis Petroleum North America, LLC			Kick-off:	11/1/2014
Well :	Wade Federal 5300 21-30 13B			Finish:	11/10/2014
County:	McKenzie	State:	ND	Directional Supervision:	
QQ:	Lot 2	Section:	30	Ryan Directional	
Township:	153	N/S:	N		
Range:	100	E/W:	W	Date:	11/20/2014
Footages:	1574	FN/SL:	N	Time:	10:53
	270	FE/WL:	W	**F9 to re-calculate**	

Minimum Curvature Method (SPE-3362) Proposed dir: 85.39

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
Tie	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1	149.00	0.80	334.90	149.00	0.94	-0.44	-0.36	0.54
2	244.00	0.90	333.80	243.98	2.21	-1.05	-0.87	0.11
3	337.00	1.20	346.40	336.97	3.81	-1.60	-1.29	0.41
4	399.00	1.20	344.80	398.96	5.07	-1.93	-1.51	0.05
5	461.00	1.30	346.30	460.94	6.38	-2.26	-1.74	0.17
6	524.00	1.20	356.60	523.93	7.73	-2.47	-1.84	0.39
7	615.00	1.10	342.80	614.91	9.52	-2.79	-2.01	0.32
8	707.00	1.10	354.90	706.89	11.24	-3.13	-2.21	0.25
9	799.00	0.80	29.20	798.88	12.68	-2.89	-1.86	0.68
10	891.00	1.30	32.80	890.86	14.12	-2.01	-0.87	0.55
11	983.00	1.10	27.60	982.84	15.78	-1.04	0.23	0.25
12	1075.00	0.90	36.30	1074.83	17.15	-0.20	1.18	0.27
13	1167.00	1.10	2.00	1166.82	18.61	0.26	1.75	0.67
14	1260.00	1.10	341.20	1259.80	20.35	0.00	1.64	0.43
15	1352.00	1.10	334.10	1351.78	21.98	-0.67	1.10	0.15
16	1444.00	1.10	322.40	1443.77	23.47	-1.59	0.30	0.24
17	1537.00	0.90	330.40	1536.75	24.81	-2.50	-0.50	0.26
18	1630.00	0.80	308.40	1629.74	25.85	-3.37	-1.28	0.36
19	1723.00	0.80	302.10	1722.73	26.60	-4.43	-2.28	0.09
20	1816.00	1.10	320.30	1815.72	27.63	-5.55	-3.31	0.45
21	1909.00	0.90	334.80	1908.71	28.98	-6.43	-4.08	0.35
22	2002.00	0.70	333.30	2001.70	30.15	-7.00	-4.55	0.22
23	2018.00	1.20	335.30	2017.69	30.39	-7.11	-4.64	3.13
24	2118.00	1.10	341.30	2117.67	32.25	-7.85	-5.24	0.16
25	2183.00	1.30	334.60	2182.66	33.51	-8.37	-5.65	0.38
26	2276.00	1.30	324.20	2275.64	35.31	-9.44	-6.57	0.25
27	2369.00	1.40	325.60	2368.61	37.11	-10.70	-7.68	0.11
28	2462.00	1.10	3.70	2461.59	38.94	-11.28	-8.12	0.93
29	2556.00	1.10	3.60	2555.57	40.74	-11.17	-7.86	0.00
30	2649.00	1.10	327.00	2648.56	42.38	-11.60	-8.16	0.74
31	2742.00	0.90	339.90	2741.54	43.81	-12.34	-8.78	0.32
32	2835.00	0.60	125.30	2834.54	44.22	-12.19	-8.60	1.54
33	2929.00	1.10	150.60	2928.53	43.15	-11.35	-7.84	0.65
34	3022.00	0.90	154.80	3021.51	41.71	-10.60	-7.21	0.23

SUNBURST CONSULTING, INC.

Operator:	Oasis Petroleum North America, LLC			Kick-off:	11/1/2014
Well :	Wade Federal 5300 21-30 13B			Finish:	11/10/2014
County:	McKenzie	State:	ND	Directional Supervision:	
QQ:	Lot 2	Section:	30	Ryan Directional	
Township:	153	N/S:	N		
Range:	100	E/W:	W	Date:	11/20/2014
Footages:	1574	FN/SL:	N	Time:	10:53
	270	FE/WL:	W	**F9 to re-calculate**	

Minimum Curvature Method (SPE-3362) Proposed dir: 85.39

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
35	3115.00	0.90	170.80	3114.50	40.32	-10.17	-6.89	0.27
36	3208.00	1.10	154.70	3207.49	38.80	-9.67	-6.52	0.37
37	3302.00	1.50	168.10	3301.46	36.78	-9.03	-6.05	0.53
38	3395.00	0.50	72.60	3394.45	35.71	-8.39	-5.50	1.75
39	3488.00	0.80	43.70	3487.45	36.30	-7.56	-4.61	0.47
40	3581.00	0.80	45.30	3580.44	37.22	-6.65	-3.63	0.02
41	3674.00	0.70	37.30	3673.43	38.13	-5.84	-2.76	0.16
42	3768.00	0.80	28.60	3767.42	39.17	-5.18	-2.01	0.16
43	3861.00	0.70	31.90	3860.41	40.22	-4.57	-1.32	0.12
44	3954.00	0.50	23.40	3953.41	41.07	-4.11	-0.79	0.23
45	4048.00	0.70	11.50	4047.40	42.01	-3.83	-0.44	0.25
46	4141.00	0.70	25.30	4140.40	43.08	-3.47	0.00	0.18
47	4234.00	0.20	64.80	4233.39	43.66	-3.08	0.44	0.60
48	4327.00	0.10	61.10	4326.39	43.77	-2.87	0.66	0.11
49	4420.00	0.10	313.70	4419.39	43.87	-2.85	0.68	0.17
50	4513.00	0.40	250.20	4512.39	43.81	-3.22	0.31	0.39
51	4607.00	0.40	239.10	4606.39	43.53	-3.81	-0.30	0.08
52	4700.00	0.60	219.40	4699.39	42.99	-4.40	-0.93	0.28
53	4793.00	0.70	224.40	4792.38	42.21	-5.10	-1.69	0.12
54	4886.00	0.50	217.10	4885.38	41.48	-5.74	-2.39	0.23
55	4979.00	0.70	218.40	4978.37	40.71	-6.34	-3.05	0.22
56	5072.00	1.20	193.80	5071.36	39.32	-6.93	-3.74	0.68
57	5165.00	1.40	192.90	5164.33	37.27	-7.41	-4.39	0.22
58	5258.00	0.60	182.40	5257.32	35.67	-7.69	-4.79	0.88
59	5351.00	0.50	158.90	5350.31	34.81	-7.56	-4.74	0.26
60	5444.00	0.70	160.60	5443.31	33.89	-7.23	-4.48	0.22
61	5537.00	1.10	161.20	5536.30	32.51	-6.75	-4.11	0.43
62	5630.00	1.80	161.80	5629.27	30.28	-6.01	-3.55	0.75
63	5723.00	1.20	154.00	5722.24	28.02	-5.12	-2.85	0.68
64	5815.00	1.20	150.40	5814.22	26.31	-4.22	-2.10	0.08
65	5909.00	1.20	144.80	5908.19	24.65	-3.17	-1.18	0.12
66	6002.00	1.50	162.80	6001.17	22.70	-2.25	-0.42	0.55
67	6090.00	0.70	216.90	6089.15	21.16	-2.23	-0.52	1.40
68	6181.00	0.60	241.20	6180.15	20.49	-2.98	-1.33	0.32
69	6274.00	0.90	60.40	6273.15	20.62	-2.77	-1.11	1.61

SUNBURST CONSULTING, INC.

Operator:	Oasis Petroleum North America, LLC			Kick-off:	11/1/2014
Well :	Wade Federal 5300 21-30 13B			Finish:	11/10/2014
County:	McKenzie	State:	ND	Directional Supervision:	
QQ:	Lot 2	Section:	30	Ryan Directional	
Township:	153	N/S:	N		
Range:	100	E/W:	W	Date:	11/20/2014
Footages:	1574	FN/SL:	N	Time:	10:53
	270	FE/WL:	W	**F9 to re-calculate**	

Minimum Curvature Method (SPE-3362) Proposed dir: 85.39

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
70	6368.00	1.00	53.90	6367.13	21.47	-1.47	0.26	0.16
71	6461.00	0.80	46.10	6460.12	22.39	-0.35	1.45	0.25
72	6554.00	0.70	49.60	6553.11	23.21	0.55	2.42	0.12
73	6647.00	0.50	32.90	6646.11	23.92	1.21	3.13	0.28
74	6740.00	0.60	37.90	6739.10	24.65	1.73	3.70	0.12
75	6833.00	0.20	4.80	6832.10	25.19	2.04	4.06	0.48
76	6926.00	0.00	288.30	6925.10	25.35	2.05	4.08	0.22
77	7019.00	1.40	3.50	7018.09	26.49	2.12	4.24	1.51
78	7112.00	1.80	12.50	7111.06	29.05	2.51	4.83	0.51
79	7205.00	1.70	7.40	7204.01	31.84	3.00	5.55	0.20
80	7298.00	1.60	3.30	7296.97	34.51	3.25	6.02	0.17
81	7391.00	1.10	11.60	7389.95	36.68	3.51	6.44	0.58
82	7484.00	1.00	12.30	7482.93	38.34	3.86	6.93	0.11
83	7578.00	0.80	10.30	7576.92	39.79	4.15	7.34	0.22
84	7671.00	0.50	357.10	7669.91	40.83	4.25	7.52	0.36
85	7764.00	0.40	32.50	7762.91	41.51	4.40	7.72	0.31
86	7857.00	0.30	284.10	7855.91	41.85	4.34	7.69	0.61
87	7951.00	0.30	243.80	7949.91	41.80	3.88	7.23	0.22
88	8044.00	0.50	221.90	8042.91	41.39	3.39	6.71	0.27
89	8137.00	0.40	172.20	8135.90	40.76	3.16	6.43	0.42
90	8230.00	0.20	225.20	8228.90	40.33	3.09	6.32	0.35
91	8324.00	0.30	217.40	8322.90	40.02	2.83	6.03	0.11
92	8417.00	0.20	223.30	8415.90	39.71	2.57	5.75	0.11
93	8510.00	0.10	290.00	8508.90	39.62	2.38	5.56	0.20
94	8603.00	0.10	343.50	8601.90	39.72	2.28	5.47	0.10
95	8696.00	0.40	217.90	8694.90	39.54	2.06	5.23	0.50
96	8790.00	0.30	257.80	8788.90	39.23	1.62	4.76	0.27
97	8883.00	0.50	260.70	8881.90	39.12	0.98	4.12	0.22
98	8976.00	0.30	281.40	8974.89	39.10	0.34	3.48	0.26
99	9069.00	0.40	280.90	9067.89	39.21	-0.22	2.93	0.11
100	9163.00	0.20	302.20	9161.89	39.36	-0.68	2.49	0.24
101	9256.00	0.50	305.30	9254.89	39.68	-1.15	2.04	0.32
102	9349.00	0.40	290.50	9347.89	40.03	-1.78	1.44	0.16
103	9442.00	0.40	278.40	9440.88	40.19	-2.41	0.83	0.09
104	9535.00	0.40	267.30	9533.88	40.22	-3.05	0.19	0.08

SUNBURST CONSULTING, INC.

Operator:	Oasis Petroleum North America, LLC			Kick-off:	11/1/2014
Well :	Wade Federal 5300 21-30 13B			Finish:	11/10/2014
County:	McKenzie	State:	ND	Directional Supervision:	
QQ:	Lot 2	Section:	30	Ryan Directional	
Township:	153	N/S:	N		
Range:	100	E/W:	W	Date:	11/20/2014
Footages:	1574	FN/SL:	N	Time:	10:53
	270	FE/WL:	W	**F9 to re-calculate**	

Minimum Curvature Method (SPE-3362) Proposed dir: 85.39

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
105	9629.00	0.40	312.90	9627.88	40.43	-3.62	-0.36	0.33
106	9722.00	0.50	278.70	9720.88	40.71	-4.26	-0.98	0.30
107	9815.00	0.40	263.50	9813.87	40.73	-4.98	-1.69	0.17
108	9908.00	0.40	265.70	9906.87	40.67	-5.63	-2.34	0.02
109	10001.00	0.40	272.00	9999.87	40.66	-6.28	-2.99	0.05
110	10063.00	0.40	301.50	10061.87	40.78	-6.68	-3.38	0.33
111	10095.00	0.40	312.70	10093.87	40.91	-6.86	-3.55	0.24
112	10126.00	2.20	44.40	10124.86	41.41	-6.52	-3.17	7.25
113	10157.00	6.10	53.60	10155.77	42.82	-4.78	-1.32	12.72
114	10188.00	10.20	56.10	10186.45	45.33	-1.17	2.47	13.27
115	10220.00	13.90	56.30	10217.74	49.04	4.38	8.31	11.56
116	10250.00	14.90	55.30	10246.80	53.24	10.55	14.79	3.43
117	10281.00	16.20	54.60	10276.66	58.01	17.35	21.96	4.24
118	10312.00	19.90	54.60	10306.13	63.57	25.18	30.21	11.94
119	10343.00	23.50	55.60	10334.93	70.12	34.58	40.11	11.67
120	10374.00	27.60	56.40	10362.89	77.59	45.67	51.76	13.27
121	10405.00	31.70	57.70	10389.83	85.92	58.54	65.26	13.39
122	10436.00	35.30	58.20	10415.68	95.00	73.04	80.44	11.65
123	10468.00	39.00	56.80	10441.18	105.39	89.33	97.51	11.86
124	10499.00	43.00	57.20	10464.57	116.46	106.38	115.40	12.93
125	10530.00	46.50	56.40	10486.58	128.41	124.64	134.56	11.44
126	10561.00	47.80	52.40	10507.67	141.64	143.11	154.03	10.35
127	10592.00	51.00	51.60	10527.84	156.13	161.65	173.68	10.51
128	10623.00	54.70	52.50	10546.56	171.32	181.13	194.32	12.16
129	10654.00	58.10	55.30	10563.71	186.52	202.00	216.34	13.30
130	10685.00	62.20	57.50	10579.14	201.39	224.39	239.85	14.59
131	10716.00	66.50	58.20	10592.56	216.25	248.05	264.63	14.02
132	10747.00	69.10	58.50	10604.27	231.31	272.48	290.19	8.43
133	10778.00	69.20	58.60	10615.30	246.43	297.19	316.04	0.44
134	10809.00	70.10	58.50	10626.08	261.59	321.99	341.97	2.92
135	10840.00	73.50	57.90	10635.76	277.11	347.01	368.16	11.12
136	10871.00	76.00	57.10	10643.92	293.18	372.23	394.59	8.44
137	10903.00	78.20	56.90	10651.06	310.16	398.39	422.03	6.90
138	10934.00	82.30	57.00	10656.31	326.82	424.00	448.89	13.23
139	10965.00	86.90	57.40	10659.23	343.54	449.93	476.09	14.89

SUNBURST CONSULTING, INC.

Operator:	Oasis Petroleum North America, LLC			Kick-off:	11/1/2014
Well :	Wade Federal 5300 21-30 13B			Finish:	11/10/2014
County:	McKenzie	State:	ND	Directional Supervision:	
QQ:	Lot 2	Section:	30	Ryan Directional	
Township:	153	N/S:	N		
Range:	100	E/W:	W	Date:	11/20/2014
Footages:	1574	FN/SL:	N	Time:	10:53
	270	FE/WL:	W	**F9 to re-calculate**	

Minimum Curvature Method (SPE-3362) Proposed dir: 85.39

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
140	10990.00	89.50	57.20	10660.01	357.03	470.96	498.13	10.43
141	11051.00	89.90	56.00	10660.33	390.61	521.88	551.59	2.07
142	11082.00	90.30	56.00	10660.28	407.95	547.58	578.60	1.29
143	11112.00	90.20	56.30	10660.15	424.66	572.49	604.77	1.05
144	11143.00	89.70	57.50	10660.17	441.59	598.46	632.02	4.19
145	11174.00	89.50	57.30	10660.39	458.29	624.58	659.39	0.91
146	11205.00	89.40	57.90	10660.69	474.90	650.75	686.82	1.96
147	11236.00	88.50	59.60	10661.26	490.98	677.25	714.52	6.20
148	11267.00	88.70	59.70	10662.01	506.64	703.99	742.43	0.72
149	11298.00	89.30	60.50	10662.55	522.09	730.86	770.46	3.23
150	11329.00	90.00	62.40	10662.74	536.90	758.09	798.79	6.53
151	11360.00	90.00	63.10	10662.74	551.09	785.65	827.40	2.26
152	11391.00	90.20	63.60	10662.69	565.00	813.36	856.13	1.74
153	11421.00	89.60	65.30	10662.74	577.94	840.42	884.15	6.01
154	11452.00	89.80	65.80	10662.90	590.77	868.64	913.31	1.74
155	11483.00	89.80	65.90	10663.01	603.45	896.93	942.53	0.32
156	11514.00	88.60	68.00	10663.44	615.59	925.45	971.93	7.80
157	11545.00	88.70	67.90	10664.18	627.22	954.17	1001.50	0.46
158	11576.00	88.80	68.50	10664.85	638.73	982.95	1031.10	1.96
159	11607.00	88.60	70.60	10665.55	649.56	1011.98	1060.92	6.80
160	11638.00	88.70	70.60	10666.29	659.85	1041.22	1090.88	0.32
161	11669.00	88.40	71.00	10667.07	670.04	1070.48	1120.87	1.61
162	11701.00	87.80	72.90	10668.13	679.95	1100.89	1151.98	6.22
163	11732.00	88.00	72.70	10669.27	689.11	1130.48	1182.21	0.91
164	11762.00	88.40	73.50	10670.21	697.83	1159.17	1211.51	2.98
165	11793.00	88.70	75.10	10670.99	706.22	1189.01	1241.92	5.25
166	11824.00	88.80	75.70	10671.67	714.03	1219.00	1272.44	1.96
167	11855.00	89.30	76.40	10672.18	721.50	1249.08	1303.03	2.77
168	11886.00	89.20	78.00	10672.59	728.37	1279.31	1333.71	5.17
169	11916.00	89.10	77.90	10673.03	734.63	1308.64	1363.45	0.47
170	11947.00	89.00	78.80	10673.55	740.89	1339.00	1394.21	2.92
171	11978.00	88.80	79.60	10674.14	746.70	1369.44	1425.03	2.66
172	12009.00	89.10	79.90	10674.71	752.21	1399.94	1455.87	1.37
173	12040.00	89.50	79.90	10675.09	757.65	1430.46	1486.73	1.29
174	12071.00	89.70	79.50	10675.31	763.19	1460.96	1517.57	1.44

SUNBURST CONSULTING, INC.

Operator:	Oasis Petroleum North America, LLC			Kick-off:	11/1/2014
Well :	Wade Federal 5300 21-30 13B			Finish:	11/10/2014
County:	McKenzie	State:	ND	Directional Supervision:	
QQ:	Lot 2	Section:	30	Ryan Directional	
Township:	153	N/S:	N		
Range:	100	E/W:	W	Date:	11/20/2014
Footages:	1574	FN/SL:	N	Time:	10:53
	270	FE/WL:	W	**F9 to re-calculate**	

Minimum Curvature Method (SPE-3362) Proposed dir: 85.39

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
175	12101.00	89.60	79.60	10675.49	768.63	1490.46	1547.42	0.47
176	12133.00	89.30	80.20	10675.80	774.24	1521.96	1579.27	2.10
177	12164.00	88.40	81.10	10676.42	779.28	1552.55	1610.16	4.11
178	12194.00	88.90	80.50	10677.13	784.07	1582.15	1640.05	2.60
179	12225.00	89.10	81.60	10677.67	788.90	1612.77	1670.96	3.61
180	12256.00	87.90	82.90	10678.48	793.08	1643.47	1701.90	5.71
181	12286.00	87.80	82.80	10679.60	796.81	1673.22	1731.85	0.47
182	12318.00	87.90	83.50	10680.80	800.62	1704.97	1763.80	2.21
183	12349.00	87.40	85.70	10682.08	803.54	1735.80	1794.77	7.27
184	12380.00	87.10	86.10	10683.56	805.75	1766.69	1825.73	1.61
185	12411.00	87.40	86.40	10685.05	807.77	1797.59	1856.69	1.37
186	12441.00	88.80	87.30	10686.05	809.42	1827.52	1886.67	5.55
187	12473.00	88.80	86.90	10686.72	811.04	1859.47	1918.64	1.25
188	12504.00	88.70	88.10	10687.39	812.39	1890.44	1949.62	3.88
189	12536.00	89.00	89.70	10688.03	813.01	1922.42	1981.55	5.09
190	12631.00	88.80	89.40	10689.86	813.75	2017.40	2076.28	0.38
191	12725.00	90.90	89.90	10690.10	814.33	2111.40	2170.01	2.30
192	12819.00	89.70	89.10	10689.61	815.15	2205.39	2263.77	1.53
193	12913.00	90.10	89.40	10689.78	816.38	2299.38	2357.56	0.53
194	13007.00	90.80	90.50	10689.04	816.46	2393.37	2451.25	1.39
195	13101.00	89.70	90.10	10688.63	815.97	2487.37	2544.91	1.25
196	13195.00	87.20	91.30	10691.17	814.82	2581.32	2638.46	2.95
197	13288.00	87.20	91.90	10695.71	812.23	2674.17	2730.80	0.64
198	13382.00	86.80	92.00	10700.63	809.03	2767.99	2824.06	0.44
199	13476.00	88.30	91.00	10704.65	806.57	2861.87	2917.44	1.92
200	13570.00	88.40	90.50	10707.36	805.34	2955.82	3010.99	0.54
201	13663.00	89.00	89.90	10709.47	805.02	3048.79	3103.63	0.91
202	13757.00	89.80	89.10	10710.45	805.84	3142.78	3197.38	1.20
203	13851.00	90.30	88.50	10710.37	807.81	3236.76	3291.22	0.83
204	13945.00	90.10	90.00	10710.04	809.04	3330.75	3385.00	1.61
205	14039.00	90.20	90.00	10709.80	809.04	3424.75	3478.70	0.11
206	14134.00	89.30	89.30	10710.21	809.62	3519.75	3573.43	1.20
207	14229.00	89.50	89.30	10711.20	810.78	3614.73	3668.21	0.21
208	14323.00	90.60	89.00	10711.12	812.17	3708.72	3762.00	1.21
209	14418.00	90.20	88.90	10710.46	813.92	3803.70	3856.82	0.43

SUNBURST CONSULTING, INC.

Operator:	Oasis Petroleum North America, LLC			Kick-off:	11/1/2014
Well :	Wade Federal 5300 21-30 13B			Finish:	11/10/2014
County:	McKenzie	State:	ND	Directional Supervision:	
QQ:	Lot 2	Section:	30	Ryan Directional	
Township:	153	N/S:	N		
Range:	100	E/W:	W	Date:	11/20/2014
Footages:	1574	FN/SL:	N	Time:	10:53
	270	FE/WL:	W	**F9 to re-calculate**	

Minimum Curvature Method (SPE-3362) Proposed dir: 85.39

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
210	14513.00	86.20	88.80	10713.44	815.82	3898.62	3951.58	4.21
211	14608.00	88.00	89.70	10718.25	817.06	3993.48	4046.23	2.12
212	14703.00	89.00	90.90	10720.74	816.56	4088.45	4140.85	1.64
213	14797.00	89.70	90.00	10721.80	815.83	4182.44	4234.48	1.21
214	14892.00	91.40	90.20	10720.89	815.66	4277.43	4329.15	1.80
215	14986.00	89.60	89.90	10720.07	815.58	4371.42	4422.83	1.94
216	15081.00	86.50	89.40	10723.30	816.16	4466.35	4517.50	3.31
217	15176.00	88.60	89.90	10727.36	816.74	4561.26	4612.15	2.27
218	15271.00	88.50	89.30	10729.77	817.40	4656.23	4706.86	0.64
219	15365.00	91.00	89.60	10730.18	818.30	4750.21	4800.61	2.68
220	15460.00	90.50	90.00	10728.93	818.63	4845.20	4895.32	0.67
221	15555.00	89.50	90.00	10728.93	818.63	4940.20	4990.02	1.05
222	15649.00	89.30	90.80	10729.92	817.98	5034.19	5083.65	0.88
223	15744.00	90.70	90.40	10729.92	816.98	5129.19	5178.26	1.53
224	15839.00	91.30	89.70	10728.26	816.90	5224.17	5272.93	0.97
225	15934.00	90.80	90.90	10726.52	816.40	5319.15	5367.56	1.37
226	16028.00	91.00	90.00	10725.04	815.66	5413.14	5461.18	0.98
227	16123.00	88.40	89.30	10725.54	816.25	5508.12	5555.91	2.83
228	16218.00	89.50	89.60	10727.28	817.16	5603.10	5650.65	1.20
229	16313.00	89.10	90.00	10728.44	817.49	5698.09	5745.36	0.60
230	16407.00	90.20	90.10	10729.02	817.41	5792.09	5839.05	1.18
231	16502.00	89.60	90.90	10729.18	816.58	5887.09	5933.67	1.05
232	16596.00	88.40	90.70	10730.82	815.27	5981.06	6027.24	1.29
233	16690.00	88.40	90.70	10733.45	814.12	6075.02	6120.80	0.00
234	16785.00	88.70	89.40	10735.85	814.03	6169.98	6215.45	1.40
235	16880.00	88.90	89.10	10737.84	815.28	6264.96	6310.21	0.38
236	16975.00	89.10	90.10	10739.50	815.94	6359.94	6404.94	1.07
237	17069.00	89.30	89.70	10740.81	816.10	6453.93	6498.64	0.48
238	17164.00	89.20	91.30	10742.05	815.28	6548.91	6593.25	1.69
239	17258.00	89.60	90.70	10743.04	813.64	6642.89	6686.80	0.77
240	17353.00	90.20	89.60	10743.20	813.39	6737.89	6781.47	1.32
241	17448.00	89.60	89.40	10743.37	814.22	6832.89	6876.22	0.67
242	17543.00	90.10	88.80	10743.62	815.71	6927.87	6971.02	0.82
243	17637.00	89.60	89.40	10743.87	817.18	7021.86	7064.82	0.83
244	17732.00	90.00	89.40	10744.20	818.18	7116.85	7159.59	0.42

SUNBURST CONSULTING, INC.

Operator:	Oasis Petroleum North America, LLC			Kick-off:	11/1/2014
Well :	Wade Federal 5300 21-30 13B			Finish:	11/10/2014
County:	McKenzie	State:	ND	Directional Supervision:	
QQ:	Lot 2	Section:	30	Ryan Directional	
Township:	153	N/S:	N		
Range:	100	E/W:	W	Date:	11/20/2014
Footages:	1574	FN/SL:	N	Time:	10:53
	270	FE/WL:	W	**F9 to re-calculate**	

Minimum Curvature Method (SPE-3362) Proposed dir: 85.39

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
245	17826.00	90.70	90.20	10743.62	818.51	7210.85	7253.31	1.13
246	17921.00	90.60	90.90	10742.55	817.60	7305.84	7347.92	0.74
247	18015.00	91.00	90.40	10741.23	816.53	7399.82	7441.51	0.68
248	18110.00	90.30	91.00	10740.15	815.37	7494.81	7536.10	0.97
249	18205.00	91.20	90.50	10738.91	814.13	7589.79	7630.67	1.08
250	18300.00	89.60	90.70	10738.25	813.13	7684.78	7725.27	1.70
251	18394.00	90.50	89.40	10738.17	813.05	7778.78	7818.96	1.68
252	18489.00	89.10	89.90	10738.50	813.63	7873.77	7913.69	1.56
253	18584.00	89.10	90.20	10739.99	813.55	7968.76	8008.37	0.32
254	18679.00	90.10	91.50	10740.65	812.14	8063.75	8102.93	1.73
255	18773.00	90.20	91.50	10740.41	809.68	8157.71	8196.40	0.11
256	18868.00	90.20	91.10	10740.08	807.52	8252.69	8290.89	0.42
257	18963.00	89.50	91.40	10740.32	805.45	8347.66	8385.39	0.80
258	19058.00	90.30	91.20	10740.49	803.29	8442.64	8479.89	0.87
259	19152.00	90.00	90.70	10740.24	801.73	8536.63	8573.45	0.62
260	19247.00	89.00	91.30	10741.07	800.08	8631.61	8667.99	1.23
261	19342.00	89.60	90.50	10742.23	798.58	8726.59	8762.54	1.05
262	19437.00	90.50	90.00	10742.15	798.17	8821.58	8857.20	1.08
263	19532.00	88.30	88.00	10743.15	799.83	8916.55	8951.99	3.13
264	19626.00	88.50	87.40	10745.77	803.60	9010.44	9045.88	0.67
265	19721.00	89.10	97.70	10747.77	799.38	9105.20	9139.99	10.86
266	19816.00	89.80	87.00	10748.68	795.49	9199.97	9234.15	11.29
267	19911.00	90.40	87.50	10748.51	800.05	9294.86	9329.10	0.82
268	20005.00	87.60	90.00	10750.15	802.10	9388.81	9422.90	3.99
269	20100.00	88.20	89.80	10753.64	802.26	9483.75	9517.55	0.67
270	20195.00	89.60	89.40	10755.46	802.93	9578.72	9612.27	1.53
271	20290.00	90.50	89.50	10755.38	803.84	9673.72	9707.03	0.95
272	20384.00	89.30	87.40	10755.54	806.38	9767.68	9800.89	2.57
273	20479.00	89.00	85.90	10756.95	811.93	9862.50	9895.85	1.61
274	20498.00	88.70	85.50	10757.33	813.36	9881.44	9914.85	2.63
275	20560.00	88.70	85.50	10758.74	818.22	9943.24	9976.83	0.00

FORMATION TOPS & STRUCTURAL RELATIONSHIPS

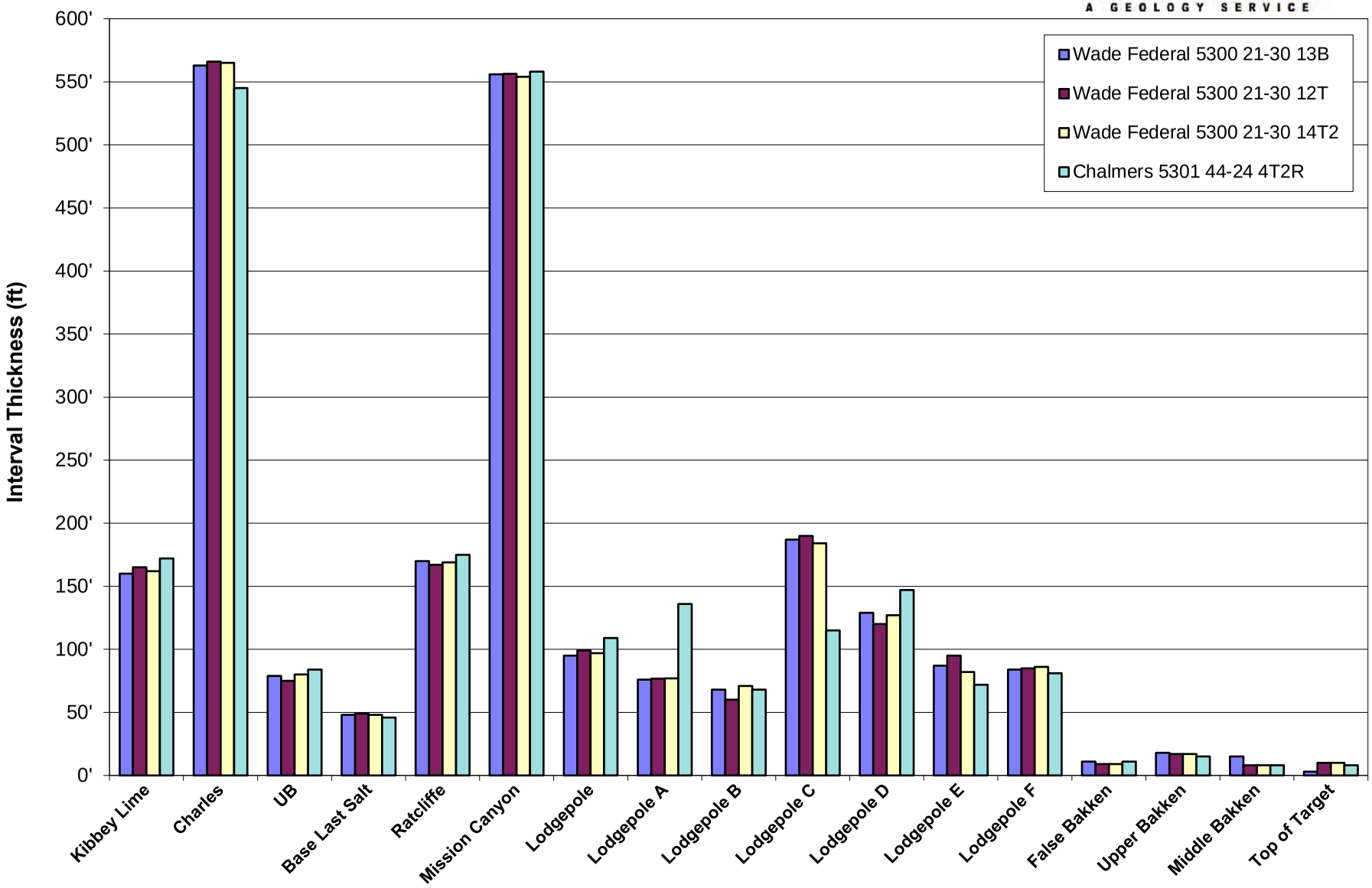
Operator:	Subject Well:								Offset Wells:		
Well Name:	Oasis Petroleum N. A., LLC										
	Wade Federal 5300 21-30 13B										
Location:	1,574' FNL & 270' FWL										
	Lot 2 Section 30, T153N, R100W										
Elevation:	GL: 1,999'	Sub: 25'	KB: 2,024'								
Formation/ Marker	Prog. Top	Prog. Datum (MSL)	Driller's Depth Top (MD)	Driller's Depth Top (TVD)	Datum (MSL)	Interval Thickness	Thickness to Target	Dip To Prog.	Dip To Wade Federal 5300 21-30 12T	Dip To Wade Federal 5300 21-30 14T2	Dip To Chalmers 5301 44-24 4T2R
Kibbey Lime	8,308'	-6,284'	8,312'	8,311'	-6,287'	160'	2,349'	-3'	-3'	-3'	11'
Charles	8,470'	-6,446'	8,472'	8,471'	-6,447'	563'	2,189'	-1'	2'	-1'	23'
UB	9,035'	-7,011'	9,035'	9,034'	-7,010'	79'	1,626'	1'	5'	1'	5'
Base Last Salt	9,115'	-7,091'	9,114'	9,113'	-7,089'	48'	1,547'	2'	1'	2'	10'
Ratcliffe	9,163'	-7,139'	9,162'	9,161'	-7,137'	170'	1,499'	2'	2'	2'	8'
Mission Canyon	9,332'	-7,308'	9,332'	9,331'	-7,307'	556'	1,329'	1'	-1'	1'	13'
Lodgepole	9,886'	-7,862'	9,888'	9,887'	-7,863'	95'	773'	-1'	-1'	-1'	15'
Lodgepole A	9,983'	-7,959'	9,983'	9,982'	-7,958'	76'	678'	1'	3'	1'	29'
Lodgepole B	10,060'	-8,036'	10,059'	10,058'	-8,034'	68'	602'	2'	4'	2'	89'
Lodgepole C	10,131'	-8,107'	10,127'	10,126'	-8,102'	187'	534'	5'	-4'	5'	89'
Lodgepole D	10,315'	-8,291'	10,319'	10,313'	-8,289'	129'	347'	2'	-1'	2'	17'
Lodgepole E	10,442'	-8,418'	10,471'	10,442'	-8,418'	87'	218'	0'	-10'	0'	35'
Lodgepole F	10,524'	-8,500'	10,594'	10,529'	-8,505'	84'	131'	-5'	-2'	-5'	20'
False Bakken	10,610'	-8,586'	10,771'	10,613'	-8,589'	11'	47'	-3'	-1'	-3'	17'
Upper Bakken	10,619'	-8,595'	10,802'	10,624'	-8,600'	18'	36'	-5'	-3'	-5'	17'
Middle Bakken	10,636'	-8,612'	10,861'	10,642'	-8,618'	15'	18'	-6'	-4'	-6'	14'
Top of Target	10,647'	-8,620'	10,937'	10,657'	-8,633'	3'	3'	-10'	-11'	-13'	7'
Landing Target	10,660'	-8,630'	11,051'	10,660'	-8,636'	-	0'	0'	-4'	-6'	12'

CONTROL DATA

Operator:	Oasis Petroleum North America				Oasis Petroleum North America				Oasis Petroleum North America			
Well Name:	Wade Federal 5300 21-30 12T				Wade Federal 5300 21-30 14T2				Chalmers 5301 44-24 4T2R			
Location:	SW NW Section 30, T153N, R100W McKenzie Co., ND Shares pad with subject well				SW NW Section 30, T153N, R100W McKenzie Co., ND Shares pad with subject well				SE SE Section 24, T153N, R101W McKenzie Co., ND 0.50 mile north of subject well			
Elevation:	KB: 2,024'				KB: 2,024'				KB: 1,968'			
Formation/ Zone	Top	Datum (MSL)	Interval Thickness	Thickness to Target	Top	Datum (MSL)	Interval Thickness	Thickness to Target	Top	Datum (MSL)	Interval Thickness	Thickness to Target
Kibbey Lime	8,308'	-6,284'	165'	2,348'	8,308'	-6,284'	162'	2,346'	8,266'	-6,298'	172'	2,350'
Charles	8,473'	-6,449'	566'	2,183'	8,470'	-6,446'	565'	2,184'	8,438'	-6,470'	545'	2,178'
UB	9,039'	-7,015'	75'	1,617'	9,035'	-7,011'	80'	1,619'	8,983'	-7,015'	84'	1,633'
Base Last Salt	9,114'	-7,090'	49'	1,542'	9,115'	-7,091'	48'	1,539'	9,067'	-7,099'	46'	1,549'
Ratcliffe	9,163'	-7,139'	167'	1,493'	9,163'	-7,139'	169'	1,491'	9,113'	-7,145'	175'	1,503'
Mission Canyon	9,330'	-7,306'	556'	1,326'	9,332'	-7,308'	554'	1,322'	9,288'	-7,320'	558'	1,328'
Lodgepole	9,886'	-7,862'	99'	770'	9,886'	-7,862'	97'	768'	9,846'	-7,878'	109'	770'
Lodgepole A	9,985'	-7,961'	77'	671'	9,983'	-7,959'	77'	671'	9,955'	-7,987'	136'	661'
Lodgepole B	10,062'	-8,038'	60'	594'	10,060'	-8,036'	71'	594'	10,091'	-8,123'	68'	525'
Lodgepole C	10,122'	-8,098'	190'	534'	10,131'	-8,107'	184'	523'	10,159'	-8,191'	115'	457'
Lodgepole D	10,312'	-8,288'	120'	344'	10,315'	-8,291'	127'	339'	10,274'	-8,306'	147'	342'
Lodgepole E	10,432'	-8,408'	95'	224'	10,442'	-8,418'	82'	212'	10,421'	-8,453'	72'	195'
Lodgepole F	10,527'	-8,503'	85'	129'	10,524'	-8,500'	86'	130'	10,493'	-8,525'	81'	123'
False Bakken	10,612'	-8,588'	9'	44'	10,610'	-8,586'	9'	44'	10,574'	-8,606'	11'	42'
Upper Bakken	10,621'	-8,597'	17'	35'	10,619'	-8,595'	17'	35'	10,585'	-8,617'	15'	31'
Middle Bakken	10,638'	-8,614'	8'	18'	10,636'	-8,612'	8'	18'	10,600'	-8,632'	8'	16'
Top of Target	10,646'	-8,622'	10'	10'	10,644'	-8,620'	10'	10'	10,608'	-8,640'	8'	8'
Landing Target	10,656'	-8,632'	-	0'	10,654'	-8,630'	-	0'	10,616'	-8,648'	-	0'

INTERVAL THICKNESS

Oasis Petroleum N. A., LLC - Wade Federal 5300 21-30 13B

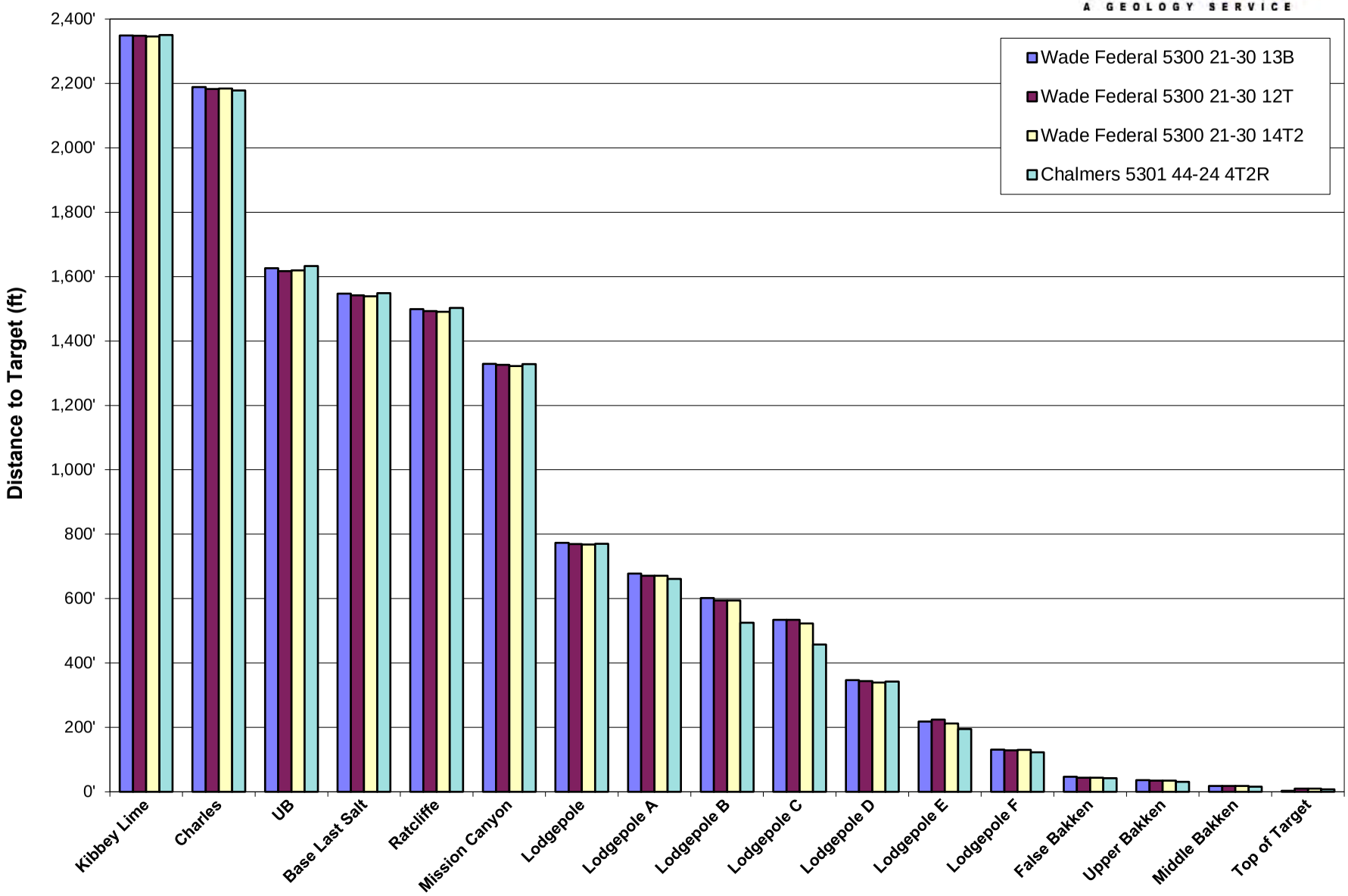


LANDING PROJECTION

Formation/ Zone:	Proposed Top of Target From:			
	Wade Federal 5300 21-30 12T	Wade Federal 5300 21-30 14T2	Chalmers 5301 44-24 4T2R	Average of Offset Wells
Kibbey Lime	10,659'	10,657'	10,661'	10,659'
Charles	10,654'	10,655'	10,649'	10,653'
UB	10,651'	10,653'	10,667'	10,657'
Base Last Salt	10,655'	10,652'	10,662'	10,656'
Ratcliffe	10,654'	10,652'	10,664'	10,657'
Mission Canyon	10,657'	10,653'	10,659'	10,656'
Lodgepole	10,657'	10,655'	10,657'	10,656'
Lodgepole A	10,653'	10,653'	10,643'	10,650'
Lodgepole B	10,652'	10,652'	10,583'	10,629'
Lodgepole C	10,660'	10,649'	10,583'	10,631'
Lodgepole D	10,657'	10,652'	10,655'	10,655'
Lodgepole E	10,666'	10,654'	10,637'	10,652'
Lodgepole F	10,658'	10,659'	10,652'	10,656'
False Bakken	10,657'	10,657'	10,655'	10,656'
Upper Bakken	10,659'	10,659'	10,655'	10,658'
Middle Bakken	10,660'	10,660'	10,658'	10,659'
Top of Target	10,667'	10,667'	10,665'	10,666'
Landing Target	10,660'	10,660'	10,660'	10,660'

ISOPACH TO TARGET

Oasis Petroleum North America, LLC - Wade Federal 5300 21-30 13B



LITHOLOGY
Wade Federal 5300 21-30 13B

Rig crews caught lagged samples in 30' intervals under the supervision of Sunburst geologists. A detailed list of sampling intervals is included in the well data summary page. Sample or gamma ray marker tops have been inserted in the sample descriptions below for reference. Samples were examined wet and dry under a binocular microscope. Drilling fluid was diesel-based invert from surface casing to intermediate casing, while salt water drilling fluid was used throughout the lateral. Sample collection began at 8,200' MD.

Logging began in the Kibbey Formation

8,200-8,230 SILTSTONE: dark orange, light brown, tan, pink, soft, sub-blocky, calcite cemented, poorly cemented; rare ANHYDRITE: off white, soft, amorphous texture; trace SILTY SANDSTONE: tan-off white, very fine grained, sub-rounded, moderately sorted, calcite cemented, poorly cemented

8,230-8,260 SILTSTONE: dark orange, light brown, tan, pink, soft, sub-blocky, calcite cemented, poorly cemented; rare ANHYDRITE: off white, soft, amorphous texture; trace SILTY SANDSTONE: as above

8,260-8,290 SILTSTONE: dark orange, light brown, tan pink, soft, sub-blocky, calcite cemented, poorly cemented; occasional ANHYDRITE: off white, amorphous texture; trace SILTY SANDSTONE: as above

Kibbey "Lime" [8,312' MD, 8,311' TVD (-6,287')]

8,290-8,320 SILTSTONE: dark orange, light brown, tan pink, soft, sub-blocky, calcite cemented, poorly cemented; occasional ANHYDRITE: off white, amorphous texture; trace SILTY SANDSTONE: as above

8,320-8,350 SILTSTONE: dark orange, light brown, tan pink, soft, sub-blocky, calcite cemented, poorly cemented; occasional ANHYDRITE: off white, amorphous texture; trace SILTY SANDSTONE: as above

8,350-8,380 SILTSTONE: dark orange, light brown, tan pink, soft, sub-blocky, calcite cemented, poorly cemented; occasional ANHYDRITE: off white, amorphous texture; trace SILTY SANDSTONE: as above

8,380-8,410 SILTSTONE: dark orange, light brown, tan pink, soft, sub-blocky, calcite cemented, poorly cemented; occasional ANHYDRITE: off white, amorphous texture; trace SILTY SANDSTONE: as above

8,410-8,440 SILTSTONE: dark orange, light brown, tan pink, soft, sub-blocky, calcite cemented, poorly cemented; occasional ANHYDRITE: off white, amorphous texture; trace SILTY SANDSTONE: as above

8,440-8,470 SILTSTONE: dark orange, light brown, tan pink, soft, sub-blocky, calcite cemented, poorly cemented; occasional ANHYDRITE: off white, amorphous texture; trace SILTY SANDSTONE: as above

Charles Formation: 1st Salt [8,472' MD, 8,471' TVD (-6,447')]

8,470-8,500 SILTSTONE: orange, brown, trace pink, soft to friable, sub-blocky, moderately calcite cemented; occasional ANHYDRITE: off-white, soft, amorphous

8,500-8,530 SALT: translucent, frosted, microcrystalline, anhedral, crystalline texture, hard; occasional ANHYDRITE: off-white, soft, amorphous

8,530-8,560 SALT: translucent, frosted, milky, microcrystalline, sub-euhedral, occasional SILTSTONE: orange, brown, tan, pink, soft, sub-blocky, moderately calcite cemented

8,560-8,590 SALT: translucent, frosted, microcrystalline, anhedral, crystalline, hard, occasional LIMESTONE: mudstone, medium gray, microcrystalline, friable-firm, dense, earthy, no visible porosity

8,590-8,620 SALT: translucent, frosted-white, microcrystalline, anhedral, crystalline, hard

8,620-8,650 LIMESTONE: mudstone, light to medium gray, occasional orange-brown, microcrystalline, soft to friable, argillaceous in part, chalky in part, no visible porosity; occasional SALT: translucent, frosted-white, microcrystalline, anhedral, crystalline, hard; occasional ANHYDRITE: off-white, microcrystalline, very soft, chalky, no visible porosity

8,650-8,680 SALT: translucent, microcrystalline, anhedral, crystalline, hard; occasional LIMESTONE: mudstone, light to medium gray, occasional orange-brown, microcrystalline, soft to friable, argillaceous in part, chalky in part, no visible porosity

8,680-8,710 LIMESTONE: mudstone, light to medium gray, tan-brown, friable-firm, argillaceous in part, earthy texture, no visible porosity; common ANHYDRITE: white, microcrystalline, soft, anhedral, chalky, no visible porosity

8,710-8,740 LIMESTONE: mudstone, light to medium gray, tan-brown, firm-hard, earthy texture, no visible porosity; white, off-white, microcrystalline, soft, anhedral, chalky, no visible porosity; occasional SALT: frosted, microcrystalline, anhedral, hard

8,740-8,770 LIMESTONE: mudstone, light to medium gray, tan-brown, firm-hard, earthy texture, no visible porosity; white, off-white, microcrystalline, soft, anhedral, chalky, no visible porosity

8,770-8,800 LIMESTONE: mudstone, light to medium gray, frequent brown, firm-hard, earthy texture, no visible porosity; common SALT: translucent, off-white, crystalline, anhedral, hard

8,800-8,830 LIMESTONE: mudstone, light to medium gray, common brown, microcrystalline, hard, earthy texture, no visible porosity

8,830-8,860 LIMESTONE: mudstone, light to medium gray, common brown, microcrystalline, hard, earthy texture, no visible porosity

8,860-8,890 LIMESTONE: mudstone, medium gray, medium brown, occasional tan, microcrystalline, friable-firm, earthy texture, dolomitic, no visible porosity; trace SALT: translucent, off-white, crystalline, anhedral, hard

8,890-8,920 LIMESTONE: mudstone, medium gray, medium gray, tan, microcrystalline, firm, earthy texture, dolomitic, no visible porosity

8,920-8,950 ARGILLACEOUS LIMESTONE: mudstone, medium gray, tan, microcrystalline, friable-firm, dense, earthy texture, no visible porosity; white, off-white, soft, amorphous, no visible porosity

8,950-8,980 ARGILLACEOUS LIMESTONE: mudstone, medium gray, light brown, tan, microcrystalline, friable-firm, dense, earthy texture, no visible porosity; white, off-white, soft to friable, amorphous, no visible porosity

8,980-9,010 LIMESTONE: mudstone, medium gray, light brown, dark brown, tan in part, microcrystalline, firm, dense, earthy texture, no visible porosity; white, off-white, soft, amorphous, no visible porosity

9,010-9,040 LIMESTONE: mudstone, light gray, light brown, tan, microcrystalline, friable-firm, earthy texture, no visible porosity; off-white, soft, amorphous, anhedral, no visible porosity; trace SALT: white, off-white, frosted, crystalline, hard, amorphous, no visible porosity

UB [9,035’ MD, 9,034’ TVD (-7,010’)]

9,040-9,070 SALT: white, translucent, microcrystalline, firm, anhedral, crystalline, frosted; ARGILLACEOUS LIMESTONE: mudstone, tan, light gray, light brown, microcrystalline, friable, earthy texture, no visible porosity

9,070-9,100 ARGILLACEOUS LIMESTONE: mudstone, tan, light gray, light brown, microcrystalline, friable, earthy texture, no visible porosity; occasional SALT: white, microcrystalline, hard, anhedral

Base Last Salt [9,114’ MD, 9,113’ TVD (-7,089’)]

9,130-9,160 ARGILLACEOUS LIMESTONE: mudstone, light gray, light to medium brown, microcrystalline, friable-firm, earthy texture, no visible porosity; off-white, microcrystalline, soft, amorphous, no visible porosity

Ratcliffe [9,162’ MD, 9,161’ TVD (-7,137’)]

9,160-9,190 ARGILLACEOUS LIMESTONE: mudstone, light gray, medium brown, microcrystalline, firm, earthy texture, no visible porosity; off-white, microcrystalline, soft, amorphous, no visible porosity

9,190-9,220 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense-earthy texture, no visible porosity; off white, soft, amorphous, no visible porosity

9,220-9,250 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense-earthy texture, no visible porosity; off white, soft, amorphous, no visible porosity

9,250-9,280 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense-earthy texture, no visible porosity; off white, soft, amorphous, no visible porosity

9,280-9,310 ARGILLACEOUS LIMESTONE: mudstone, tan, light gray, medium gray, microcrystalline, friable-firm, dense, earthy texture, no visible porosity

9,310-9,340 ARGILLACEOUS LIMESTONE: mudstone, tan, light gray, medium gray, microcrystalline, friable-firm, dense, earthy texture, no visible porosity

Mission Canyon Formation [9,332’ MD, 9,331’ TVD (-7,307’)]

9,340-9,370 ARGILLACEOUS LIMESTONE: mudstone, tan, light gray, medium gray, microcrystalline, friable-firm, dense, earthy texture, no visible porosity

9,370-9,400 ARGILLACEOUS LIMESTONE: mudstone, tan, light gray, medium gray, microcrystalline, friable-firm, dense, earthy texture, no visible porosity

9,400-9,430 ARGILLACEOUS LIMESTONE: mudstone, medium gray, light brown, microcrystalline, firm, dense, earthy texture, occasional lamination in part, no visible porosity

9,430-9,460 ARGILLACEOUS LIMESTONE: mudstone, medium gray, light brown, microcrystalline, firm, dense, earthy texture, occasional lamination in part, no visible porosity

9,460-9,490 ARGILLACEOUS LIMESTONE: mudstone, medium brown, medium gray, dark gray to black, microcrystalline, friable-firm, earthy texture, laminated in part, no visible porosity

9,490-9,520 ARGILLACEOUS LIMESTONE: mudstone, medium brown, medium gray, dark gray to black, microcrystalline, friable-firm, earthy texture, laminated in part, no visible porosity

9,520-9,550 ARGILLACEOUS LIMESTONE: mudstone, medium brown, medium gray, dark gray to black, microcrystalline, friable-firm, earthy texture, laminated in part, no visible porosity

9,550-9,580 ARGILLACEOUS LIMESTONE: mudstone, light brown, dark brown, microcrystalline, firm, dense, earthy texture, lamination visible in part, no visible porosity

9,580-9,610 ARGILLACEOUS LIMESTONE: mudstone, light brown, dark brown, microcrystalline, firm, dense, earthy texture, lamination visible in part, no visible porosity

9,610-9,640 ARGILLACEOUS LIMESTONE: mudstone, light brown, dark brown, microcrystalline, firm, dense, earthy texture, lamination visible in part, no visible porosity

9,640-9,670 ARGILLACEOUS LIMESTONE: mudstone, medium brown, medium gray, dark gray to black, microcrystalline, friable-firm, earthy texture, laminated in part, no visible porosity

9,670-9,700 ARGILLACEOUS LIMESTONE: mudstone, medium brown, medium gray, dark gray to black, microcrystalline, friable-firm, earthy texture, laminated in part, no visible porosity

9,700-9,730 ARGILLACEOUS LIMESTONE: mudstone, medium brown, medium gray, dark gray to black, microcrystalline, friable-firm, earthy texture, laminated in part, no visible porosity

9,730-9,760 ARGILLACEOUS LIMESTONE: mudstone, medium brown, medium gray, dark gray to black, microcrystalline, friable-firm, earthy texture, laminated in part, no visible porosity

9,760-9,790 ARGILLACEOUS LIMESTONE: mudstone, medium brown, medium gray, dark gray, occasional black, microcrystalline, friable-firm, earthy, occasional laminated in part, no visible porosity

9,790-9,820 ARGILLACEOUS LIMESTONE: mudstone, light gray, medium gray, microcrystalline, firm, earthy texture, occasional laminated in part, no visible porosity

9,820-9,850 ARGILLACEOUS LIMESTONE: mudstone, light gray, medium gray, occasional brown, microcrystalline, friable, earthy texture, no visible porosity

9,850-9,880 ARGILLACEOUS LIMESTONE: mudstone, light to medium gray, medium brown, microcrystalline, friable-fair, earthy texture, no visible porosity

Lodgepole Formation [9,888’ MD, 9,887’ TVD (-7,863’)]

9,880-9,910 ARGILLACEOUS LIMESTONE: mudstone, medium gray-brown, occasional dark brown, occasional light brown-tan, microcrystalline, dark sample are firm, light brown friable, earthy texture, no visible porosity

9,910-9,940 ARGILLACEOUS LIMESTONE: mudstone, medium gray, brown, dark brown, microcrystalline, firm, earthy texture, no visible porosity

9,940-9,970 ARGILLACEOUS LIMESTONE: mudstone, light to medium gray, brown, microcrystalline, friable-firm, earthy texture, no visible porosity

9,970-10,000 ARGILLACEOUS LIMESTONE: mudstone, medium gray, medium-dark brown, microcrystalline, firm, earthy texture, no visible porosity; trace disseminated pyrite

10,000-10,030 ARGILLACEOUS LIMESTONE: mudstone, light gray, medium gray, dark brown, microcrystalline, firm, earthy texture, no visible porosity

10,030-10,060 ARGILLACEOUS LIMESTONE: mudstone, medium brown, dark brown, microcrystalline, friable, earthy texture, no visible porosity

10,060-10,090 ARGILLACEOUS LIMESTONE: mudstone, light gray, medium gray, brown, microcrystalline, friable-firm, earthy texture, no visible porosity

10,090-10,118 ARGILLACEOUS LIMESTONE: mudstone, light gray, medium gray, brown, microcrystalline, friable-firm, earthy texture, no visible porosity

10,118-10,150 ARGILLACEOUS LIMESTONE: mudstone, light to medium brown, common dark brown, microcrystalline, friable, firm in part, dense, no visible porosity

10,150-10,180 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, dense, trace disseminated pyrite, no visible porosity

10,180-10,210 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, dense, trace disseminated pyrite, no visible porosity

10,210-10,240 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, dense, trace disseminated pyrite, no visible porosity

10,240-10,270 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, dense, trace disseminated pyrite, no visible porosity

10,270-10,300 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, dense, trace disseminated pyrite, no visible porosity

10,300-10,330 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, dense, trace disseminated pyrite, no visible porosity

10,330-10,360 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, dense, trace disseminated pyrite, no visible porosity

10,360-10,390 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, dense, trace disseminated pyrite, no visible porosity

10,390-10,420 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, dense, trace disseminated pyrite, no visible porosity

10,420-10,450 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, dense, trace disseminated pyrite, no visible porosity

10,450-10,480 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, dense, trace disseminated pyrite, no visible porosity

10,480-10,510 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, dense, trace disseminated pyrite, no visible porosity

10,510-10,540 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, dense, trace disseminated pyrite, no visible porosity

10,540-10,570 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, dense, trace disseminated pyrite, no visible porosity

10,570-10,600 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, dense, trace disseminated pyrite, no visible porosity

10,600-10,630 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, dense, trace disseminated pyrite, no visible porosity

10,630-10,660 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, dense, trace disseminated pyrite, no visible porosity

10,660-10,690 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, dense, trace disseminated pyrite, no visible porosity

10,690-10,720 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, dense, trace disseminated pyrite, no visible porosity

10,720-10,750 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, dense, trace disseminated pyrite, no visible porosity

10,750-10,780 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, dense, trace disseminated pyrite, no visible porosity

False Bakken [10,771’ MD, 10,613’ TVD (-8,589’)]

10,780-10,810 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, dense, trace disseminated pyrite, no visible porosity; trace SHALE: black-very dark brown, firm, blocky, earthy texture, no visible porosity, abundant brown even oil stain

Bakken Formation: Upper Bakken Shale [10,802’ MD, 10,624’ TVD (-8,600’)]

10,810-10,820 LIMESTONE: mudstone, gray to dark gray, microcrystalline, firm, dense, trace disseminated pyrite, no visible porosity; occasional SHALE: as above

10,820-10,830 SHALE: black-very dark brown, firm, blocky, earthy texture, petroliferous, carbonaceous, no visible porosity, abundant brown even oil stain

10,830-10,840 SHALE: black-very dark brown, firm, blocky, earthy texture, petroliferous, carbonaceous, no visible porosity, abundant brown even oil stain

10,840-10,850 SHALE: black-very dark brown, firm, blocky, earthy texture, petroliferous, carbonaceous, no visible porosity, abundant brown even oil stain

10,850-10,860 SHALE: black-very dark brown, firm, blocky, earthy texture, petroliferous, carbonaceous, no visible porosity, abundant brown even oil stain

Middle Bakken [10,861’ MD, 10,642’ TVD (-8,618’)]

10,860-10,870 SHALE: black-very dark brown, firm, blocky, earthy texture, petroliferous, carbonaceous, no visible porosity, common brown even oil stain

10,870-10,880 SILTSTONE: medium gray, medium-dark brown, sub-blocky, moderately calcite cemented, earthy, calcite cemented; occasional SHALE: black, dark brown, blocky, earthy, even oil stain

10,880-10,890 SILTSTONE: medium gray, dark brown, sub-blocky, moderately calcite cemented, earthy, calcite cemented; rare disseminated pyrite

10,890-10,900 SILTSTONE: medium gray, medium brown, sub-blocky, moderately calcite cemented, earthy, moderately calcite cemented; rare disseminated pyrite

10,900-10,910 SILTSTONE: medium gray, medium brown, occasional dark brown, sub-blocky, earthy, moderately calcite cemented, trace oil stain; rare disseminated pyrite

10,910-10,920 SILTSTONE: medium gray, medium brown, sub-blocky, moderately calcite cemented, earthy, trace oil stain; rare disseminated pyrite; occasional SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain

10,920-10,930 SILTSTONE: medium gray, dark gray, brown, sub-blocky, moderately calcite cemented, earthy, trace oil stain; rare disseminated pyrite; occasional SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain

10,930-10,940 SILTSTONE: light gray, medium gray, friable-firm, sub-blocky, moderately calcite cemented, earthy, trace oil stain; rare disseminated pyrite; occasional SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain

10,940-10,950 SILTSTONE: light gray, medium gray, friable-firm, sub-blocky, moderately calcite cemented, earthy, trace oil stain; rare disseminated pyrite; occasional SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain

10,950-10,960 SILTSTONE: medium gray, sub-blocky, moderately calcite cemented, earthy, trace oil stain; occasional SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain

10,960-10,970 SILTSTONE: light gray, medium gray, medium brown, friable-firm, sub-blocky, moderately calcite cemented, earthy, trace oil stain; occasional SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain

10,970-10,980 SILTSTONE: medium gray, light brown, sub-blocky, friable, moderately calcite cemented, earthy, trace oil stain; common SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain

10,980-10,990 SILTSTONE: light gray, medium gray, sub-blocky, friable, moderately calcite cemented, earthy, rare trace oil stain; SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain

10,990-11,000 SILTSTONE: medium gray, sub-blocky, friable-firm, calcite cemented, earthy, visible oil stain; common SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain

11,000-11,010 SILTSTONE: light to medium gray, sub-blocky, friable-firm, calcite cemented, earthy, visible oil stain; common SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain

11,010-11,020 SILTSTONE: light to medium gray, sub-blocky, friable-firm, calcite cemented, earthy, visible oil stain; common SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain

11,020-11,030 SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain

11,030-11,040 SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain

11,040-11,050 SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain

11,080-11,110 SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak diffuse pale yellow cut fluorescence

11,110-11,140 SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak diffuse pale yellow cut fluorescence

11,140-11,170 SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak diffuse pale yellow cut fluorescence

11,170-11,200 SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak diffuse pale yellow cut fluorescence

11,200-11,230 SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak diffuse pale yellow cut fluorescence

11,230-11,260 SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak diffuse pale yellow cut fluorescence

11,260-11,290 SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak diffuse pale yellow cut fluorescence

11,290-11,320 SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak diffuse pale yellow cut fluorescence

11,320-11,350 SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak diffuse pale yellow cut fluorescence

11,350-11,380 SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak diffuse pale yellow cut fluorescence

11,380-11,410 SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak cloudy pale yellow cut fluorescence

11,410-11,440 SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak cloudy pale yellow cut fluorescence

11,440-11,470 SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak cloudy pale yellow cut fluorescence

11,470-11,500 SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak cloudy pale yellow cut fluorescence

11,500-11,530 SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak cloudy pale yellow cut fluorescence

11,530-11,560 SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak cloudy pale yellow cut fluorescence

11,560-11,590 SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak cloudy pale yellow cut fluorescence

11,590-11,620 SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak cloudy pale yellow cut fluorescence

11,620-11,650 SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak cloudy pale yellow cut fluorescence

11,650-11,680 SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak cloudy pale yellow cut fluorescence

11,680-11,710 SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak cloudy pale yellow cut fluorescence

11,710-11,740 SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak cloudy pale yellow cut fluorescence

11,740-11,770 SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak cloudy pale yellow cut fluorescence

11,770-11,800 SILTY SANDSTONE: medium to light gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak cloudy pale yellow cut fluorescence

11,800-11,830 SILTY SANDSTONE: medium to light gray, occasional light brown, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak cloudy pale yellow cut fluorescence

11,830-11,860 SILTY SANDSTONE: light gray, rare light brown, very fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, occasional disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak diffuse cloudy pale yellow cut fluorescence

11,860-11,890 SILTY SANDSTONE: light to medium gray, very fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, occasional disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak diffuse cloudy pale yellow cut fluorescence

11,890-11,920 SILTY SANDSTONE: light to medium gray, very fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, occasional disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak diffuse cloudy pale yellow cut fluorescence

11,920-11,950 SILTY SANDSTONE: light to medium gray, trace light-brown, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak diffuse cloudy pale yellow cut fluorescence

11,950-11,980 SILTY SANDSTONE: light gray, light-brown, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak diffuse cloudy pale yellow cut fluorescence

11,980-12,010 SILTY SANDSTONE: light gray, light-brown, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak diffuse cloudy pale yellow cut fluorescence

12,010-12,040 SILTY SANDSTONE: light gray, light-brown, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain, fast weak diffuse cloudy pale yellow cut fluorescence

12,040-12,070 SILTY SANDSTONE: light to medium gray, light-brown, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, occasional disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast weak diffuse cloudy pale yellow cut fluorescence

12,070-12,100 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast weak diffuse cloudy pale yellow cut fluorescence

12,100-12,130 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast weak diffuse cloudy pale yellow cut fluorescence

12,130-12,160 SILTY SANDSTONE: light to medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, moderately cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast weak diffuse cloudy pale yellow cut fluorescence

12,160-12,190 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, moderately well sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately weak diffuse cloudy pale yellow cut fluorescence

12,190-12,220 SILTY SANDSTONE: light gray, medium gray, fine grained, sub-rounded, firm, moderately well sorted, calcite cemented, poorly cemented, trace disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately weak diffuse cloudy pale yellow cut fluorescence

12,220-12,250 SILTY SANDSTONE: light gray, medium gray, fine grained, sub-rounded, firm, moderately well sorted, calcite cemented, poorly cemented, trace disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately weak diffuse cloudy pale yellow cut fluorescence

12,250-12,280 SILTY SANDSTONE: light gray, medium gray, fine grained, sub-rounded, firm, moderately well sorted, calcite cemented, poorly cemented, trace disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately weak diffuse cloudy pale yellow cut fluorescence

12,280-12,310 SILTY SANDSTONE: light gray, very fine grained, sub-rounded, firm, well sorted, moderately calcite cemented, poorly cemented, trace disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, fast weak diffuse cloudy pale yellow cut fluorescence

12,310-12,340 SILTY SANDSTONE: light gray, very fine grained, sub-rounded, firm, well sorted, moderately calcite cemented, poorly cemented, trace disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, fast weak diffuse cloudy pale yellow cut fluorescence

12,340-12,370 SILTY SANDSTONE: light gray, very fine grained, sub-rounded, firm, well sorted, moderately calcite cemented, poorly cemented, trace disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, fast weak diffuse cloudy pale yellow cut fluorescence

12,370-12,400 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, trace disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

12,400-12,430 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, trace disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

12,430-12,460 SILTY SANDSTONE: light gray, occasional medium gray, very fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

12,460-12,490 SILTY SANDSTONE: light gray, occasional medium gray, very fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

12,490-12,520 SILTY SANDSTONE: light gray, medium gray, fine grained, sub-rounded, firm, very well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

12,520-12,550 SILTY SANDSTONE: light gray, medium gray, fine grained, sub-rounded, firm, very well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

12,550-12,580 SILTY SANDSTONE: light gray, medium gray, fine grained, sub-rounded, firm, very well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

12,580-12,610 SILTY SANDSTONE: light to medium gray, fine grained, sub-rounded, firm, very well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

12,610-12,640 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

12,640-12,670 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

12,670-12,700 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

12,700-12,730 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

12,730-12,760 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

12,760-12,790 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

12,790-12,820 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

12,820-12,850 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

12,850-12,880 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

12,880-12,910 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

12,910-12,940 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

12,940-12,970 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

12,970-13,000 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

13,000-13,030 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

13,030-13,060 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

13,060-13,090 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

13,090-13,120 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

13,120-13,150 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

13,150-13,180 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

13,180-13,210 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

13,210-13,240 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

13,240-13,270 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

13,270-13,300 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

13,300-13,330 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

13,330-13,360 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

13,360-13,390 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

13,390-13,420 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

13,420-13,450 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

13,450-13,480 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

13,480-13,510 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

13,510-13,540 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

13,540-13,570 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

13,570-13,600 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

13,600-13,630 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

13,630-13,660 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

13,660-13,690 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

13,690-13,720 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

13,720-13,750 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, common spotty light brown oil stain, fast moderately diffuse cloudy pale yellow cut fluorescence

13,750-13,780 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, trace disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

13,780-13,810 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, trace disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

13,810-13,840 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, trace disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

13,840-13,870 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, trace disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

13,870-13,900 SILTY SANDSTONE: light gray, rare medium gray, very fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

13,900-13,930 SILTY SANDSTONE: light gray, very fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

13,930-13,960 SILTY SANDSTONE: light gray, rare medium gray, very fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

13,960-13,990 SILTY SANDSTONE: light gray, rare medium gray, very fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

13,990-14,020 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

14,020-14,050 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, moderately calcite cemented, poorly cemented, disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately streaming cloudy pale yellow cut fluorescence

14,050-14,080 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, moderately calcite cemented, poorly cemented, disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately streaming cloudy pale yellow cut fluorescence

14,080-14,110 SILTY SANDSTONE: light gray, medium gray, fine grained, sub-rounded, firm, well sorted, moderately calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately streaming cloudy pale yellow cut fluorescence

14,120-14,140 SILTY SANDSTONE: light gray, medium gray, fine grained, sub-rounded, firm, well sorted, moderately calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately streaming cloudy pale yellow cut fluorescence

14,140-14,170 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, moderately calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately streaming cloudy pale yellow cut fluorescence

14,170-14,200 SILTY SANDSTONE: light gray, common medium gray, very fine grained, sub-rounded, firm, very well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately streaming cloudy pale yellow cut fluorescence

14,200-14,230 SILTY SANDSTONE: light gray, common medium gray, very fine grained, sub-rounded, firm, very well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

14,230-14,260 SILTY SANDSTONE: light gray, common medium gray, very fine grained, sub-rounded, firm, very well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

14,260-14,290 SILTY SANDSTONE: light gray, medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

14,290-14,320 SILTY SANDSTONE: light gray, medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

14,320-14,350 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, fast diffuse cloudy pale yellow cut fluorescence

14,350-14,380 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, fast diffuse cloudy pale yellow cut fluorescence

14,380-14,410 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

14,410-14,440 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

14,440-14,470 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite AND rare nodules pyrite, trace intergranular porosity, rare spotty light brown oil stain, fast diffuse cloudy pale yellow cut fluorescence

14,470-14,500 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain on light gray sample, moderately diffuse cloudy pale yellow cut fluorescence

14,500-14,530 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain on light gray sample, moderately diffuse cloudy pale yellow cut fluorescence

14,530-14,560 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately streaming cloudy pale yellow cut fluorescence

14,560-14,590 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately streaming cloudy pale yellow cut fluorescence

14,590-14,620 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately streaming cloudy pale yellow cut fluorescence

14,620-14,650 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately streaming cloudy pale yellow cut fluorescence

14,650-14,680 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

14,680-14,710 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

14,710-14,740 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

14,740-14,770 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

14,770-14,800 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

14,800-14,830 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

14,830-14,860 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

14,860-14,890 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

14,890-14,920 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

14,920-14,950 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

14,950-14,980 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

14,980-15,010 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,010-15,040 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,040-15,070 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,070-15,100 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,100-15,130 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,130-15,160 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,160-15,190 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,190-15,220 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,220-15,250 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,250-15,280 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,280-15,310 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,310-15,340 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,340-15,370 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,370-15,400 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,400-15,430 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,430-15,460 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,460-15,490 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,490-15,520 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,520-15,550 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,550-15,580 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,580-15,610 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,610-15,640 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,640-15,670 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,670-15,700 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,700-15,730 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,730-15,760 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,760-15,790 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,790-15,820 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,820-15,850 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,850-15,880 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,880-15,910 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,910-15,940 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,940-15,970 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

15,970-16,000 SILTY SANDSTONE: light gray, trace medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

16,000-16,030 SILTY SANDSTONE: light gray, trace medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

16,030-16,060 SILTY SANDSTONE: light gray, trace medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

16,060-16,090 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, very well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

16,090-16,120 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

16,120-16,150 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

16,150-16,180 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

16,180-16,210 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, trace disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, slow streaming cloudy pale yellow cut fluorescence

16,210-16,240 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, trace disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, slow streaming cloudy pale yellow cut fluorescence

16,240-16,270 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, trace disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, slow streaming cloudy pale yellow cut fluorescence

16,270-16,300 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, trace disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, slow streaming cloudy pale yellow cut fluorescence

16,300-16,330 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, trace disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

16,330-16,360 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, trace disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

16,360-16,390 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, trace disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, slow diffuse cloudy pale yellow cut fluorescence

16,390-16,420 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, trace disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, slow diffuse cloudy pale yellow cut fluorescence

16,420-16,450 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, trace disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, slow diffuse cloudy pale yellow cut fluorescence

16,450-16,480 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, trace disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, slow diffuse cloudy pale yellow cut fluorescence

16,480-16,510 SILTY SANDSTONE: light gray, medium gray, fine grained, sub-rounded, firm, moderately well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

16,510-16,540 SILTY SANDSTONE: light gray, medium gray, fine grained, sub-rounded, firm, moderately well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

16,540-16,570 SILTY SANDSTONE: light gray, medium gray, fine grained, sub-rounded, firm, moderately well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

16,570-16,600 SILTY SANDSTONE: light gray, medium gray, occasional dark gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

16,600-16,630 SILTY SANDSTONE: light gray, medium gray, occasional dark gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

16,630-16,660 SILTY SANDSTONE: light gray, medium gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, slow diffuse cloudy pale yellow cut fluorescence

16,660-16,690 SILTY SANDSTONE: light gray, medium gray, fine grained, sub-rounded, firm, moderately sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, slow diffuse cloudy pale yellow cut fluorescence

16,690-16,720 SILTY SANDSTONE: light gray, common medium gray, rare dark gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, slow diffuse cloudy pale yellow cut fluorescence

16,720-16,750 SILTY SANDSTONE: light gray, common medium gray, rare dark gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, slow diffuse cloudy pale yellow cut fluorescence

16,750-16,780 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, slow diffuse cloudy pale yellow cut fluorescence

16,780-16,810 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, slow diffuse cloudy pale yellow cut fluorescence

16,810-16,840 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

16,840-16,870 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

16,870-16,900 SILTY SANDSTONE: light gray, medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

16,900-16,930 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

16,930-16,960 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain, moderately diffuse cloudy pale yellow cut fluorescence

16,960-16,990 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

16,990-17,020 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,020-17,050 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,050-17,080 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,080-17,110 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,110-17,140 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,140-17,170 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,170-17,200 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,200-17,230 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,230-17,260 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,260-17,290 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,290-17,320 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,320-17,350 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,350-17,380 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,380-17,410 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,410-17,440 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,440-17,470 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,470-17,500 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,500-17,530 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,530-17,560 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,560-17,590 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,590-17,620 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,620-17,650 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,650-17,680 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,680-17,710 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,710-17,740 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,740-17,770 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,770-17,800 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,800-17,830 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,830-17,860 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,860-17,890 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,890-17,920 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,920-17,950 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,950-17,980 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,950-17,980 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

17,980-18,010 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

18,010-18,040 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

18,040-18,070 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

18,070-18,100 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

18,100-18,130 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, common disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

18,130-18,160 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, common disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

18,160-18,190 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, common disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

18,190-18,220 SILTY SANDSTONE: light gray, medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain

18,220-18,250 SILTY SANDSTONE: light gray, medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain

18,250-18,280 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, occasional spotty light brown oil stain

18,280-18,310 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

18,310-18,340 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

18,340-18,370 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

18,370-18,400 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

18,400-18,430 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

18,430-18,460 SILTY SANDSTONE: light gray, occasional medium gray, very fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

18,460-18,490 SILTY SANDSTONE: light gray, occasional medium gray, very fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

18,490-18,520 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

18,520-18,550 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

18,550-18,580 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

18,580-18,610 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

18,610-18,640 SILTY SANDSTONE: light gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain

18,640-18,670 SILTY SANDSTONE: light gray, common medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain,

18,670-18,700 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain,

18,700-18,730 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, rare spotty light brown oil stain,

18,730-18,760 SILTY SANDSTONE: medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

18,760-18,790 SILTY SANDSTONE: medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

18,790-18,820 SILTY SANDSTONE: medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

18,820-18,850 SILTY SANDSTONE: light to medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

18,850-18,880 SILTY SANDSTONE: light to medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

18,880-18,910 SILTY SANDSTONE: light to medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

18,910-18,940 SILTY SANDSTONE: light gray, medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, sample contaminated with lube

18,940-18,970 SILTY SANDSTONE: light gray, medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, sample contaminated with lube

18,970-19,000 SILTY SANDSTONE: light gray, medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,000-19,030 SILTY SANDSTONE: light gray, medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,030-19,060 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,060-19,090 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,090-19,120 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,120-19,150 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,150-19,180 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,180-19,210 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,210-19,240 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,240-19,270 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,270-19,300 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,300-19,330 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,330-19,360 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,360-19,390 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,390-19,420 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,420-19,450 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,450-19,480 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,480-19,510 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,510-19,540 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,540-19,570 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,570-19,600 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,600-19,630 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,630-19,660 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,660-19,690 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,690-19,720 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,720-19,750 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,750-19,780 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,780-19,810 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,810-19,840 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,840-19,870 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,870-19,900 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,900-19,930 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,930-19,960 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,960-19,990 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

19,990-20,020 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

20,020-20,050 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

20,050-20,080 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

20,080-20,110 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

20,110-20,140 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

20,140-20,170 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

20,170-20,200 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

20,200-20,230 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

20,230-20,260 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

20,260-20,290 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

20,290-20,320 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

20,320-20,350 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

20,350-20,380 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

20,380-20,410 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

20,410-20,440 SILTY SANDSTONE: light gray, occasional medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, rare disseminated pyrite, trace intergranular porosity, sample contaminated with lube

20,440-20,470 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, sample contaminated with lube

20,470-20,500 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, sample contaminated with lube

20,500-20,530 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, sample contaminated with lube

20,530-20,560 SILTY SANDSTONE: light gray, rare medium gray, fine grained, sub-rounded, firm, well sorted, calcite cemented, poorly cemented, occasional disseminated pyrite, trace intergranular porosity, sample contaminated with lube

SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

AUG 2014 RECEIVED ND OIL & GAS DIVISION

Well File No. **28978**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☑ Notice of Intent	Approximate Start Date **October 16, 2014**	☐ Drilling Prognosis	☐ Spill Report
☐ Report of Work Done	Date Work Completed	☐ Redrilling or Repair	☐ Shooting
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date	☐ Casing or Liner	☐ Acidizing
		☐ Plug Well	☐ Fracture Treatment
		☐ Supplemental History	☐ Change Production Method
		☐ Temporarily Abandon	☑ Reclamation
		☐ Other **offsite pit**	

Well Name and Number				
Wade Federal 5300 21-30 13B				
Footages	Qtr-Qtr	Section	Township	Range
1574 F **N** L **270** F **W** L	**LOT2**	**30**	**153 N**	**100 W**
Field	Pool	County		
Baker	**Bakken**	**McKenzie**		

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum North America LLC respectfully requests to use an offsite pit for this well. The following wells will also use this pit:

Wade Federal 5300 31-30 2B
Wade Federal 5300 41-30 3T2
Wade Federal 5300 41-30 4T
Wade Federal 5300 41-30 5T2
Wade Federal 5300 41-30 6B
Wade Federal 5300 41-30 7T
Wade Federal 5300 41-30 8T2
Wade Federal 5300 41-30 9B
Wade Federal 5300 21-30 12T
Wade Federal 5300 21-30 14T2

Attached are the plats for the offsite pit location.

Company		Telephone Number
Oasis Petroleum North America LLC		**281-404-9589**
Address		
1001 Fannin, Suite 1500		
City	State	Zip Code
Houston	**TX**	**77002**
Signature Sonja Rolfes	Printed Name **Sonja Rolfes**	
Title **Regulatory Analyst**	Date **August 20, 2014**	
Email Address **srolfes@oasispetroleum.com**		

FOR STATE USE ONLY

☐ Received	☒ Approved
Date 10-1-14	
By	
Title	

OFF-SITE PIT AGREEMENT

In consideration of the sum of [redacted] paid by Oasis Petroleum North America LLC ("Oasis") the undersigned surface owners, Wesley Lindvig and Barbara Lindvig, for themselves and their heirs, successors, administrators and assigns, hereby acknowledge the receipt and sufficiency of said payment in full and complete settlement for and as a release of all claim for loss, damage or injury to the hereafter described surface property arising out of the off-site cuttings pit, in which the cuttings from the **Wade Federal 5300 21-30 13B, Wade Federal 5300 21-30 14T2** wells will be buried, located on the approximately two (2.0) acre tract of land identified on the plat attached hereto as Exhibit "A" and which is situated on the following described real property located in McKenzie County, State of North Dakota, towit:

Township 153 North, Range 100 West, 5th P.M.
Section 30: Lots 3 & 4 a/k/a W½SW¼

The undersigned knows that Oasis Petroleum North America LLC is the operator and will be drilling the **Wade Federal 5300 21-30 13B, Wade Federal 5300 21-30 14T2** wells. The undersigned further states that they are fully aware that the cuttings generated from the drilling of the **Wade Federal 5300 21-30 13B, Wade Federal 5300 21-30 14T2** wells will be buried in the pit on the above described location.

Dated this 19 day of May, 2014.

SURFACE OWNER(S)

Wesley Lindvig

Wesley Lindvig

Barbara J. Lindvig

Barbara Lindvig

BJL W.G.L.

Location will be fenced after Construction.

Pit will be reclaimed to Owners Satisfaction

BJL W.G.L.

ACKNOWLEDGMENT INDIVIDUAL

State of North Dakota)
)
County of McKenzie)

BE IT REMEMBERED, That on this 19 day of May, 2014 before me, a Notary Public, in and for said County and State, personally appeared Wesley Lindvig and Barbara Lindvig, to me known to be the identical persons described in and who executed the within and foregoing instrument and acknowledged to me to that they executed the same as their free and voluntary act and deed for the uses and purposes therein set forth.

IN WITNESS WHEREOF, I have hereunto set my official signature and affixed my notarial seal, the day and year last above written.

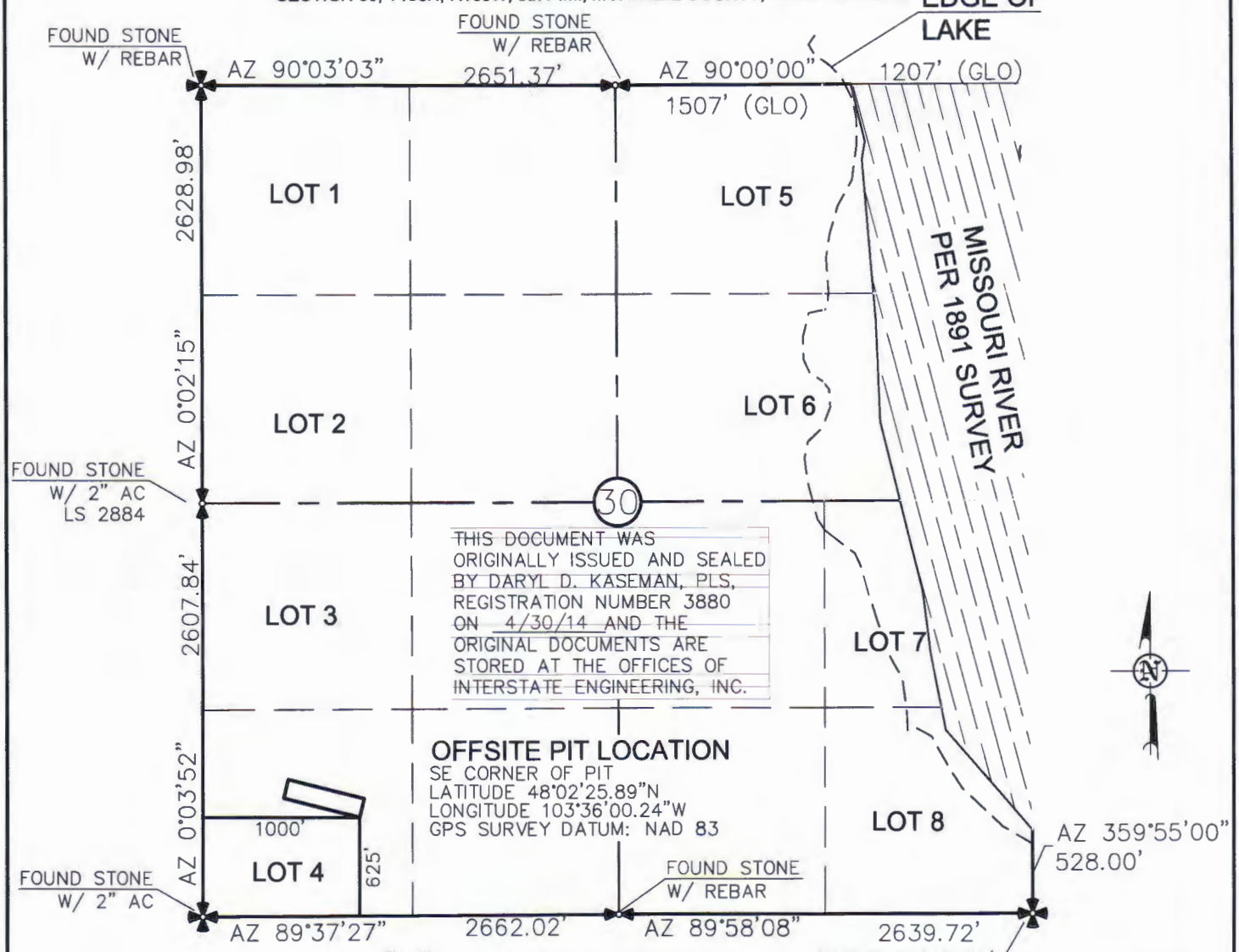
My Commission expires:

JAMES DEMORRETT
Notary Public
State of North Dakota
My Commission Expires Aug. 16, 2018

Notary Public

OFFSITE PIT LOCATION PLAT

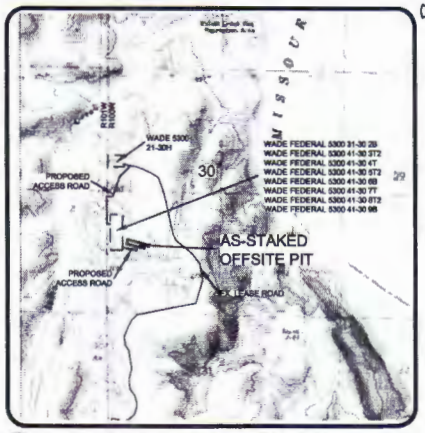
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"AS-STAKED OFFSITE PIT FOR WADE FEDERAL 5300 31-30 2B, WADE FEDERAL 5300 41-30 3T2, WADE FEDERAL 5300 41-30 4T, WADE FEDERAL 5300 41-30 5T2, WADE FEDERAL 5300 41-30 6B, WADE FEDERAL 5300 41-30 7T, WADE FEDERAL 5300 41-30 8T2, & WADE FEDERAL 5300 41-30 9B"
625 FEET FROM SOUTH LINE AND 1000 FEET FROM WEST LINE
SECTION 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



MONUMENT – RECOVERED
MONUMENT – NOT RECOVERED

0 1000
1" = 1000'

VICINITY MAP



STAKED ON 4/30/14
VERTICAL CONTROL DATUM WAS BASED UPON CONTROL POINT 705 WITH AN ELEVATION OF 2158.3'

THIS SURVEY AND PLAT IS BEING PROVIDED AT THE REQUEST OF ERIC BAYES OF OASIS PETROLEUM. I CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS WORK PERFORMED BY ME OR UNDER MY SUPERVISION AND IS TRUE AND CORRECT TO THE BEST OF MY KNOWLEDGE AND BELIEF.

DARYL D. KASEMAN LS-3880

REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE: 4/30/14
NORTH DAKOTA

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1/3
SHEET NO.

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Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com
Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
OFFSITE PIT LOCATION PLAT
SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By:	B.H.H.	Project No.:	S13-09-381.09
Checked By:	D.D.K.	Date:	APRIL 2014

Revision No.	Date	By	Description

PAD LAYOUT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"AS-STAKED OFFSITE PIT FOR WADE FEDERAL 5300 31-30 2B, WADE FEDERAL 5300 41-30 3T2, WADE FEDERAL 5300 41-30 4T, WADE FEDERAL 5300 41-30 5T2, WADE FEDERAL 5300 41-30 6B, WADE FEDERAL 5300 41-30 7T, WADE FEDERAL 5300 41-30 8T2, & WADE FEDERAL 5300 41-30 9B"
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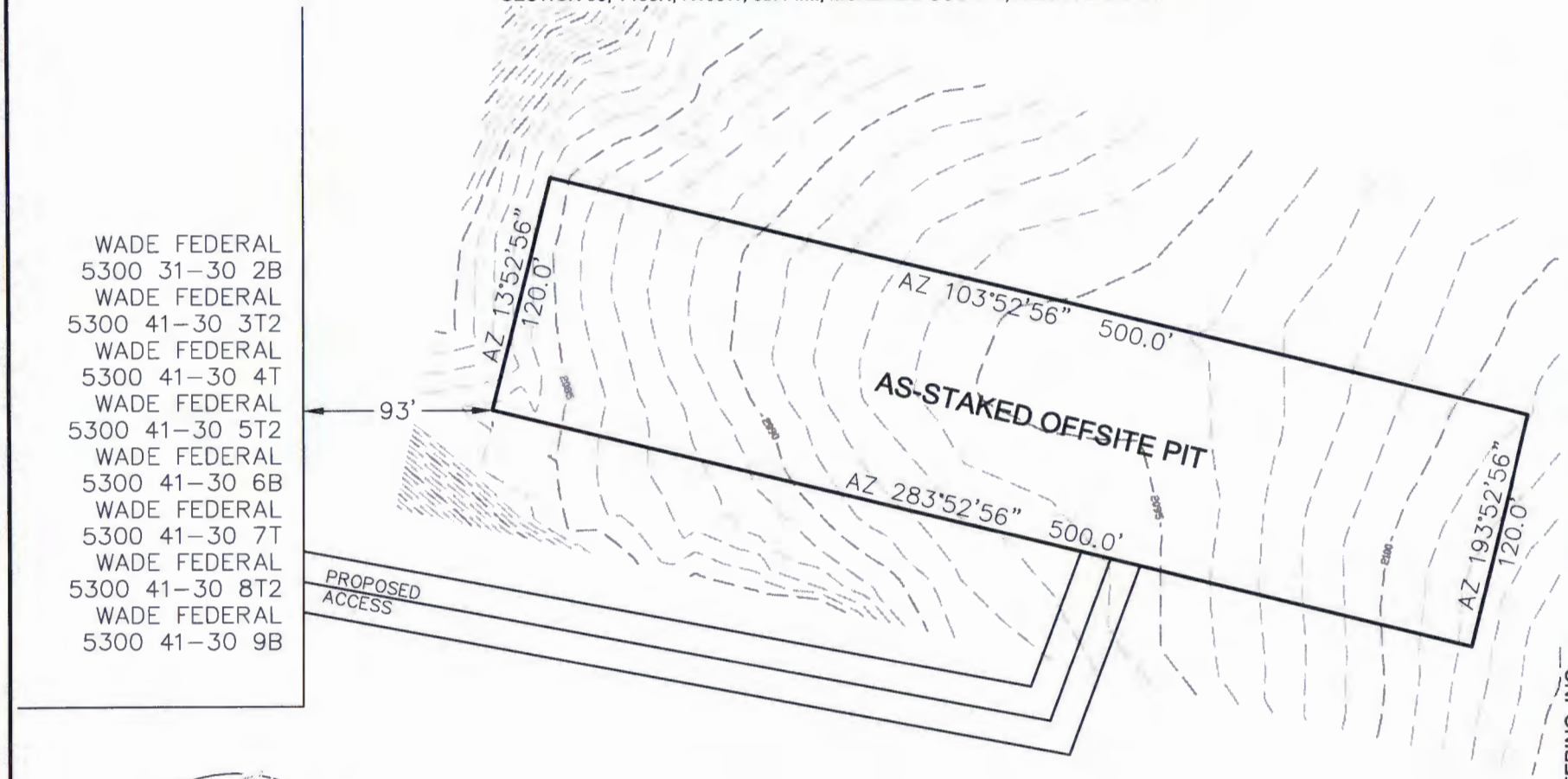

REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE: 4/30/14
NORTH DAKOTA

THIS DOCUMENT WAS ORIGINALLY ISSUED AND SEALED BY DARYL D. KASEMAN, PLS, REGISTRATION NUMBER 3880 ON 4/30/14 AND THE ORIGINAL DOCUMENTS ARE STORED AT THE OFFICES OF INTERSTATE ENGINEERING, INC.

NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

0 80
1" = 80'

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Revision No.	Date	By	Description

OASIS PETROLEUM NORTH AMERICA, LLC
PAD LAYOUT
SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S13-09-381.09
Checked By: D.D.K. Date: APRIL 2014

Interstate Engineering, Inc.
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425 East Main Street
Sidney, Montana 59270
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Fax (406) 433-5618
www.interstateeng.com
Other offices in Minnesota, North Dakota and South Dakota

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ACCESS APPROACH

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"AS-STAKED OFFSITE PIT FOR WADE FEDERAL 5300 31-30 2B, WADE FEDERAL 5300 41-30 3T2, WADE FEDERAL 5300 41-30 4T, WADE FEDERAL 5300 41-30 5T2, WADE FEDERAL 5300 41-30 6B, WADE FEDERAL 5300 41-30 7T, WADE FEDERAL 5300 41-30 8T2, & WADE FEDERAL 5300 41-30 9B"
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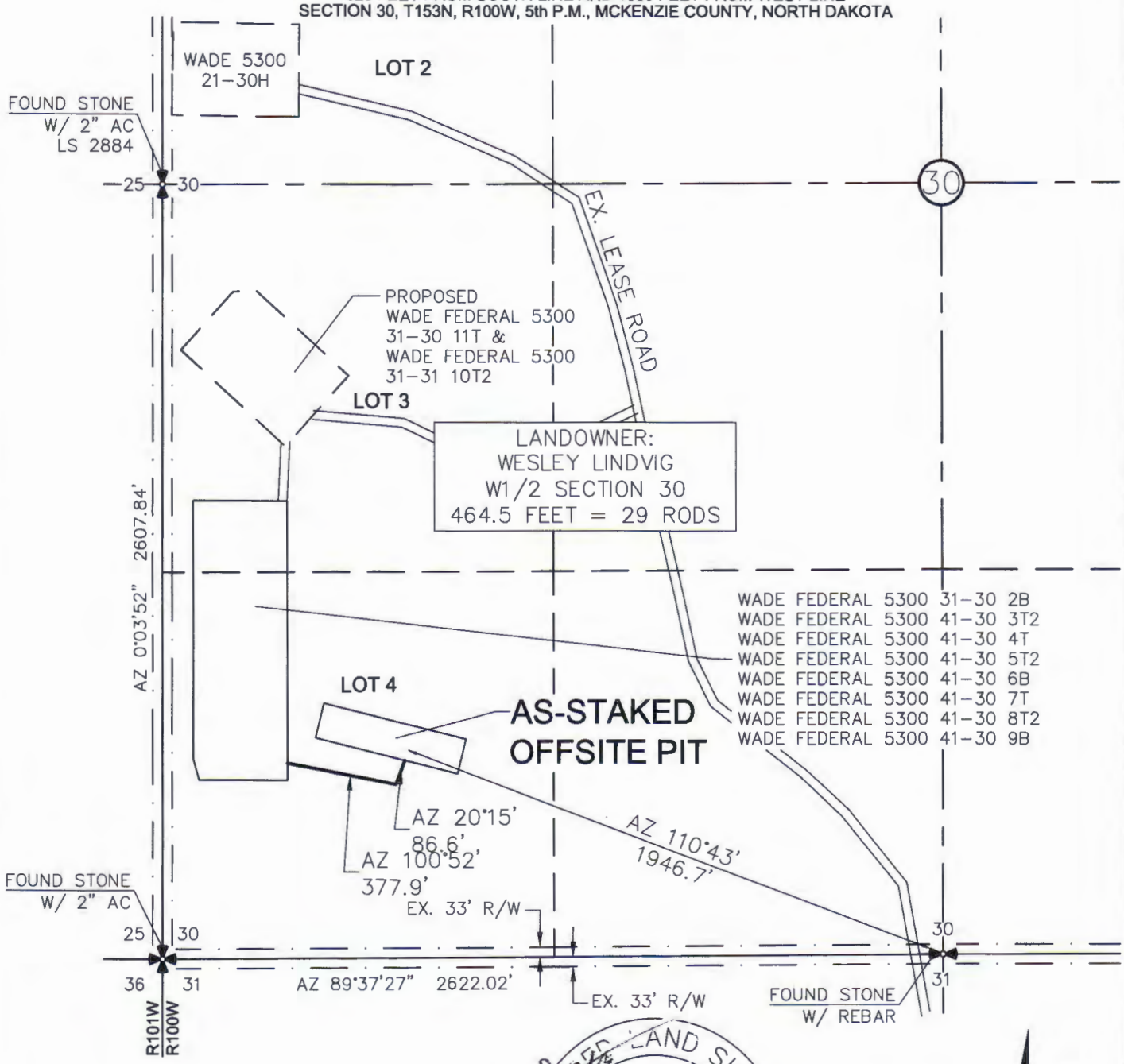
SECTION 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE: 4/30/14
NORTH DAKOTA

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OASIS PETROLEUM NORTH AMERICA, LLC
ACCESS APPROACH
SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S13-09-381.09
Checked By: D.D.K. Date: APRIL 2014

Revision No.	Date	By	Description

SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

SEP 2014 RECEIVED ND OIL & GAS DIVISION

Well File No. **28978**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

- [x] Notice of Intent — Approximate Start Date: **September 14, 2014**
- [ ] Report of Work Done — Date Work Completed:
- [ ] Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. — Approximate Start Date:

- [ ] Drilling Prognosis
- [ ] Redrilling or Repair
- [x] Casing or Liner
- [ ] Plug Well
- [ ] Supplemental History
- [ ] Temporarily Abandon
- [ ] Other
- [ ] Spill Report
- [ ] Shooting
- [ ] Acidizing
- [ ] Fracture Treatment
- [ ] Change Production Method
- [ ] Reclamation

Well Name and Number					
Wade Federal 5300 21-30 13B					
Footages	Qtr-Qtr	Section	Township	Range	
1574 F N L 270 F W L	**LOT2**	**30**	**153 N**	**100 W**	
Field	Pool	County			
Baker	**Bakken**	**McKenzie**			

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum North America LLC requests permission to make the following changes to the above referenced well:

Surface casing changed to 13 3/8"
Contingency 9 5/8" casing added
7" casing changed to all 32#

Attached are revised drill plan, plot and well summary.

Company		Telephone Number
Oasis Petroleum North America LLC		**281-404-9563**
Address		
1001 Fannin, Suite 1500		
City	State	Zip Code
Houston	**TX**	**77002**
Signature: Sonja Rolfes	Printed Name: **Sonja Rolfes**	
Title: **Regulatory Specialist**	Date: **September 4, 2014**	
Email Address: **srolfes@oasispetroleum.com**		

FOR STATE USE ONLY

- [ ] Received
- [x] Approved

Date: 9-9-14

By: Nathaniel Erbele

Title: Petroleum Resource Specialist

DRILLING PLAN

OPERATOR	Oasis Petroleum	COUNTY/STATE	McKenzie Co., ND
WELL NAME	Wade Federal 5300 21-30 13B	RIG	B25
WELL TYPE	Horizontal Middle Bakken		
LOCATION	SW NW 30-153N-100W	Surface Location (survey plat): 1574' FNL	270' FWL
EST. T.D.	20,674'	GROUND ELEV: 1,999'	Sub Height: 25'
	TOTAL LATERAL: 9,751'	KB ELEV: 2,024'	

MARKER		TVD	Subsea TVD
Pierre	NDIC MAP	1,874	150
Greenhorn		4,521	-2,497
Mowry		4,923	-2,899
Dakota		5,347	-3,323
Rierdon		6,360	-4,336
Dunham Salt		6,851	-4,827
Dunham Salt Base		6,901	-4,877
Pine Salt		7,161	-5,137
Pine Salt Base		7,179	-5,155
Opeche Salt		7,249	-5,225
Opeche Salt Base		7,328	-5,304
Amsden		7,569	-5,545
Tyler		7,730	-5,706
Otter/Base Minnelusa		7,957	-5,933
Kibbey Lime		8,299	-6,275
Charles Salt		8,441	-6,417
Base Last Salt		9,115	-7,091
Mission Canyon		9,331	-7,307
Lodgepole		9,881	-7,857
False Bakken		10,612	-8,588
Upper Bakken Shale		10,624	-8,600
Middle Bakken (Top of Target)		10,649	-8,625
Middle Bakken (Base of target)		10,660	-8,636
Lower Bakken Shale		10,674	-8,650
Threeforks		10,698	-8,674

	Type	Interval
LOGS:	OH Logs: Request Log waiver based on the Wade Federal 5300 21-30H 600' S of surface location	
	CBL/GR: Above top of cement/GR to base of casing	
	MWD GR: KOP to lateral TD	
DEVIATION:	Surf:	3 deg. max., 1 deg / 100'; srvy every 500'
	Prod:	5 deg. max., 1 deg / 100'; srvy every 100'
CST'S:	None planned	
CORES:	None planned	
MUDLOGGING:	Two-Man:	Begin 200' above Kibbey 30' samples in curve and lateral
BOP:	11" 5000 psi blind, pipe & annular	

Est. Dip Rate: [illegible]

Max. Anticipated BHP: 4825

Surface Formation: *Glacial till*

MUD:	Interval		Type	WT	Vis	WL	Remarks
Surface:	0' -	1,980'	FW/Gel - Lime Sweeps	8.4-9.0	28-32	NC	Circ Mud Tanks
Intermediate:	1,980' -	10,923'	Invert	9.5-10.4	40-50	30+HtHp	Circ Mud Tanks
Lateral:	10,923' -	20,674'	Salt Water	9.8-10.2	28-32	NC	Circ Mud Tanks

CASING:	Size	Wt ppf	Hole	Depth		Cement	WOC	Remarks
Surface:	13-3/8"	54.5#	17-1/2"	1,980'		To Surface	12	100' into Pierre
Inermediate (Dakota):	9-5/8"	40#	12-1/4"	6,400'		To Surface	24	Set Casing across Dakota
Intermediate:	7"	32#	8-3/4"	10,923'		4847	24	500' above Dakota
Production Liner:	4.5"	13.5#	6"	20,674'	TOL @ 10,128'			50' above KOP

PROBABLE PLUGS, IF REQ'D:

OTHER:

	MD	TVD	FNL/FSL	FEL/FWL	S-T-R	AZI	Survey Company:	
Surface:	1,980	1,980	1574' FNL	270' FWL	T150N R100W Sec. 30			
KOP:	10,178'	10,178'	1524' FNL	270' FWL	T150N R100W Sec. 30	0.0	Build Rate:	12 deg /100'
EOC:	10,923'	10,655'	1258' FNL	661' FWL	T150N R100W Sec. 30	55.9		
Casing Point:	10,923'	10,655'	1258' FNL	661' FWL	T150N R100W Sec. 30	55.9		
Middle Bakken Lateral TD:	20,674'	10,749'	785' FNL	200' FEL	T150N R100W Sec. 29	90.0		

Comments:

Request Log waiver based on the Wade Federal 5300 21-30H 600' S of surface location

No frac string planned

35 packers and 20 sleeves planned 3.6MM lbs 30% ceramic

Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.

68334-30-5 (Primary Name: Fuels, diesel) 68476-34-6 (Primary Name: Fuels, diesel, No. 2) 68476-30-2 (Primary Name: Fuel oil No. 2)

68476-31-3 (Primary Name: Fuel oil, No. 4) 8008-20-6 (Primary Name: Kerosene)

OASIS PETROLEUM

Geology: N. Gabelman 3/20/2014 Engineering: e m p 4.8.14

Azimuths to True North
Magnetic North: 8.15°

Magnetic Field
Strength: 56473.1snT
Dip Angle: 72.95°
Date: 4/4/2014
Model: IGRF200510

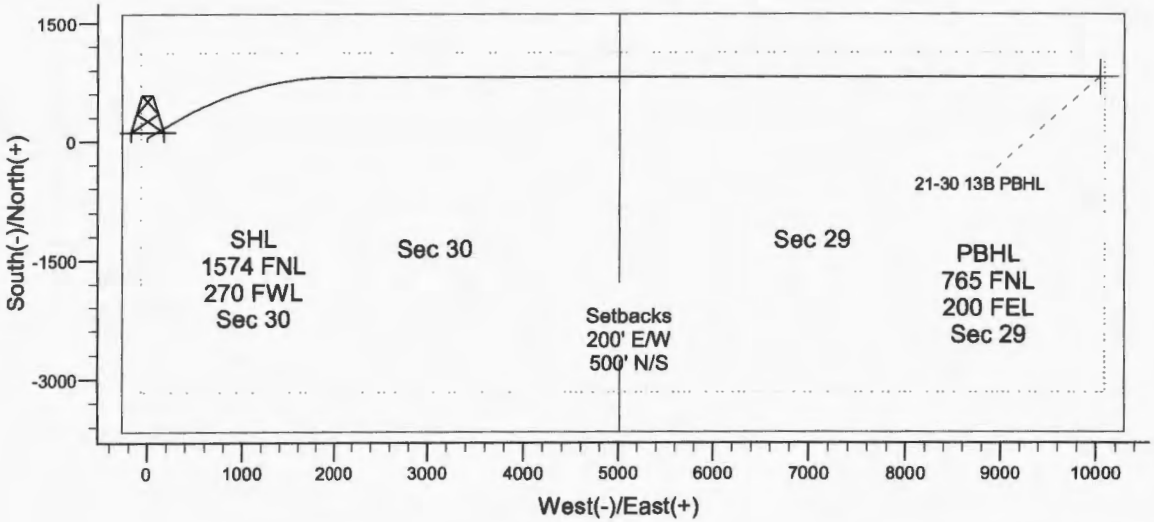
OASIS PETROLEUM

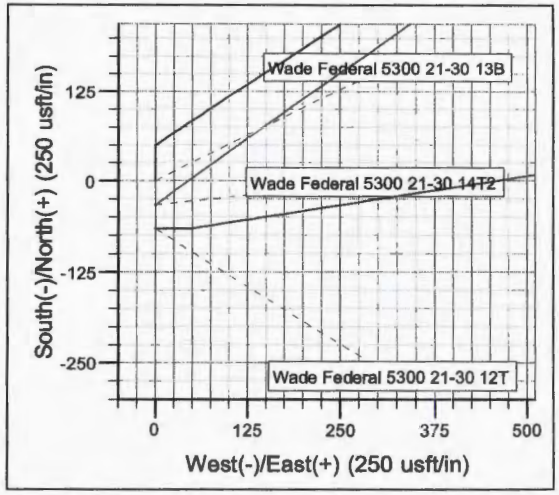
Project: Indian Hills
Site: 153N-100W-29/30
Well: Wade Federal 5300 21-30 13B
Wellbore: Wade Federal 13B
Design: Dising#2

WELL DETAILS: Wade Federal 5300 21-30 13B

Ground Level: 1999.0

Northing	Easting	Latitude	Longitude
397868.63	1209741.76	48° 2' 55.790 N	103° 36' 10.970 W



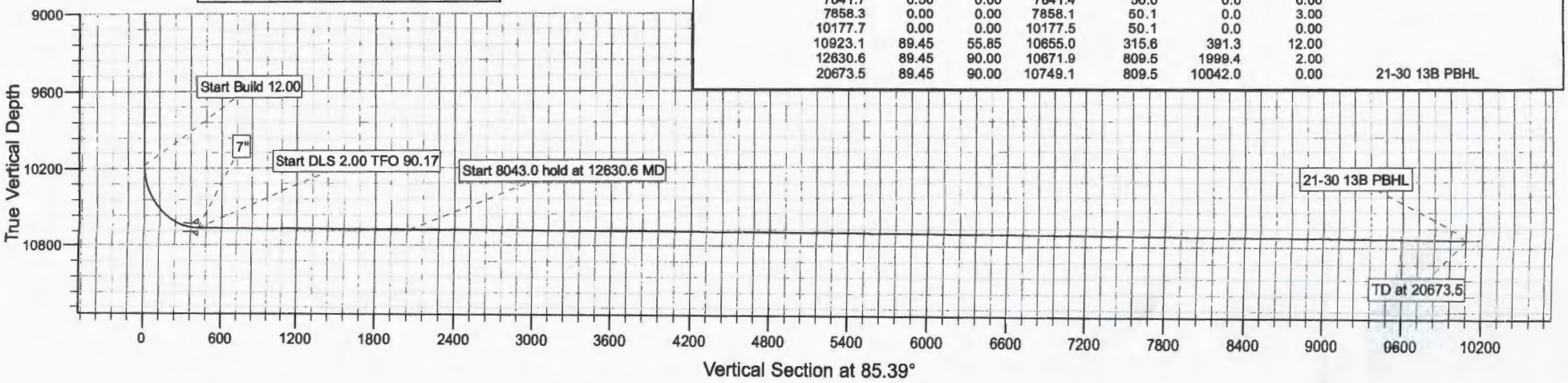


CASING DETAILS

TVD	MD	Name	Size
1980.0	1980.0	13 1/8"	13 3/8
10655.0	10924.2	7"	7

SECTION DETAILS

MD	Inc	Azi	TVD	+N/-S	+E/-W	Dleg	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	
2100.0	0.00	0.00	2100.0	0.0	0.0	0.00	
2116.7	0.50	0.00	2116.7	0.1	0.0	3.00	
7841.7	0.50	0.00	7841.4	50.0	0.0	0.00	
7858.3	0.00	0.00	7858.1	50.1	0.0	3.00	
10177.7	0.00	0.00	10177.5	50.1	0.0	0.00	
10923.1	89.45	55.85	10655.0	315.6	391.3	12.00	
12630.6	89.45	90.00	10671.9	809.5	1999.4	2.00	
20673.5	89.45	90.00	10749.1	809.5	10042.0	0.00	21-30 13B PBHL



Oasis Petroleum
Well Summary
Wade Federal 5300 21-30 13B
Section 30 T153N R100W
McKenzie County, ND

SURFACE CASING AND CEMENT DESIGN

							Make-up Torque (ft-lbs)		
Size	**Interval**	**Weight**	**Grade**	**Coupling**	**I.D.**	**Drift**	**Minimum**	**Optimum**	**Max**
13-3/8"	0' to 1,980'	54.5	J-55	STC	12.615"	12.459"	4,100	5,470	6,840

Interval	Description	Collapse	Burst	Tension
		(psi) a	(psi) b	(1000 lbs) c
0' to 1,980'	13-3/8", 54.5#, J-55, STC, 8rd	1130 / 1.22	2730 / 2.95	514 / 2.66

API Rating & Safety Factor

a) Based on full casing evacuation with 9 ppg fluid on backside (1,980' setting depth).

b) Burst pressure based on 9 ppg fluid with no fluid on backside (1,980' setting depth).

c) Based on string weight in 9 ppg fluid at 1,980' TVD plus 100k# overpull. (Buoyed weight equals 93k lbs.)

Cement volumes are based on 13-3/8" casing set in 17-1/2" hole with 50% excess to circulate cement back to surface. Mix and pump the following slurry.

Pre-flush (Spacer): **20 bbls** fresh water

Lead Slurry: **569 sks** (294 bbls) 2.9 yield conventional system with 94 lb/sk cement, .25 lb/sk D130 Lost Circulation Control Agent, 2% $CaCl_2$, 4% D079 Extender and 2% D053 Expanding Agent.

Tail Slurry: **349 sks** (72 bbls) 1.16 yield conventional system with 94 lb/sk cement, .25% $CaCl_2$ and 0.25 lb/sk Lost Circulation Control Agent

Oasis Petroleum
Well Summary
Wade Federal 5300 21-30 13B
Section 30 T153N R100W
McKenzie County, ND

CONTINGENCY SURFACE CASING AND CEMENT DESIGN

							Make-up Torque (ft-lbs)		
Size	**Interval**	**Weight**	**Grade**	**Coupling**	**I.D.**	**Drift**	**Minimum**	**Optimum**	**Max**
9-5/8"	0' to 6,400'	40	L-80	LTC	8.835"	8.75"	5,450	7,270	9,090

Interval	Description	Collapse	Burst	Tension
		(psi) a	(psi) b	(1000 lbs) c
0' to 6,400'	9-5/8", 40#, L-80, LTC, 8rd	3090 / 3.71	5750 / 1.24	837 / 3.86

API Rating & Safety Factor

a) Collapse pressure based on 11.5 ppg fluid on the backside and 9 ppg fluid inside of casing.

b) Burst pressure calculated from a gas kick coming from the production zone (Bakken Pool) at 9,000 psi and a subsequent breakdown at the 9-5/8" shoe, based on a 13.5#/ft facture gradient. Backup of 9 ppg fluid.

c) Yield based on string weight in 10 ppg fluid, (217k lbs buoyed weight) plus 100k lbs overpull.

Cement volumes are based on 9-5/8" casing set in 12-1/4" hole with 10% excess in OH and 0% excess inside surface casing. TOC at surface.

Pre-flush (Spacer): **20 bbls** Chem wash

Lead Slurry: **592 sks** (210 bbls) Conventional system with 75 lb/sk cement, 0.5 lb/sk lost circulation, 10% expanding agent, 2% extender, 2% $CaCl_2$, 0.2% anti-foam and 0.4% fluid loss agent.

Tail Slurry: **521 sks** (108 bbls) Conventional system with 94 lb/sk cement, 0.3% anti-settling agent, 0.3% fluid loss agent, 0.3 lb/sk lost circulation control agent, 0.2% anti-foam and 0.1% retarder.

Oasis Petroleum
Well Summary
Wade Federal 5300 21-30 13B
Section 30 T153N R100W
McKenzie County, ND

INTERMEDIATE CASING AND CEMENT DESIGN

							Make-up Torque (ft-lbs)		
Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Minimum	Optimum	Max
7"	0' – 10,928'	32	HCP-110	LTC	6.094"	6.000"	6730	8970	9870

***Special drift

Interval	Length	Description	Collapse	Burst	Tension
			(psi) a	(psi) b	(1000 lbs) c
0' – 10,923'	10,923'	7", 32#, HCP-110, LTC, 8rd	11820 / 2.13*	12460 / 1.29	897 / 2.27
6,701' – 10,178'	3,477'	7", 32#, HCP-110, LTC, 8rd	11820 / 1.38*	12460 / 1.29	

API Rating & Safety Factor

a) *Assume full casing evacuation with 10 ppg fluid on backside. **Assume full casing evacuation with 1.2 psi/ft equivalent fluid gradient across salt intervals.

b) Burst pressure based on 9000 psig max press for stimulation plus 10.2 ppg fluid in casing and 9 ppg fluid on backside-to 10,655' TVD.

c) Based on string weight in 10 ppg fluid, (296k lbs buoyed weight) plus 100k

Cement volumes are estimates based on 7" casing set in an 8-3/4" hole with 30% excess.

Pre-flush (Spacer): **50 bbls Saltwater**
40 bbls Weighted MudPush Express

Lead Slurry: **229 sks** (92 bbls) 2.21 yield conventional system with 47 lb/sk cement, 37 lb/sk D035 extender, 3.0% KCl, 3.0% D154 extender, 0.3% D208 viscosifier, 0.07% retarder, 0.2% anti-foam, 0.5 lb/sk, D130 LCM.

Tail Slurry: **581 sks** (160 bbls) 1.54 yield conventional system with 94 lb/sk cement, 3.0% KCl, 35.0% Silica, 0.5% retarder, 0.2% fluid loss, 0.2% anti-foam and 0.5 lb/sk LCM.

Oasis Petroleum
Well Summary
Wade Federal 5300 21-30 13B
Section 30 T153N R100W
McKenzie County, ND

PRODUCTION LINER

Size	**Interval**	**Weight**	**Grade**	**Coupling**	**I.D.**	**Drift**	**Estimated Torque**
4-1/2"	10,128' – 20,674'	13.5	P-110	BTC	3.92"	3.795"	4,500

Interval	**Description**	**Collapse**	**Burst**	**Tension**
		(psi) **a**	(psi) **b**	(1000 lbs) **c**
10,128' – 20,674'	4-1/2", 13.5 lb, P-110, BTC, 8rd	10670 / 1.99	12410 / 1.28	443 / 2.02

API Rating & Safety Factor

a) Based on full casing evacuation with 9.5 ppg fluid on backside @ 10,749' TVD.

b) Burst pressure based on 9000 psi treating pressure with 10.2 ppg internal fluid gradient and 9 ppg external fluid gradient @ 10,749' TVD.

c) Based on string weight in 9.5 ppg fluid (Buoyed weight: 105k lbs.) plus 100k lbs overpull.

Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.

68334-30-5 (Primary Name: Fuels, diesel)
68476-34-6 (Primary Name: Fuels, diesel, No. 2)
68476-30-2 (Primary Name: Fuel oil No. 2)
68476-31-3 (Primary Name: Fuel oil, No. 4)
8008-20-6 (Primary Name: Kerosene)

NORTH DAKOTA
OIL AND GAS DIVISION

Oil and Gas Division

Lynn D. Helms - Director Bruce E. Hicks - Assistant Director
Department of Mineral Resources
Lynn D. Helms - Director
North Dakota Industrial Commission
www.dmr.nd.gov/oilgas

BRANDI TERRY
OASIS PETROLEUM NORTH AMERICA LLC
1001 FANNIN STE 1500
HOUSTON, TX 77002 USA

Date: **7/28/2014**

RE: CORES AND SAMPLES

Well Name: **WADE FEDERAL 5300 21-30 13B** Well File No.: **28978**
Location: **LOT2 30-153-100** County: MCKENZIE
Permit Type: **Development - HORIZONTAL**
Field: **BAKER** Target Horizon: BAKKEN

Dear BRANDI TERRY:

North Dakota Century Code Section 38-08-04 provides for the preservation of cores and samples and their shipment to the State Geologist when requested. The following is required on the above referenced well:

1) All cores, core chips and samples must be submitted to the State Geologist as provided for under North Dakota Century Code: Section 38-08-04 and North Dakota Administrative Code: Section 43-02-03-38.1.

2) Samples: The Operator is to begin collecting sample drill cuttings no lower than the:
Base of the Last Charles Salt
   - Sample cuttings shall be collected at:
     - 30' maximum intervals through all vertical and build sections.
     - 100' maximum intervals through any horizontal sections.
   - Samples must be washed, dried, placed in standard sample envelopes (3" x 4.5"), packed in the correct order into standard sample boxes (3.5" x 5.25" x 15.25").
   - Samples boxes are to be carefully identified with a label that indicates the operator, well name, well file number, American Petroleum Institute (API) number, location and depth of samples; and forwarded in to the state core and sample library within 30 days of the completion of drilling operations.

3) Cores: Any cores cut shall be preserved in correct order, boxed in standard core boxes (4.5", 4.5", 35.75"), and the entire core forwarded to the state core and samples library within 180 days of completion of drilling operations. Any extension of time must have approval on a Form 4 Sundry Notice.

All cores, core chips, and samples must be shipped, prepaid, to the state core and samples library at the following address:

ND Geological Survey Core Library
2835 Campus Road, Stop 8156
Grand Forks, ND 58202

North Dakota Century Code Section 38-08-16 allows for a civil penalty for any violation of Chapter 38 08 not to exceed $12,500 for each offense, and each day's violation is a separate offense.

Sincerely

Stephen Fried
Geologist

SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

JUN 2014 RECEIVED ND OIL & GAS DIVISION

Well File No. 28978

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☑ Notice of Intent	Approximate Start Date **May 1, 2014**	☐ Drilling Prognosis	☐ Spill Report
☐ Report of Work Done	Date Work Completed	☐ Redrilling or Repair	☐ Shooting
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date	☐ Casing or Liner	☐ Acidizing
		☐ Plug Well	☐ Fracture Treatment
		☐ Supplemental History	☐ Change Production Method
		☐ Temporarily Abandon	☐ Reclamation
		☑ Other **Waiver to rule Rule 43-02-03-31**	

Well Name and Number					
Wade Federal 5300 21-30 13B					
Footages	Qtr-Qtr	Section	Township	Range	
1574 F N L 270 F W L	**LOT2**	**30**	**153 N**	**100 W**	
Field	Pool	County			
	Bakken	**McKenzie**			

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum respectfully requests a waiver to Rule 43-02-03-31 in regards to running open hole logs for the above referenced well. Justification for this request is as follows:

The Oasis Petroleum/ Wade Federal 5300 21-30H #20197 located within a mile of subject location

If this exception is approved, Oasis Petroleum will run a CBL on the intermediate string, and we will also run GR to surface. Oasis Petroleum will also submit two digital copies of each cased hole log and a copy of the mud log containing MWD gamma ray.

Company **Oasis Petroleum North America LLC**		Telephone Number **281-404-9491**
Address **1001 Fannin, Suite 1500**		
City **Houston**	State **TX**	Zip Code **77002**
Signature Brandi Terry	Printed Name **Brandi Terry**	
Title **Regulatory Specialist**	Date **April 14, 2014**	
Email Address **bterry@oasispetroleum.com**		

FOR STATE USE ONLY

☐ Received	☒ Approved
Date 7-25-2014	
By	
Title Stephen Fried Geologist	

Oil and Gas Division

Lynn D. Helms - Director Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.oilgas.nd.gov

July 25, 2014

Brandi Terry
Regulatory Specialist
OASIS PETROLEUM NORTH AMERICA LLC
1001 Fannin Suite 1500
Houston, TX 77002

RE: HORIZONTAL WELL
WADE FEDERAL 5300 21-30 13B
LOT2 Section 30-153N-100W
McKenzie County
Well File # 28978

Dear Brandi:

Pursuant to Commission Order No. 23752, approval to drill the above captioned well is hereby given. The approval is granted on the condition that all portions of the well bore not isolated by cement, be no closer than the **500'** setback from the north & south boundaries and **200'** setback from the east & west boundaries within the 2560 acre spacing unit consisting of Sections 29 & 30 T153N R100W. **Tool error is not required pursuant to order.**

PERMIT STIPULATIONS: Effective June 1, 2014, a covered leak-proof container (with placard) for filter sock disposal must be maintained on the well site beginning when the well is spud, and must remain on-site during clean-out, completion, and flow-back whenever filtration operations are conducted. Due to drainage adjacent to the well site, a dike is required surrounding the entire location. OASIS PETRO NO AMER must contact NDIC Field Inspector Richard Dunn at 701-770-3554 prior to location construction.

Drilling pit
NDAC 43-02-03-19.4 states that "a pit may be utilized to bury drill cuttings and solids generated during well drilling and completion operations, providing the pit can be constructed, used and reclaimed in a manner that will prevent pollution of the land surface and freshwaters. Reserve and circulation of mud system through earthen pits are prohibited. All pits shall be inspected by an authorized representative of the director prior to lining and use. Drill cuttings and solids must be stabilized in a manner approved by the director prior to placement in a cuttings pit."

Form 1 Changes & Hard Lines
Any changes, shortening of casing point or lengthening at Total Depth must have prior approval by the NDIC. The proposed directional plan is at a legal location. Based on the azimuth of the proposed lateral the maximum legal coordinate from the well head is: 1074'N and 10043'E.

Location Construction Commencement (Three Day Waiting Period)
Operators shall not commence operations on a drill site until the 3rd business day following publication of the approved drilling permit on the NDIC - OGD Daily Activity Report. If circumstances require operations to commence before the 3rd business day following publication on the Daily Activity Report, the waiting period may be waived by the Director. Application for a waiver must be by sworn affidavit providing the information necessary to evaluate the extenuating circumstances, the factors of NDAC 43-02-03-16.2 (1), (a)-(f), and any other information that would allow the Director to conclude that in the event another owner seeks revocation of the drilling permit, the applicant should retain the permit.

Permit Fee & Notification
Payment was received in the amount of $100 via credit card .The permit fee has been received. It is requested that notification be given immediately upon the spudding of the well. This information should be relayed to the Oil & Gas Division, Bismarck, via telephone. The following information must be included: Well name, legal location, permit number, drilling contractor, company representative, date and time of spudding. Office hours are 8:00 a.m. to 12:00 p.m. and 1:00 p.m. to 5:00 p.m. Central Time. Our telephone number is (701) 328-8020, leave a message if after hours or on the weekend.

Survey Requirements for Horizontal, Horizontal Re-entry, and Directional Wells

NDAC Section 43-02-03-25 (Deviation Tests and Directional Surveys) states in part (that) the survey contractor shall file a certified copy of all surveys with the director free of charge within thirty days of completion. Surveys must be submitted as one electronic copy, or in a form approved by the director. However, the director may require the directional survey to be filed immediately after completion if the survey is needed to conduct the operation of the director's office in a timely manner. Certified surveys must be submitted via email in one adobe document, with a certification cover page to certsurvey@nd.gov.
Survey points shall be of such frequency to accurately determine the entire location of the well bore.
Specifically, the Horizontal and Directional well survey frequency is 100 feet in the vertical, 30 feet in the curve (or when sliding) and 90 feet in the lateral.

Surface casing cement

Tail cement utilized on surface casing must have a minimum compressive strength of 500 psi within 12 hours, and tail cement utilized on production casing must have a minimum compressive strength of 500 psi before drilling the plug or initiating tests.

Logs

NDAC Section 43-02-03-31 requires the running of (1) a suite of open hole logs from which formation tops and porosity zones can be determined, (2) a Gamma Ray Log run from total depth to ground level elevation of the well bore, and (3) a log from which the presence and quality of cement can be determined (Standard CBL or Ultrasonic cement evaluation log) in every well in which production or intermediate casing has been set, this log must be run prior to completing the well.
All logs run must be submitted free of charge, as one digital TIFF (tagged image file format) copy and one digital LAS (log ASCII) formatted copy. Digital logs may be submitted on a standard CD, DVD, or attached to an email sent to digitallogs@nd.gov
Thank you for your cooperation.

Sincerely,

Alice Webber
Engineering Tech

APPLICATION FOR PERMIT TO DRILL HORIZONTAL WELL - FORM 1H

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 54269 (08-2005)

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Type of Work	Type of Well	Approximate Date Work Will Start	Confidential Status
New Location	Oil & Gas	05 / 01 / 2014	No

Operator	Telephone Number
OASIS PETROLEUM NORTH AMERICA LLC	281-404-9491

Address	City	State	Zip Code
1001 Fannin Suite 1500	Houston	TX	77002

☑ Notice has been provided to the owner of any permanently occupied dwelling within 1,320 feet.

☑ This well is not located within five hundred feet of an occupied dwelling.

WELL INFORMATION (If more than one lateral proposed, enter data for additional laterals on page 2)

Well Name	Well Number
WADE FEDERAL	5300 21-30 13B

Surface Footages	Qtr-Qtr	Section	Township	Range	County
1574 F N L 270 F W L	LOT2	30	153 N	100 W	McKenzie

Longstring Casing Point Footages	Qtr-Qtr	Section	Township	Range	County
1258 F N L 662 F W L	LOT1	30	153 N	100 W	McKenzie

Longstring Casing Point Coordinates From Well Head	Azimuth	Longstring Total Depth
316 N From WH 392 E From WH	55.85 º	10924 Feet MD 10655 Feet TVD

Bottom Hole Footages From Nearest Section Line	Qtr-Qtr	Section	Township	Range	County
764 F N L 201 F E L	NENE	29	153 N	100 W	Williams

Bottom Hole Coordinates From Well Head	KOP Lateral 1	Azimuth Lateral 1	Estimated Total Depth Lateral 1
810 N From WH 10042 E From WH	10178 Feet MD	90.0 º	20674 Feet MD 10749 Feet TVD

Latitude of Well Head	Longitude of Well Head	NAD Reference	Description of (Subject to NDIC Approval)
48 ° 02 ' 55.79 "	-103 ° 36 ' 10.97 "	NAD83	Spacing Unit: Sections 29 & 30 T153N R100W

Ground Elevation	Acres in Spacing/Drilling Unit	Spacing/Drilling Unit Setback Requirement	Industrial Commission Order
2008 Feet Above S.L.	2560	500 Feet N/S 200 Feet E/W	23752

North Line of Spacing/Drilling Unit	South Line of Spacing/Drilling Unit	East Line of Spacing/Drilling Unit	West Line of Spacing/Drilling Unit
10513 Feet	10522 Feet	5082 Feet	5236 Feet

Objective Horizons	Pierre Shale Top
Bakken	1874

Proposed	Size	Weight	Depth	Cement Volume	
Surface Casing	9 - 5/8 "	36 Lb./Ft.	1980 Feet	586 Sacks	NOTE: Surface hole must be drilled with fresh water and surface casing must be cemented back to surface.

Proposed	Size	Weight(s)	Longstring Total Depth	Cement Volume	Cement Top	Top Dakota Sand
Longstring Casing	7 - "	29/32 Lb./Ft.	10924 Feet MD 10655 Feet TVD	800 Sacks	4847 Feet	5347 Feet

Base Last Charles Salt (If Applicable)	
9115 Feet	NOTE: Intermediate or longstring casing string must be cemented above the top Dakota Group Sand.

Proposed Logs
Triple Combo: KOP to Kibby GR/Res to BSC GR to surf CND through the Dakota

Drilling Mud Type (Vertical Hole - Below Surface Casing)	Drilling Mud Type (Lateral)
Invert	Salt Water Gel

Survey Type in Vertical Portion of Well	Survey Frequency: Build Section	Survey Frequency: Lateral	Survey Contractor
MWD Every 100 Feet	30 Feet	90 Feet	Ryan

NOTE: A Gamma Ray log must be run to ground surface and a CBL must be run on intermediate or longstring casing string if set.

Surveys are required at least every 30 feet in the build section and every 90 feet in the lateral section of a horizontal well. Measurement inaccuracies are not considered when determining compliance with the spacing/drilling unit boundary setback requirement except in the following scenarios: 1) When the angle between the well bore and the respective boundary is 10 degrees or less; or 2) If Industry standard methods and equipment are not utilized. Consult the applicable field order for exceptions.

If measurement inaccuracies are required to be considered, a 2° MWD measurement inaccuracy will be applied to the horizontal portion of the well bore. This measurement inaccuracy is applied to the well bore from KOP to TD.

REQUIRED ATTACHMENTS: Certified surveyor's plat, horizontal section plat, estimated geological tops, proposed mud/cementing plan, directional plot/plan, $100 fee.

See Page 2 for Comments section and signature block.

COMMENTS, ADDITIONAL INFORMATION, AND/OR LIST OF ATTACHMENTS

Documents forwarded by email: Drill plan with drilling fluids, Well Summary with casing/cement plans, Directional Plan & Plot, Plats

Lateral 2

KOP Lateral 2 Feet MD	Azimuth Lateral 2 °	Estimated Total Depth Lateral 2 Feet MD / Feet TVD	KOP Coordinates From Well Head From WH / From WH	
Formation Entry Point Coordinates From Well Head From WH / From WH		Bottom Hole Coordinates From Well Head From WH / From WH		
KOP Footages From Nearest Section Line F L / F L	Qtr-Qtr	Section	Township N	Range W / County
Bottom Hole Footages From Nearest Section Line F L / F L	Qtr-Qtr	Section	Township N	Range W / County

Lateral 3

KOP Lateral 3 Feet MD	Azimuth Lateral 3 °	Estimated Total Depth Lateral 3 Feet MD / Feet TVD	KOP Coordinates From Well Head From WH / From WH	
Formation Entry Point Coordinates From Well Head From WH / From WH		Bottom Hole Coordinates From Well Head From WH / From WH		
KOP Footages From Nearest Section Line F L / F L	Qtr-Qtr	Section	Township N	Range W / County
Bottom Hole Footages From Nearest Section Line F L / F L	Qtr-Qtr	Section	Township N	Range W / County

Lateral 4

KOP Lateral 4 Feet MD	Azimuth Lateral 4 °	Estimated Total Depth Lateral 4 Feet MD / Feet TVD	KOP Coordinates From Well Head From WH / From WH	
Formation Entry Point Coordinates From Well Head From WH / From WH		Bottom Hole Coordinates From Well Head From WH / From WH		
KOP Footages From Nearest Section Line F L / F L	Qtr-Qtr	Section	Township N	Range W / County
Bottom Hole Footages From Nearest Section Line F L / F L	Qtr-Qtr	Section	Township N	Range W / County

Lateral 5

KOP Lateral 5 Feet MD	Azimuth Lateral 5 °	Estimated Total Depth Lateral 5 Feet MD / Feet TVD	KOP Coordinates From Well Head From WH / From WH	
Formation Entry Point Coordinates From Well Head From WH / From WH		Bottom Hole Coordinates From Well Head From WH / From WH		
KOP Footages From Nearest Section Line F L / F L	Qtr-Qtr	Section	Township N	Range W / County
Bottom Hole Footages From Nearest Section Line F L / F L	Qtr-Qtr	Section	Township N	Range W / County

I hereby swear or affirm the information provided is true, complete and correct as determined from all available records.		Date 04 / 14 / 2014
ePermit	Printed Name **Brandi Terry**	Title **Regulatory Specialist**

FOR STATE USE ONLY

Permit and File Number	API Number
28978	**33 - 053 - 06131**
Field **BAKER**	
Pool **BAKKEN**	Permit Type **DEVELOPMENT**

FOR STATE USE ONLY

Date Approved
7 / 25 / 2014
By **Alice Webber**
Title **Engineering Tech**

Oil and Gas Division

Lynn D. Helms - Director Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.oilgas.nd.gov

April 9, 2014

RE: **Filter Socks and Other Filter Media**
Leakproof Container Required
Oil and Gas Wells

Dear Operator,

North Dakota Administrative Code Section 43-02-03-19.2 states in part that all waste material associated with exploration or production of oil and gas must be properly disposed of in an authorized facility in accord with all applicable local, state, and federal laws and regulations.

Filtration systems are commonly used during oil and gas operations in North Dakota. The Commission is very concerned about the proper disposal of used filters (including filter socks) used by the oil and gas industry.

Effective June 1, 2014, a container must be maintained on each well drilled in North Dakota beginning when the well is spud and must remain on-site during clean-out, completion, and flow-back whenever filtration operations are conducted. The on-site container must be used to store filters until they can be properly disposed of in an authorized facility. Such containers must be:

- leakproof to prevent any fluids from escaping the container
- covered to prevent precipitation from entering the container
- placard to indicate only filters are to be placed in the container

If the operator will not utilize a filtration system, a waiver to the container requirement will be considered, but only upon the operator submitting a Sundry Notice (Form 4) justifying their request.

As previously stated in our March 13, 2014 letter, North Dakota Administrative Code Section 33-20-02.1-01 states in part that every person who transports solid waste (which includes oil and gas exploration and production wastes) is required to have a valid permit issued by the North Dakota Department of Health, Division of Waste Management. Please contact the Division of Waste Management at (701) 328-5166 with any questions on the solid waste program. Note oil and gas exploration and production wastes include produced water, drilling mud, invert mud, tank bottom sediment, pipe scale, filters, and fly ash.

Thank you for your cooperation.

Sincerely,

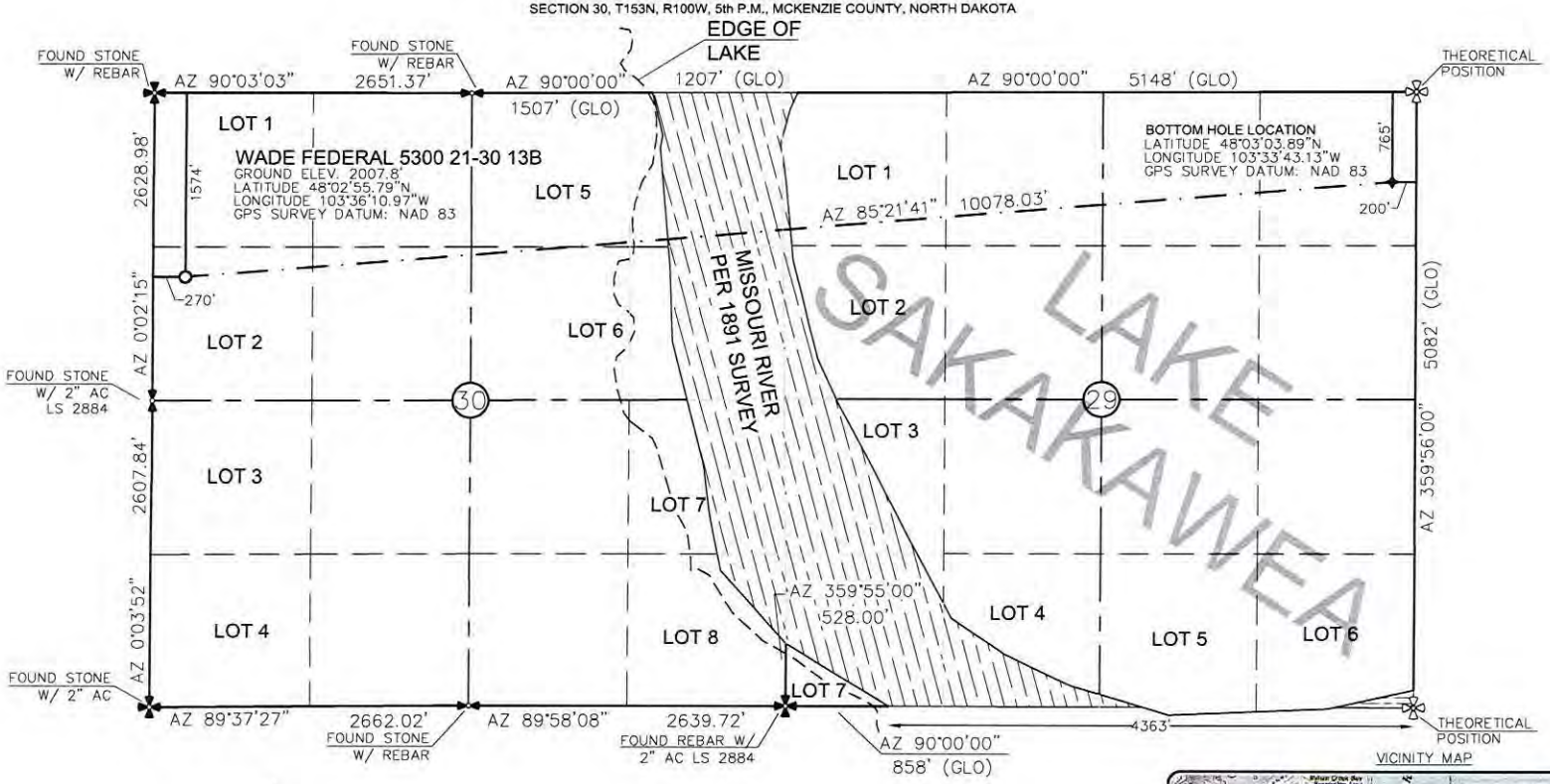
Bruce E. Hicks

Assistant Director

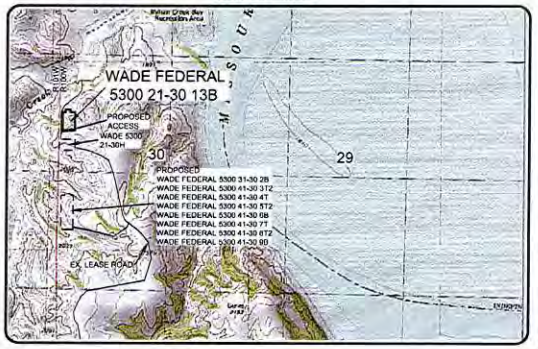
600 E Boulevard Ave - Dept 405, Bismarck, North Dakota 58505-0840 Phone (701)328-8020 Fax (701)328-8022

WELL LOCATION PLAT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"WADE FEDERAL 5300 21-30 13B"
1574 FEET FROM NORTH LINE AND 270 FEET FROM WEST LINE
SECTION 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



VICINITY MAP



THIS DOCUMENT WAS ORIGINALLY ISSUED AND SEALED BY DARYL D. KASEMAN, PLS, REGISTRATION NUMBER 3880 ON 3/19/14 AND THE ORIGINAL DOCUMENTS ARE STORED AT THE OFFICES OF INTERSTATE ENGINEERING, INC.

0 1000
1" = 1000'

✠ - MONUMENT - RECOVERED
✠ - MONUMENT - NOT RECOVERED

STAKED ON 3/17/14
VERTICAL CONTROL DATUM WAS BASED UPON CONTROL POINT 705 WITH AN ELEVATION OF 2158.3'

THIS SURVEY AND PLAT IS BEING PROVIDED AT THE REQUEST OF ERIC BAYES OF OASIS PETROLEUM. I CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS WORK PERFORMED BY ME OR UNDER MY SUPERVISION AND IS TRUE AND CORRECT TO THE BEST OF MY KNOWLEDGE AND BELIEF.

DARYL D. KASEMAN LS-3880

REGISTERED PROFESSIONAL LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE: 3/19/14
NORTH DAKOTA

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Revision No.	Date	By	Description

OASIS PETROLEUM NORTH AMERICA, LLC
WELL LOCATION PLAT
SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.K. Project No.: S14-09-019.01
Checked By: D.D.K. Date: MAR. 2014

Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph: (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com
Other offices in Minnesota, North Dakota and South Dakota

INTERSTATE ENGINEERING
Professionals you need, people you trust

SHEET NO. 1/8

DRILLING PLAN

OPERATOR	Oasis Petroleum	**COUNTY/STATE**	McKenzie Co., ND
WELL NAME	Wade Federal 5300 21-30 13B	**RIG**	B25
WELL TYPE	Horizontal Middle Bakken		
LOCATION	SW NW 30-153N-100W Surface Location (survey plat): 1574' FNL 270' FWL		
EST. T.D.	20,774'	**GROUND ELEV:**	1,999' **Sub Height:** 25'
	TOTAL LATERAL: 9,851'	**KB ELEV:**	2,024'

MARKER		TVD	Subsea TVD
Pierre	NDIC MAP	1,874	150
Greenhorn		4,521	-2,497
Mowry		4,923	-2,899
Dakota		5,347	-3,323
Rierdon		6,360	-4,336
Dunham Salt		6,851	-4,827
Dunham Salt Base		6,901	-4,877
Pine Salt		7,161	-5,137
Pine Salt Base		7,179	-5,155
Opeche Salt		7,249	-5,225
Opeche Salt Base		7,328	-5,304
Amsden		7,569	-5,545
Tyler		7,730	-5,706
Otter/Base Minnelusa		7,957	-5,933
Kibbey Lime		8,299	-6,275
Charles Salt		8,441	-6,417
Base Last Salt		9,115	-7,091
Mission Canyon		9,331	-7,307
Lodgepole		9,881	-7,857
False Bakken		10,612	-8,588
Upper Bakken Shale		10,624	-8,600
Middle Bakken (Top of Target)		10,649	-8,625
Middle Bakken (Base of target)		10,660	-8,636
Lower Bakken Shale		10,674	-8,650
Threeforks		10,698	-8,674

Est. Dip Rate: -0.55

Max. Anticipated BHP: 4825

LOGS:	Type	Interval
	OH Logs: Request Log waiver based on the Wade Federal 5300 21-30H 600' S of surface location	
	CBL/GR: Above top of cement/GR to base of casing	
	MWD GR: KOP to lateral TD	

DEVIATION:	Surf:	3 deg. max., 1 deg / 100'; srvy every 500'
	Prod:	5 deg. max., 1 deg / 100'; srvy every 100'
DST'S:	None planned	
CORES:	None planned	
MUDLOGGING:	Two-Man:	Begin 200' above Kibbey 30' samples in curve and lateral
BOP:	11" 5000 psi blind, pipe & annular	

Surface Formation: *Glacial till*

MUD:	Interval		Type	WT	Vis	WL	Remarks
Surface:	0' -	1,974'	FW/Gel - Lime Sweeps	8.4-9.0	28-32	NC	Circ Mud Tanks
Intermediate:	1,974' -	10,923'	Invert	9.5-10.4	40-50	30+HtHp	Circ Mud Tanks
Lateral:	10,923' -	20,774'	Salt Water	9.8-10.2	28-32	NC	Circ Mud Tanks

CASING:	Size	Wt ppf	Hole	Depth		Cement	WOC	Remarks
Surface:	9-5/8"	36#	13-1/2"	1,974'		To Surface	12	100' into Pierre
Intermediate:	7"	29/32#	8-3/4"	10,923'		4847	24	500' above Dakota
Production Liner:	4.5"	11.6#	6"	20,774'	TOL @ 10,128'			50' above KOP

PROBABLE PLUGS, IF REQ'D:

OTHER:

	MD	TVD	FNL/FSL	FEL/FWL	S-T-R	AZI
Surface:	1,974	1,974	1574' FNL	270' FWL	T150N R100W Sec. 30	Survey Company:
KOP:	10,178'	10,178'	1524' FNL	270' FWL	T150N R100W Sec. 30	0.0 Build Rate: 12 deg /100'
EOC:	10,923'	10,655'	1258' FNL	661' FWL	T150N R100W Sec. 30	55.9
Casing Point:	10,923'	10,655'	1258' FNL	661' FWL	T150N R100W Sec. 30	55.9
Middle Bakken Lateral TD:	20,774'	10,750'	765' FNL	200' FEL	T150N R100W Sec. 29	90.0

Comments:

Request Log waiver based on the Wade Federal 5300 21-30H 600' S of surface location

No frac string planned

35 packers and 20 sleeves planned 3.6MM lbs 30% ceramic

Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.
68334-30-5 (Primary Name: Fuels, diesel) 68476-34-6 (Primary Name: Fuels, diesel, No. 2) 68476-30-2 (Primary Name: Fuel oil No. 2)
68476-31-3 (Primary Name: Fuel oil, No. 4) 8008-20-6 (Primary Name: Kerosene)

OASIS PETROLEUM

Geology: N. Gabelman 3/20/2014 **Engineering:** smg 4.8.14

Oasis Petroleum
Well Summary
WADE FEDERAL 5300 21-30 13B
Section 30 T153N R100W
Mckenzie Co. ND

SURFACE CASING AND CEMENT DESIGN

							Make-up Torque (ft-lbs)		
Size	**Interval**	**Weight**	**Grade**	**Coupling**	**I.D.**	**Drift**	**Minimum**	**Optimum**	**Max**
9-5/8"	0' - 1980'	36	J-55	LTC	8.921"	8.765"	3400	4530	5660

Interval	Description	Collapse	Burst	Tension
		(psi) / a	(psi) / b	(1000 lbs) / c
0' - 1980'	9-5/8", 36#, J-55, LTC, 8rd	2020 / 2.17	3520 / 3.79	453 / 2.80

API Rating & Safety Factor

a) Based on full casing evacuation with 9 ppg fluid on backside (1980' setting depth).

b) Burst pressure based on 9 ppg fluid with no fluid on backside (1980' setting depth).

c) Based on string weight in 9 ppg fluid at 1980' TVD plus 100k# overpull. (Buoyed weight equals 61k lbs.)

Cement volumes are based on 9-5/8" casing set in 13-1/2 " hole with 60% excess to circulate cement back to surface.
Mix and pump the following slurry.

Pre-flush (Spacer): **20 bbls** fresh water

Lead Slurry: **425 sks** (219 bbls), 2.90 ft3/sk, 11.5 lb/gal Conventional system with 94 lb/sk cement, 4% D079 extender, 2% D053 expanding agent, 2% CaCl2 and 0.250 lb/sk D130 lost circulation control agent.

Tail Slurry: **300 sks** (62 bbls), 1.16 ft3/sk 15.8 lb/gal Conventional system with 94 lb/sk cement, 0.25% CaCl2, and 0.250 lb/sk lost circulation control agent

Oasis Petroleum
Well Summary
WADE FEDERAL 5300 21-30 13B
Section 30 T153N R100W
Mckenzie Co. ND

INTERMEDIATE CASING AND CEMENT DESIGN

							Make-up Torque (ft-lbs)		
Size	**Interval**	**Weight**	**Grade**	**Coupling**	**I.D.**	**Drift****	**Minimum**	**Optimum**	**Max**
7"	0' - 6701'	29	P-110	LTC	6.184"	6.059"	5980	7970	8770
7"	6701' - 10178'	32	HCP-110	LTC	6.094"	6.000"**	6730	8970	9870
7"	10178' - 10923'	29	P-110	LTC	6.184"	6.059"	5980	7970	8770

****Special Drift 7" 32# to 6.0"**

Interval	Length	Description	Collapse	Burst	Tension
			(psi) a	(psi) b	(1000 lbs) c
0' - 6701'	6701'	7", 29#, P-110, LTC, 8rd	8530 / 2.44*	11220 / 1.19	797 / 2.11
6701' - 10178'	3477'	7", 32#, HCP-110, LTC, 8rd	11820 / 2.23*	12460 / 1.29	
6701' - 10178'	3477'	7", 32#, HCP-110, LTC, 8rd	11820 / 1.08**	12460 / 1.29	
10178' - 10923'	745'	7", 29#, P-110, LTC, 8rd	8530 / 1.53*	11220 / 1.16	

API Rating & Safety Factor

a) *Assume full casing evacuation with 10 ppg fluid on backside. **Assume full casing evacuation with 1.2 psi/ft equivalent fluid gradient across salt intervals.

b) Burst pressure based on 9000 psig max press for stimulation plus 10.2 ppg fluid in casing and 9 ppg fluid on backside-to 10655' TVD.

c) Based on string weight in 10 ppg fluid, (277k lbs buoyed weight) plus 100k lbs overpull.

Cement volumes are estimates based on 7" casing set in an 8-3/4" hole with 30% excess.
Mix and pump the following slurry

Pre-flush (Spacer): **100 bbls** Saltwater
20bbls CW8
20bbls Fresh Water

Lead Slurry: **189 sks** (87 bbls), 11.8 lb/gal, 2.59 ft3/sk yield, Conventional system with 61 lb/sk cement, 10% NaCl, 23 lb/sk extender, 0.2% D046 Anti Foam, 0.25lb/sk D130 Lost Circulation, 0.8% D112 Fluid Loss, 6% D035 Extender.

Tail Slurry: **589 sks** (163 bbls), 15.8 lb/gal, 1.55 ft3/sk yield, Conventional system with 94 lb/sk cement, 10% NaCl, 35% Silica, 0.2% fluid loss agent D167, 0.27% Retarder D198, 0.25 lb/sk D130 lost circulation control, 0.2% Anti Foam D046.

Oasis Petroleum
Well Summary
WADE FEDERAL 5300 21-30 13B
Section 30 T153N R100W
Mckenzie Co. ND

PRODUCTION LINER

							Make-up Torque (ft-lbs)		
Size	**Interval**	**Weight**	**Grade**	**Coupling**	**I.D.**	**Drift**	**Minimum**	**Optimum**	**Max**
4-1/2"	10128' - 20674'	11.6	P-110	BTC	4.000"	3.875"	2270	3020	3780

Interval	Length	Desctiption	Collapse	Burst	Tension
			(psi) a	(psi) b	(1000 lbs) c
10128' - 20674'	10546	4-1/2", 11.6 lb, P-110, BTC	7560 / 1.42	10690 / 1.10	385 / 1.88

API Rating & Safety Factor

a) Based on full casing evacuation with 9.5 ppg fluid on backside @ 10749' TVD.

b) Burst pressure based on 9000 psi treating pressure with 10.2 ppg internal fluid gradient and 9 ppg external fluid gradient @ 10749' TVD.

c) Based on string weight in 9.5 ppg fluid (Buoyed weight: 105k lbs.) plus 100k lbs overpull.

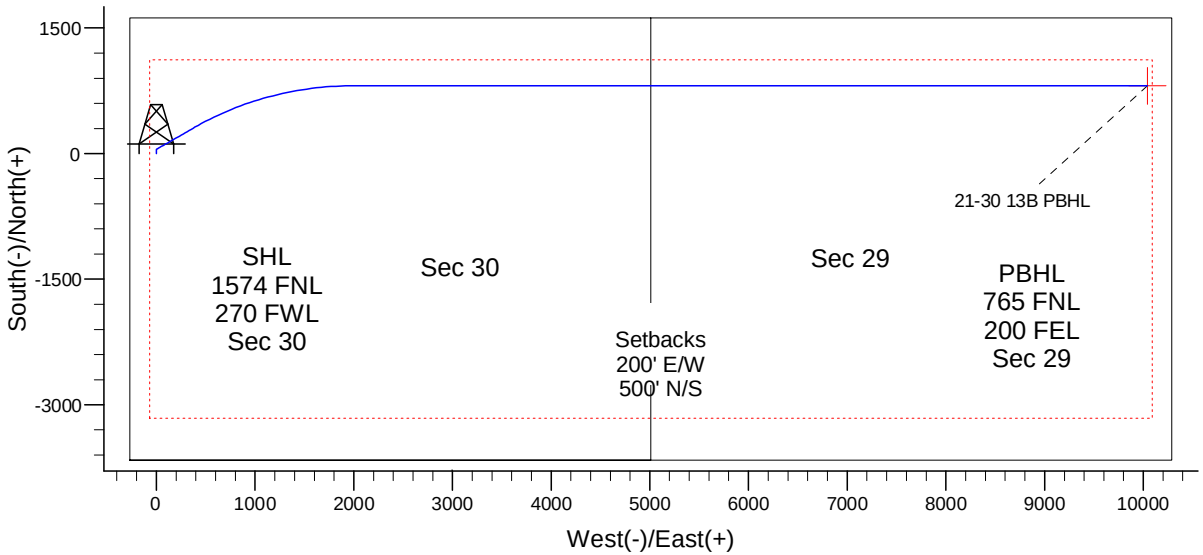
Azimuths to True North
Magnetic North: 8.15°

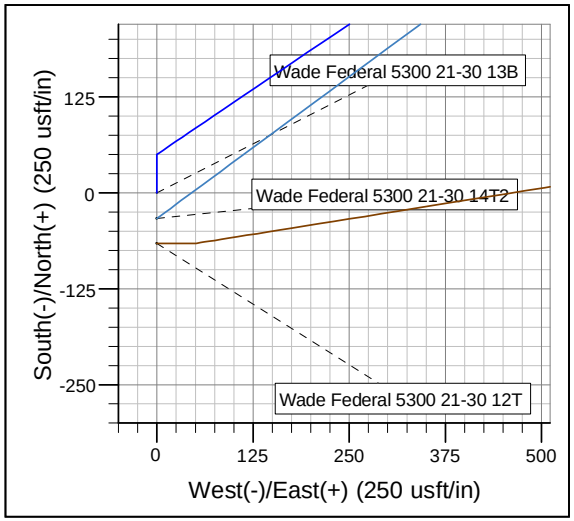
Magnetic Field
Strength: 56473.1snT
Dip Angle: 72.95°
Date: 4/4/2014
Model: IGRF200510

OASIS PETROLEUM

WELL DETAILS: Wade Federal 5300 21-30 13B			
	Ground Level:	1999.0	
Northing	Easting	Latitude	Longitude
397868.63	1209741.76	48° 2' 55.790 N	103° 36' 10.970 W

Project: Indian Hills
Site: 153N-100W-29/30
Well: Wade Federal 5300 21-30 13B
Wellbore: Wade Federal 13B
Design: Dising#2



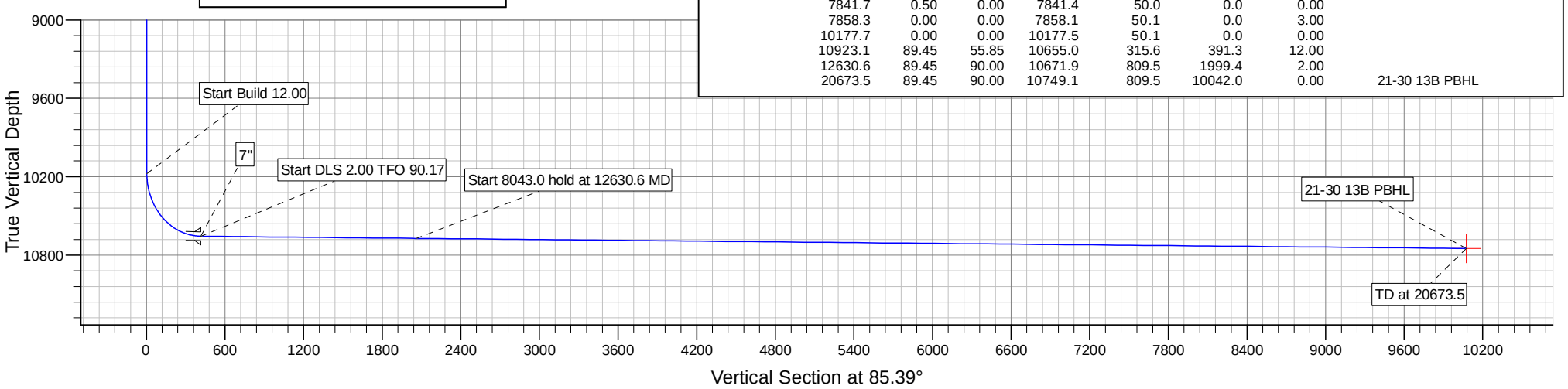


CASING DETAILS

TVD	MD	Name	Size
1974.0	1974.0	9 5/8"	9-5/8
10655.0	10924.2	7"	7

SECTION DETAILS

MD	Inc	Azi	TVD	+N/-S	+E/-W	DIeg	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	
2100.0	0.00	0.00	2100.0	0.0	0.0	0.00	
2116.7	0.50	0.00	2116.7	0.1	0.0	3.00	
7841.7	0.50	0.00	7841.4	50.0	0.0	0.00	
7858.3	0.00	0.00	7858.1	50.1	0.0	3.00	
10177.7	0.00	0.00	10177.5	50.1	0.0	0.00	
10923.1	89.45	55.85	10655.0	315.6	391.3	12.00	
12630.6	89.45	90.00	10671.9	809.5	1999.4	2.00	
20673.5	89.45	90.00	10749.1	809.5	10042.0	0.00	21-30 13B PBHL



Oasis

Indian Hills
153N-100W-29/30
Wade Federal 5300 21-30 13B

Wade Federal 13B

Plan: Dising#2

Standard Planning Report

25 July, 2014

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	**Local Co-ordinate Reference:**	Well Wade Federal 5300 21-30 13B
Company:	Oasis	**TVD Reference:**	WELL @ 2024.0usft (Original Well Elev)
Project:	Indian Hills	**MD Reference:**	WELL @ 2024.0usft (Original Well Elev)
Site:	153N-100W-29/30	**North Reference:**	True
Well:	Wade Federal 5300 21-30 13B	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Wade Federal 13B		
Design:	Dising#2		

Project Indian Hills

Map System:	US State Plane 1983	**System Datum:**	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		

Site 153N-100W-29/30

Site Position:		**Northing:**	395,519.95 usft	**Latitude:**	48° 2' 32.580 N
From:	Lat/Long	**Easting:**	1,209,617.13 usft	**Longitude:**	103° 36' 11.410 W
Position Uncertainty:	0.0 usft	**Slot Radius:**	13-3/16 "	**Grid Convergence:**	-2.31 °

Well Wade Federal 5300 21-30 13B

Well Position	**+N/-S**	2,351.8 usft	**Northing:**	397,868.63 usft	**Latitude:**	48° 2' 55.790 N
	+E/-W	29.9 usft	**Easting:**	1,209,741.75 usft	**Longitude:**	103° 36' 10.970 W
Position Uncertainty		0.0 usft	**Wellhead Elevation:**		**Ground Level:**	1,999.0 usft

Wellbore Wade Federal 13B

Magnetics	**Model Name**	**Sample Date**	**Declination (°)**	**Dip Angle (°)**	**Field Strength (nT)**
	IGRF200510	4/4/2014	8.15	72.95	56,473

Design Dising#2

Audit Notes:

Version:	**Phase:**	PROTOTYPE	**Tie On Depth:**	0.0

Vertical Section:	**Depth From (TVD) (usft)**	**+N/-S (usft)**	**+E/-W (usft)**	**Direction (°)**
	0.0	0.0	0.0	85.39

Plan Sections

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)	TFO (°)	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,100.0	0.00	0.00	2,100.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,116.7	0.50	0.00	2,116.7	0.1	0.0	3.00	3.00	0.00	0.00	
7,841.7	0.50	0.00	7,841.4	50.0	0.0	0.00	0.00	0.00	0.00	
7,858.3	0.00	0.00	7,858.1	50.1	0.0	3.00	-3.00	0.00	180.00	
10,177.7	0.00	0.00	10,177.5	50.1	0.0	0.00	0.00	0.00	0.00	
10,923.1	89.45	55.85	10,655.0	315.6	391.3	12.00	12.00	0.00	55.85	
12,630.6	89.45	90.00	10,671.9	809.5	1,999.4	2.00	0.00	2.00	90.17	
20,673.5	89.45	90.00	10,749.1	809.5	10,042.0	0.00	0.00	0.00	0.00	21-30 13B PBHL

Database:	OpenWellsCompass - EDM Prod	**Local Co-ordinate Reference:**	Well Wade Federal 5300 21-30 13B
Company:	Oasis	**TVD Reference:**	WELL @ 2024.0usft (Original Well Elev)
Project:	Indian Hills	**MD Reference:**	WELL @ 2024.0usft (Original Well Elev)
Site:	153N-100W-29/30	**North Reference:**	True
Well:	Wade Federal 5300 21-30 13B	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Wade Federal 13B		
Design:	Dising#2		

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
2,100.0	0.00	0.00	2,100.0	0.0	0.0	0.0	0.00	0.00	0.00
2,116.7	0.50	0.00	2,116.7	0.1	0.0	0.0	3.00	3.00	0.00
2,200.0	0.50	0.00	2,200.0	0.8	0.0	0.1	0.00	0.00	0.00
2,300.0	0.50	0.00	2,300.0	1.7	0.0	0.1	0.00	0.00	0.00
2,400.0	0.50	0.00	2,400.0	2.5	0.0	0.2	0.00	0.00	0.00
2,500.0	0.50	0.00	2,500.0	3.4	0.0	0.3	0.00	0.00	0.00
2,600.0	0.50	0.00	2,600.0	4.3	0.0	0.3	0.00	0.00	0.00
2,700.0	0.50	0.00	2,700.0	5.2	0.0	0.4	0.00	0.00	0.00
2,800.0	0.50	0.00	2,800.0	6.0	0.0	0.5	0.00	0.00	0.00
2,900.0	0.50	0.00	2,900.0	6.9	0.0	0.6	0.00	0.00	0.00
3,000.0	0.50	0.00	3,000.0	7.8	0.0	0.6	0.00	0.00	0.00
3,100.0	0.50	0.00	3,100.0	8.7	0.0	0.7	0.00	0.00	0.00
3,200.0	0.50	0.00	3,200.0	9.5	0.0	0.8	0.00	0.00	0.00
3,300.0	0.50	0.00	3,300.0	10.4	0.0	0.8	0.00	0.00	0.00
3,400.0	0.50	0.00	3,400.0	11.3	0.0	0.9	0.00	0.00	0.00
3,500.0	0.50	0.00	3,499.9	12.1	0.0	1.0	0.00	0.00	0.00
3,600.0	0.50	0.00	3,599.9	13.0	0.0	1.0	0.00	0.00	0.00
3,700.0	0.50	0.00	3,699.9	13.9	0.0	1.1	0.00	0.00	0.00
3,800.0	0.50	0.00	3,799.9	14.8	0.0	1.2	0.00	0.00	0.00
3,900.0	0.50	0.00	3,899.9	15.6	0.0	1.3	0.00	0.00	0.00
4,000.0	0.50	0.00	3,999.9	16.5	0.0	1.3	0.00	0.00	0.00
4,100.0	0.50	0.00	4,099.9	17.4	0.0	1.4	0.00	0.00	0.00
4,200.0	0.50	0.00	4,199.9	18.3	0.0	1.5	0.00	0.00	0.00
4,300.0	0.50	0.00	4,299.9	19.1	0.0	1.5	0.00	0.00	0.00
4,400.0	0.50	0.00	4,399.9	20.0	0.0	1.6	0.00	0.00	0.00
4,500.0	0.50	0.00	4,499.9	20.9	0.0	1.7	0.00	0.00	0.00
4,521.1	0.50	0.00	4,521.0	21.1	0.0	1.7	0.00	0.00	0.00
Greenhorn									
4,600.0	0.50	0.00	4,599.9	21.7	0.0	1.7	0.00	0.00	0.00
4,700.0	0.50	0.00	4,699.9	22.6	0.0	1.8	0.00	0.00	0.00
4,800.0	0.50	0.00	4,799.9	23.5	0.0	1.9	0.00	0.00	0.00
4,900.0	0.50	0.00	4,899.9	24.4	0.0	2.0	0.00	0.00	0.00
4,923.1	0.50	0.00	4,923.0	24.6	0.0	2.0	0.00	0.00	0.00
Mowry									
5,000.0	0.50	0.00	4,999.9	25.2	0.0	2.0	0.00	0.00	0.00
5,100.0	0.50	0.00	5,099.9	26.1	0.0	2.1	0.00	0.00	0.00
5,200.0	0.50	0.00	5,199.9	27.0	0.0	2.2	0.00	0.00	0.00
5,300.0	0.50	0.00	5,299.9	27.9	0.0	2.2	0.00	0.00	0.00
5,347.1	0.50	0.00	5,347.0	28.3	0.0	2.3	0.00	0.00	0.00
Dakota									
5,400.0	0.50	0.00	5,399.9	28.7	0.0	2.3	0.00	0.00	0.00
5,500.0	0.50	0.00	5,499.9	29.6	0.0	2.4	0.00	0.00	0.00
5,600.0	0.50	0.00	5,599.9	30.5	0.0	2.4	0.00	0.00	0.00
5,700.0	0.50	0.00	5,699.9	31.3	0.0	2.5	0.00	0.00	0.00
5,800.0	0.50	0.00	5,799.9	32.2	0.0	2.6	0.00	0.00	0.00
5,900.0	0.50	0.00	5,899.9	33.1	0.0	2.7	0.00	0.00	0.00
6,000.0	0.50	0.00	5,999.9	34.0	0.0	2.7	0.00	0.00	0.00
6,100.0	0.50	0.00	6,099.8	34.8	0.0	2.8	0.00	0.00	0.00
6,200.0	0.50	0.00	6,199.8	35.7	0.0	2.9	0.00	0.00	0.00
6,300.0	0.50	0.00	6,299.8	36.6	0.0	2.9	0.00	0.00	0.00
6,360.2	0.50	0.00	6,360.0	37.1	0.0	3.0	0.00	0.00	0.00
Rierdon									
6,400.0	0.50	0.00	6,399.8	37.5	0.0	3.0	0.00	0.00	0.00
6,500.0	0.50	0.00	6,499.8	38.3	0.0	3.1	0.00	0.00	0.00
6,600.0	0.50	0.00	6,599.8	39.2	0.0	3.1	0.00	0.00	0.00

Database:	OpenWellsCompass - EDM Prod	**Local Co-ordinate Reference:**	Well Wade Federal 5300 21-30 13B
Company:	Oasis	**TVD Reference:**	WELL @ 2024.0usft (Original Well Elev)
Project:	Indian Hills	**MD Reference:**	WELL @ 2024.0usft (Original Well Elev)
Site:	153N-100W-29/30	**North Reference:**	True
Well:	Wade Federal 5300 21-30 13B	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Wade Federal 13B		
Design:	Dising#2		

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
6,700.0	0.50	0.00	6,699.8	40.1	0.0	3.2	0.00	0.00	0.00
6,800.0	0.50	0.00	6,799.8	40.9	0.0	3.3	0.00	0.00	0.00
6,851.2	0.50	0.00	6,851.0	41.4	0.0	3.3	0.00	0.00	0.00
Dunham Salt									
6,900.0	0.50	0.00	6,899.8	41.8	0.0	3.4	0.00	0.00	0.00
6,901.2	0.50	0.00	6,901.0	41.8	0.0	3.4	0.00	0.00	0.00
Dunham Salt Base									
7,000.0	0.50	0.00	6,999.8	42.7	0.0	3.4	0.00	0.00	0.00
7,100.0	0.50	0.00	7,099.8	43.6	0.0	3.5	0.00	0.00	0.00
7,161.2	0.50	0.00	7,161.0	44.1	0.0	3.5	0.00	0.00	0.00
Pine Salt									
7,179.2	0.50	0.00	7,179.0	44.3	0.0	3.6	0.00	0.00	0.00
Pine Salt Base									
7,200.0	0.50	0.00	7,199.8	44.4	0.0	3.6	0.00	0.00	0.00
7,249.2	0.50	0.00	7,249.0	44.9	0.0	3.6	0.00	0.00	0.00
Opeche Salt									
7,300.0	0.50	0.00	7,299.8	45.3	0.0	3.6	0.00	0.00	0.00
7,328.2	0.50	0.00	7,328.0	45.6	0.0	3.7	0.00	0.00	0.00
Opeche Salt Base									
7,400.0	0.50	0.00	7,399.8	46.2	0.0	3.7	0.00	0.00	0.00
7,500.0	0.50	0.00	7,499.8	47.1	0.0	3.8	0.00	0.00	0.00
7,569.2	0.50	0.00	7,569.0	47.7	0.0	3.8	0.00	0.00	0.00
Amsden									
7,600.0	0.50	0.00	7,599.8	47.9	0.0	3.9	0.00	0.00	0.00
7,700.0	0.50	0.00	7,699.8	48.8	0.0	3.9	0.00	0.00	0.00
7,730.2	0.50	0.00	7,730.0	49.1	0.0	3.9	0.00	0.00	0.00
Tyler									
7,800.0	0.50	0.00	7,799.8	49.7	0.0	4.0	0.00	0.00	0.00
7,841.7	0.50	0.00	7,841.4	50.0	0.0	4.0	0.00	0.00	0.00
7,858.3	0.00	0.00	7,858.1	50.1	0.0	4.0	3.00	-3.00	0.00
7,900.0	0.00	0.00	7,899.8	50.1	0.0	4.0	0.00	0.00	0.00
7,957.2	0.00	0.00	7,957.0	50.1	0.0	4.0	0.00	0.00	0.00
Otter/Base Minnelusa									
8,000.0	0.00	0.00	7,999.8	50.1	0.0	4.0	0.00	0.00	0.00
8,100.0	0.00	0.00	8,099.8	50.1	0.0	4.0	0.00	0.00	0.00
8,200.0	0.00	0.00	8,199.8	50.1	0.0	4.0	0.00	0.00	0.00
8,299.2	0.00	0.00	8,299.0	50.1	0.0	4.0	0.00	0.00	0.00
Kibbey Lime									
8,300.0	0.00	0.00	8,299.8	50.1	0.0	4.0	0.00	0.00	0.00
8,400.0	0.00	0.00	8,399.8	50.1	0.0	4.0	0.00	0.00	0.00
8,441.2	0.00	0.00	8,441.0	50.1	0.0	4.0	0.00	0.00	0.00
Charles Salt									
8,500.0	0.00	0.00	8,499.8	50.1	0.0	4.0	0.00	0.00	0.00
8,600.0	0.00	0.00	8,599.8	50.1	0.0	4.0	0.00	0.00	0.00
8,700.0	0.00	0.00	8,699.8	50.1	0.0	4.0	0.00	0.00	0.00
8,800.0	0.00	0.00	8,799.8	50.1	0.0	4.0	0.00	0.00	0.00
8,900.0	0.00	0.00	8,899.8	50.1	0.0	4.0	0.00	0.00	0.00
9,000.0	0.00	0.00	8,999.8	50.1	0.0	4.0	0.00	0.00	0.00
9,100.0	0.00	0.00	9,099.8	50.1	0.0	4.0	0.00	0.00	0.00
9,115.2	0.00	0.00	9,115.0	50.1	0.0	4.0	0.00	0.00	0.00
Base Last Salt									
9,200.0	0.00	0.00	9,199.8	50.1	0.0	4.0	0.00	0.00	0.00
9,300.0	0.00	0.00	9,299.8	50.1	0.0	4.0	0.00	0.00	0.00

Database:	OpenWellsCompass - EDM Prod	**Local Co-ordinate Reference:**	Well Wade Federal 5300 21-30 13B
Company:	Oasis	**TVD Reference:**	WELL @ 2024.0usft (Original Well Elev)
Project:	Indian Hills	**MD Reference:**	WELL @ 2024.0usft (Original Well Elev)
Site:	153N-100W-29/30	**North Reference:**	True
Well:	Wade Federal 5300 21-30 13B	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Wade Federal 13B		
Design:	Dising#2		

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
9,331.2	0.00	0.00	9,331.0	50.1	0.0	4.0	0.00	0.00	0.00
Mission Canyon									
9,400.0	0.00	0.00	9,399.8	50.1	0.0	4.0	0.00	0.00	0.00
9,500.0	0.00	0.00	9,499.8	50.1	0.0	4.0	0.00	0.00	0.00
9,600.0	0.00	0.00	9,599.8	50.1	0.0	4.0	0.00	0.00	0.00
9,700.0	0.00	0.00	9,699.8	50.1	0.0	4.0	0.00	0.00	0.00
9,800.0	0.00	0.00	9,799.8	50.1	0.0	4.0	0.00	0.00	0.00
9,881.2	0.00	0.00	9,881.0	50.1	0.0	4.0	0.00	0.00	0.00
Lodgepole									
9,900.0	0.00	0.00	9,899.8	50.1	0.0	4.0	0.00	0.00	0.00
10,000.0	0.00	0.00	9,999.8	50.1	0.0	4.0	0.00	0.00	0.00
10,100.0	0.00	0.00	10,099.8	50.1	0.0	4.0	0.00	0.00	0.00
10,177.7	0.00	0.00	10,177.5	50.1	0.0	4.0	0.00	0.00	0.00
10,200.0	2.67	55.85	10,199.8	50.4	0.4	4.5	12.00	12.00	0.00
10,225.0	5.67	55.85	10,224.7	51.4	1.9	6.1	12.00	12.00	0.00
10,250.0	8.67	55.85	10,249.5	53.2	4.5	8.8	12.00	12.00	0.00
10,275.0	11.67	55.85	10,274.1	55.6	8.2	12.6	12.00	12.00	0.00
10,300.0	14.67	55.85	10,298.4	58.8	12.9	17.6	12.00	12.00	0.00
10,325.0	17.67	55.85	10,322.5	62.8	18.6	23.6	12.00	12.00	0.00
10,350.0	20.67	55.85	10,346.1	67.4	25.4	30.8	12.00	12.00	0.00
10,375.0	23.67	55.85	10,369.2	72.7	33.2	39.0	12.00	12.00	0.00
10,400.0	26.67	55.85	10,391.8	78.6	42.0	48.2	12.00	12.00	0.00
10,425.0	29.67	55.85	10,413.9	85.3	51.8	58.5	12.00	12.00	0.00
10,450.0	32.67	55.85	10,435.3	92.5	62.5	69.8	12.00	12.00	0.00
10,475.0	35.67	55.85	10,455.9	100.4	74.1	82.0	12.00	12.00	0.00
10,500.0	38.67	55.85	10,475.9	108.9	86.6	95.1	12.00	12.00	0.00
10,525.0	41.67	55.85	10,495.0	117.9	100.0	109.1	12.00	12.00	0.00
10,550.0	44.67	55.85	10,513.2	127.5	114.1	124.0	12.00	12.00	0.00
10,575.0	47.67	55.85	10,530.5	137.7	129.1	139.7	12.00	12.00	0.00
10,600.0	50.67	55.85	10,546.8	148.3	144.7	156.2	12.00	12.00	0.00
10,625.0	53.67	55.85	10,562.2	159.4	161.1	173.3	12.00	12.00	0.00
10,650.0	56.67	55.85	10,576.5	170.9	178.0	191.2	12.00	12.00	0.00
10,675.0	59.67	55.85	10,589.6	182.8	195.6	209.7	12.00	12.00	0.00
10,700.0	62.67	55.85	10,601.7	195.1	213.7	228.7	12.00	12.00	0.00
10,723.6	65.50	55.85	10,612.0	207.0	231.3	247.2	12.00	12.00	0.00
False Bakken									
10,725.0	65.67	55.85	10,612.6	207.7	232.4	248.3	12.00	12.00	0.00
10,750.0	68.67	55.85	10,622.3	220.7	251.4	268.3	12.00	12.00	0.00
10,754.8	69.25	55.85	10,624.0	223.2	255.1	272.2	12.00	12.00	0.00
Upper Bakken Shale									
10,775.0	71.67	55.85	10,630.8	233.9	270.9	288.8	12.00	12.00	0.00
10,800.0	74.67	55.85	10,638.0	247.3	290.7	309.6	12.00	12.00	0.00
10,825.0	77.67	55.85	10,644.0	260.9	310.8	330.7	12.00	12.00	0.00
10,850.0	80.67	55.85	10,648.7	274.7	331.1	352.1	12.00	12.00	0.00
10,852.1	80.92	55.85	10,649.0	275.8	332.8	353.9	12.00	12.00	0.00
Middle Bakken (Top of Target)									
10,875.0	83.67	55.85	10,652.1	288.6	351.6	373.6	12.00	12.00	0.00
10,900.0	86.67	55.85	10,654.2	302.6	372.2	395.3	12.00	12.00	0.00
10,923.1	89.45	55.85	10,655.0	315.6	391.3	415.4	12.00	12.00	0.00
10,924.2	89.45	55.87	10,655.0	316.2	392.2	416.3	2.00	-0.01	2.00
7"									
11,000.0	89.45	57.39	10,655.7	357.8	455.5	482.8	2.00	-0.01	2.00
11,100.0	89.44	59.39	10,656.7	410.3	540.7	571.9	2.00	-0.01	2.00
11,200.0	89.44	61.39	10,657.7	459.7	627.6	662.5	2.00	0.00	2.00
11,300.0	89.43	63.39	10,658.6	506.0	716.2	754.5	2.00	0.00	2.00

Database:	OpenWellsCompass - EDM Prod	**Local Co-ordinate Reference:**	Well Wade Federal 5300 21-30 13B
Company:	Oasis	**TVD Reference:**	WELL @ 2024.0usft (Original Well Elev)
Project:	Indian Hills	**MD Reference:**	WELL @ 2024.0usft (Original Well Elev)
Site:	153N-100W-29/30	**North Reference:**	True
Well:	Wade Federal 5300 21-30 13B	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Wade Federal 13B		
Design:	Dising#2		

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
11,400.0	89.43	65.39	10,659.6	549.2	806.4	847.9	2.00	0.00	2.00
11,437.1	89.43	66.13	10,660.0	564.4	840.2	882.8	2.00	0.00	2.00
Middle Bakken (Base of target)									
11,500.0	89.43	67.39	10,660.6	589.3	898.0	942.4	2.00	0.00	2.00
11,600.0	89.43	69.39	10,661.6	626.1	990.9	1,038.0	2.00	0.00	2.00
11,700.0	89.42	71.39	10,662.6	659.7	1,085.1	1,134.6	2.00	0.00	2.00
11,800.0	89.42	73.39	10,663.6	689.9	1,180.4	1,232.0	2.00	0.00	2.00
11,900.0	89.43	75.39	10,664.6	716.8	1,276.7	1,330.2	2.00	0.00	2.00
12,000.0	89.43	77.39	10,665.6	740.4	1,373.9	1,429.0	2.00	0.00	2.00
12,100.0	89.43	79.39	10,666.6	760.5	1,471.9	1,528.2	2.00	0.00	2.00
12,200.0	89.43	81.39	10,667.6	777.2	1,570.4	1,627.8	2.00	0.00	2.00
12,300.0	89.43	83.39	10,668.6	790.4	1,669.5	1,727.7	2.00	0.00	2.00
12,400.0	89.44	85.39	10,669.6	800.2	1,769.1	1,827.6	2.00	0.00	2.00
12,500.0	89.44	87.39	10,670.6	806.5	1,868.9	1,927.6	2.00	0.00	2.00
12,600.0	89.45	89.39	10,671.6	809.3	1,968.8	2,027.5	2.00	0.01	2.00
12,630.6	89.45	90.00	10,671.9	809.5	1,999.4	2,057.9	2.00	0.01	2.00
12,700.0	89.45	90.00	10,672.5	809.5	2,068.8	2,127.2	0.00	0.00	0.00
12,800.0	89.45	90.00	10,673.5	809.5	2,168.8	2,226.8	0.00	0.00	0.00
12,854.5	89.45	90.00	10,674.0	809.5	2,223.3	2,281.1	0.00	0.00	0.00
Lower Bakken Shale									
12,900.0	89.45	90.00	10,674.4	809.5	2,268.8	2,326.5	0.00	0.00	0.00
13,000.0	89.45	90.00	10,675.4	809.5	2,368.8	2,426.2	0.00	0.00	0.00
13,100.0	89.45	90.00	10,676.4	809.5	2,468.8	2,525.8	0.00	0.00	0.00
13,200.0	89.45	90.00	10,677.3	809.5	2,568.8	2,625.5	0.00	0.00	0.00
13,300.0	89.45	90.00	10,678.3	809.5	2,668.8	2,725.2	0.00	0.00	0.00
13,400.0	89.45	90.00	10,679.2	809.5	2,768.8	2,824.9	0.00	0.00	0.00
13,500.0	89.45	90.00	10,680.2	809.5	2,868.8	2,924.5	0.00	0.00	0.00
13,600.0	89.45	90.00	10,681.2	809.5	2,968.8	3,024.2	0.00	0.00	0.00
13,700.0	89.45	90.00	10,682.1	809.5	3,068.8	3,123.9	0.00	0.00	0.00
13,800.0	89.45	90.00	10,683.1	809.5	3,168.7	3,223.5	0.00	0.00	0.00
13,900.0	89.45	90.00	10,684.0	809.5	3,268.7	3,323.2	0.00	0.00	0.00
14,000.0	89.45	90.00	10,685.0	809.5	3,368.7	3,422.9	0.00	0.00	0.00
14,100.0	89.45	90.00	10,686.0	809.5	3,468.7	3,522.6	0.00	0.00	0.00
14,200.0	89.45	90.00	10,686.9	809.5	3,568.7	3,622.2	0.00	0.00	0.00
14,300.0	89.45	90.00	10,687.9	809.5	3,668.7	3,721.9	0.00	0.00	0.00
14,400.0	89.45	90.00	10,688.8	809.5	3,768.7	3,821.6	0.00	0.00	0.00
14,500.0	89.45	90.00	10,689.8	809.5	3,868.7	3,921.2	0.00	0.00	0.00
14,600.0	89.45	90.00	10,690.8	809.5	3,968.7	4,020.9	0.00	0.00	0.00
14,700.0	89.45	90.00	10,691.7	809.5	4,068.7	4,120.6	0.00	0.00	0.00
14,800.0	89.45	90.00	10,692.7	809.5	4,168.7	4,220.3	0.00	0.00	0.00
14,900.0	89.45	90.00	10,693.6	809.5	4,268.7	4,319.9	0.00	0.00	0.00
15,000.0	89.45	90.00	10,694.6	809.5	4,368.7	4,419.6	0.00	0.00	0.00
15,100.0	89.45	90.00	10,695.6	809.5	4,468.7	4,519.3	0.00	0.00	0.00
15,200.0	89.45	90.00	10,696.5	809.5	4,568.7	4,619.0	0.00	0.00	0.00
15,300.0	89.45	90.00	10,697.5	809.5	4,668.7	4,718.6	0.00	0.00	0.00
15,354.7	89.45	90.00	10,698.0	809.5	4,723.4	4,773.2	0.00	0.00	0.00
Threeforks									
15,400.0	89.45	90.00	10,698.4	809.5	4,768.7	4,818.3	0.00	0.00	0.00
15,500.0	89.45	90.00	10,699.4	809.5	4,868.7	4,918.0	0.00	0.00	0.00
15,600.0	89.45	90.00	10,700.4	809.5	4,968.7	5,017.6	0.00	0.00	0.00
15,700.0	89.45	90.00	10,701.3	809.5	5,068.7	5,117.3	0.00	0.00	0.00
15,800.0	89.45	90.00	10,702.3	809.5	5,168.7	5,217.0	0.00	0.00	0.00
15,900.0	89.45	90.00	10,703.2	809.5	5,268.7	5,316.7	0.00	0.00	0.00
16,000.0	89.45	90.00	10,704.2	809.5	5,368.6	5,416.3	0.00	0.00	0.00

Database:	OpenWellsCompass - EDM Prod	**Local Co-ordinate Reference:**	Well Wade Federal 5300 21-30 13B
Company:	Oasis	**TVD Reference:**	WELL @ 2024.0usft (Original Well Elev)
Project:	Indian Hills	**MD Reference:**	WELL @ 2024.0usft (Original Well Elev)
Site:	153N-100W-29/30	**North Reference:**	True
Well:	Wade Federal 5300 21-30 13B	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Wade Federal 13B		
Design:	Dising#2		

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
16,100.0	89.45	90.00	10,705.2	809.5	5,468.6	5,516.0	0.00	0.00	0.00
16,200.0	89.45	90.00	10,706.1	809.5	5,568.6	5,615.7	0.00	0.00	0.00
16,300.0	89.45	90.00	10,707.1	809.5	5,668.6	5,715.3	0.00	0.00	0.00
16,400.0	89.45	90.00	10,708.0	809.5	5,768.6	5,815.0	0.00	0.00	0.00
16,500.0	89.45	90.00	10,709.0	809.5	5,868.6	5,914.7	0.00	0.00	0.00
16,600.0	89.45	90.00	10,710.0	809.5	5,968.6	6,014.4	0.00	0.00	0.00
16,700.0	89.45	90.00	10,710.9	809.5	6,068.6	6,114.0	0.00	0.00	0.00
16,800.0	89.45	90.00	10,711.9	809.5	6,168.6	6,213.7	0.00	0.00	0.00
16,900.0	89.45	90.00	10,712.8	809.5	6,268.6	6,313.4	0.00	0.00	0.00
17,000.0	89.45	90.00	10,713.8	809.5	6,368.6	6,413.1	0.00	0.00	0.00
17,100.0	89.45	90.00	10,714.8	809.5	6,468.6	6,512.7	0.00	0.00	0.00
17,200.0	89.45	90.00	10,715.7	809.5	6,568.6	6,612.4	0.00	0.00	0.00
17,300.0	89.45	90.00	10,716.7	809.5	6,668.6	6,712.1	0.00	0.00	0.00
17,400.0	89.45	90.00	10,717.6	809.5	6,768.6	6,811.7	0.00	0.00	0.00
17,500.0	89.45	90.00	10,718.6	809.5	6,868.6	6,911.4	0.00	0.00	0.00
17,600.0	89.45	90.00	10,719.6	809.5	6,968.6	7,011.1	0.00	0.00	0.00
17,700.0	89.45	90.00	10,720.5	809.5	7,068.6	7,110.8	0.00	0.00	0.00
17,800.0	89.45	90.00	10,721.5	809.5	7,168.6	7,210.4	0.00	0.00	0.00
17,900.0	89.45	90.00	10,722.4	809.5	7,268.6	7,310.1	0.00	0.00	0.00
18,000.0	89.45	90.00	10,723.4	809.5	7,368.6	7,409.8	0.00	0.00	0.00
18,100.0	89.45	90.00	10,724.4	809.5	7,468.5	7,509.4	0.00	0.00	0.00
18,200.0	89.45	90.00	10,725.3	809.5	7,568.5	7,609.1	0.00	0.00	0.00
18,300.0	89.45	90.00	10,726.3	809.5	7,668.5	7,708.8	0.00	0.00	0.00
18,400.0	89.45	90.00	10,727.2	809.5	7,768.5	7,808.5	0.00	0.00	0.00
18,500.0	89.45	90.00	10,728.2	809.5	7,868.5	7,908.1	0.00	0.00	0.00
18,600.0	89.45	90.00	10,729.2	809.5	7,968.5	8,007.8	0.00	0.00	0.00
18,700.0	89.45	90.00	10,730.1	809.5	8,068.5	8,107.5	0.00	0.00	0.00
18,800.0	89.45	90.00	10,731.1	809.5	8,168.5	8,207.1	0.00	0.00	0.00
18,900.0	89.45	90.00	10,732.0	809.5	8,268.5	8,306.8	0.00	0.00	0.00
19,000.0	89.45	90.00	10,733.0	809.5	8,368.5	8,406.5	0.00	0.00	0.00
19,100.0	89.45	90.00	10,734.0	809.5	8,468.5	8,506.2	0.00	0.00	0.00
19,200.0	89.45	90.00	10,734.9	809.5	8,568.5	8,605.8	0.00	0.00	0.00
19,300.0	89.45	90.00	10,735.9	809.5	8,668.5	8,705.5	0.00	0.00	0.00
19,400.0	89.45	90.00	10,736.8	809.5	8,768.5	8,805.2	0.00	0.00	0.00
19,500.0	89.45	90.00	10,737.8	809.5	8,868.5	8,904.9	0.00	0.00	0.00
19,600.0	89.45	90.00	10,738.8	809.5	8,968.5	9,004.5	0.00	0.00	0.00
19,700.0	89.45	90.00	10,739.7	809.5	9,068.5	9,104.2	0.00	0.00	0.00
19,800.0	89.45	90.00	10,740.7	809.5	9,168.5	9,203.9	0.00	0.00	0.00
19,900.0	89.45	90.00	10,741.6	809.5	9,268.5	9,303.5	0.00	0.00	0.00
20,000.0	89.45	90.00	10,742.6	809.5	9,368.5	9,403.2	0.00	0.00	0.00
20,100.0	89.45	90.00	10,743.6	809.5	9,468.5	9,502.9	0.00	0.00	0.00
20,200.0	89.45	90.00	10,744.5	809.5	9,568.5	9,602.6	0.00	0.00	0.00
20,300.0	89.45	90.00	10,745.5	809.5	9,668.4	9,702.2	0.00	0.00	0.00
20,400.0	89.45	90.00	10,746.4	809.5	9,768.4	9,801.9	0.00	0.00	0.00
20,500.0	89.45	90.00	10,747.4	809.5	9,868.4	9,901.6	0.00	0.00	0.00
20,600.0	89.45	90.00	10,748.4	809.5	9,968.4	10,001.2	0.00	0.00	0.00
20,673.5	89.45	90.00	10,749.1	809.5	10,042.0	10,074.5	0.00	0.00	0.00
21-30 13B PBHL									

Database:	OpenWellsCompass - EDM Prod	**Local Co-ordinate Reference:**	Well Wade Federal 5300 21-30 13B
Company:	Oasis	**TVD Reference:**	WELL @ 2024.0usft (Original Well Elev)
Project:	Indian Hills	**MD Reference:**	WELL @ 2024.0usft (Original Well Elev)
Site:	153N-100W-29/30	**North Reference:**	True
Well:	Wade Federal 5300 21-30 13B	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Wade Federal 13B		
Design:	Dising#2		

Design Targets

Target Name - hit/miss target - Shape	Dip Angle (°)	Dip Dir. (°)	TVD (usft)	+N/-S (usft)	+E/-W (usft)	Northing (usft)	Easting (usft)	Latitude	Longitude
21-30 13B PBHL	0.00	0.00	10,749.1	809.0	10,042.0	398,272.37	1,219,808.20	48° 3' 3.748 N	103° 33' 43.158 W
- plan misses target center by 0.5usft at 20673.5usft MD (10749.1 TVD, 809.5 N, 10042.0 E)									
- Point									

Casing Points

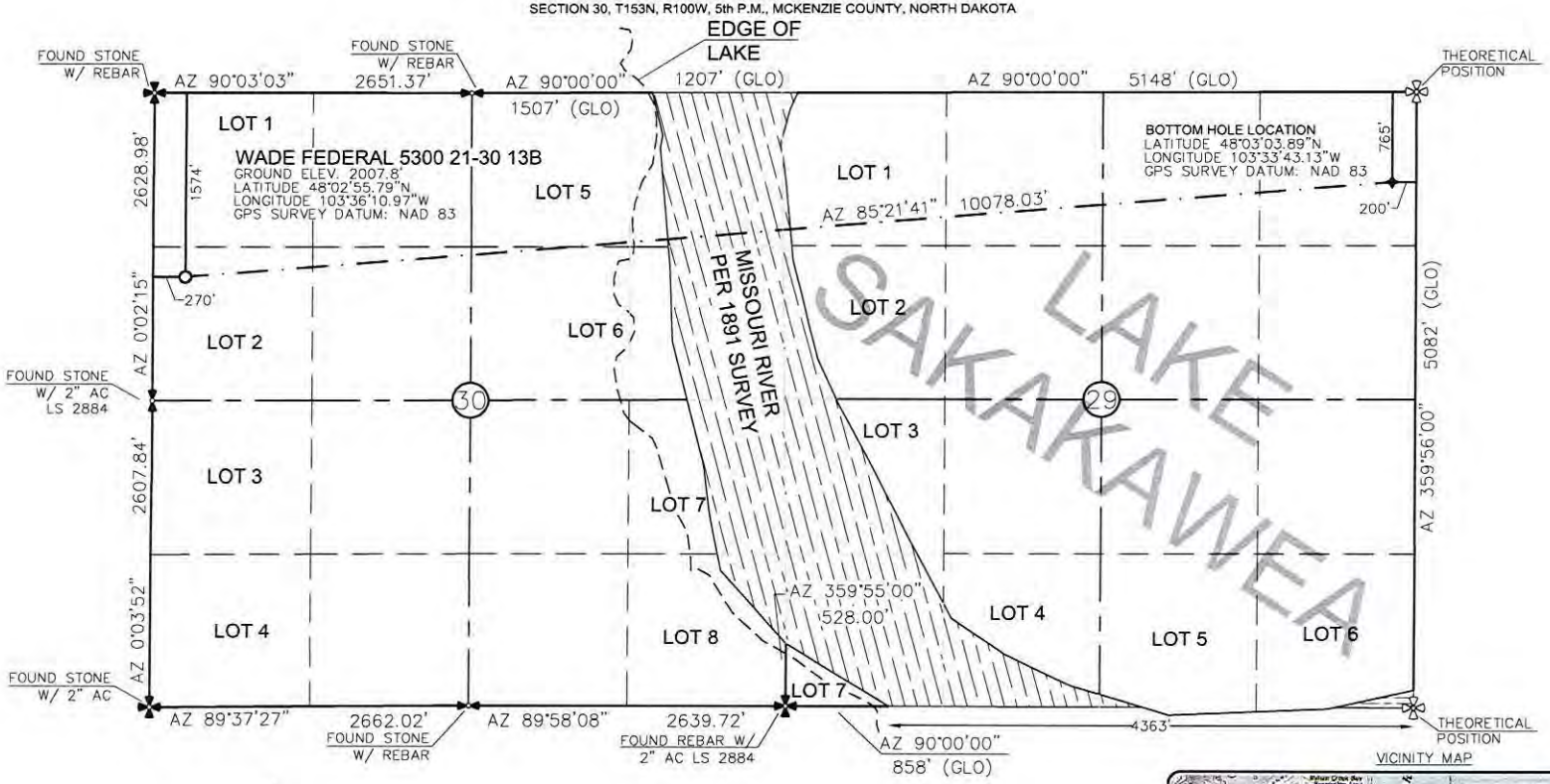
Measured Depth (usft)	Vertical Depth (usft)	Name	Casing Diameter (")	Hole Diameter (")
1,974.0	1,974.0	9 5/8"	9-5/8	13-1/2
10,924.2	10,655.0	7"	7	8-3/4

Formations

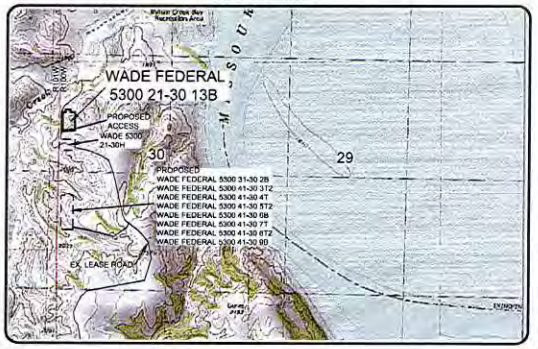
Measured Depth (usft)	Vertical Depth (usft)	Name	Lithology	Dip (°)	Dip Direction (°)
1,874.0	1,874.0	Pierre		0.00	
4,521.1	4,521.0	Greenhorn		0.00	
4,923.1	4,923.0	Mowry		0.00	
5,347.1	5,347.0	Dakota		0.00	
6,360.2	6,360.0	Rierdon		0.00	
6,851.2	6,851.0	Dunham Salt		0.00	
6,901.2	6,901.0	Dunham Salt Base		0.00	
7,161.2	7,161.0	Pine Salt		0.00	
7,179.2	7,179.0	Pine Salt Base		0.00	
7,249.2	7,249.0	Opeche Salt		0.00	
7,328.2	7,328.0	Opeche Salt Base		0.00	
7,569.2	7,569.0	Amsden		0.00	
7,730.2	7,730.0	Tyler		0.00	
7,957.2	7,957.0	Otter/Base Minnelusa		0.00	
8,299.2	8,299.0	Kibbey Lime		0.00	
8,441.2	8,441.0	Charles Salt		0.00	
9,115.2	9,115.0	Base Last Salt		0.00	
9,331.2	9,331.0	Mission Canyon		0.00	
9,881.2	9,881.0	Lodgepole		0.00	
10,723.6	10,612.0	False Bakken		0.00	
10,754.8	10,624.0	Upper Bakken Shale		0.00	
10,852.1	10,649.0	Middle Bakken (Top of Target)		0.00	
11,437.1	10,660.0	Middle Bakken (Base of target)		0.00	
12,854.5	10,674.0	Lower Bakken Shale		0.00	
15,354.7	10,698.0	Threeforks		0.00	

WELL LOCATION PLAT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"WADE FEDERAL 5300 21-30 13B"
1574 FEET FROM NORTH LINE AND 270 FEET FROM WEST LINE
SECTION 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



VICINITY MAP



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0 1000
1" = 1000'

✠ - MONUMENT - RECOVERED
✠ - MONUMENT - NOT RECOVERED

STAKED ON 3/17/14
VERTICAL CONTROL DATUM WAS BASED UPON CONTROL POINT 705 WITH AN ELEVATION OF 2158.3'

THIS SURVEY AND PLAT IS BEING PROVIDED AT THE REQUEST OF ERIC BAYES OF OASIS PETROLEUM. I CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS WORK PERFORMED BY ME OR UNDER MY SUPERVISION AND IS TRUE AND CORRECT TO THE BEST OF MY KNOWLEDGE AND BELIEF.

DARYL D. KASEMAN LS-3880

REGISTERED PROFESSIONAL LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE: 3/19/14
NORTH DAKOTA

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Revision No.	Date	By	Description

OASIS PETROLEUM NORTH AMERICA, LLC
WELL LOCATION PLAT
SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.K. Project No.: S14-09-019.01
Checked By: D.D.K. Date: MAR. 2014

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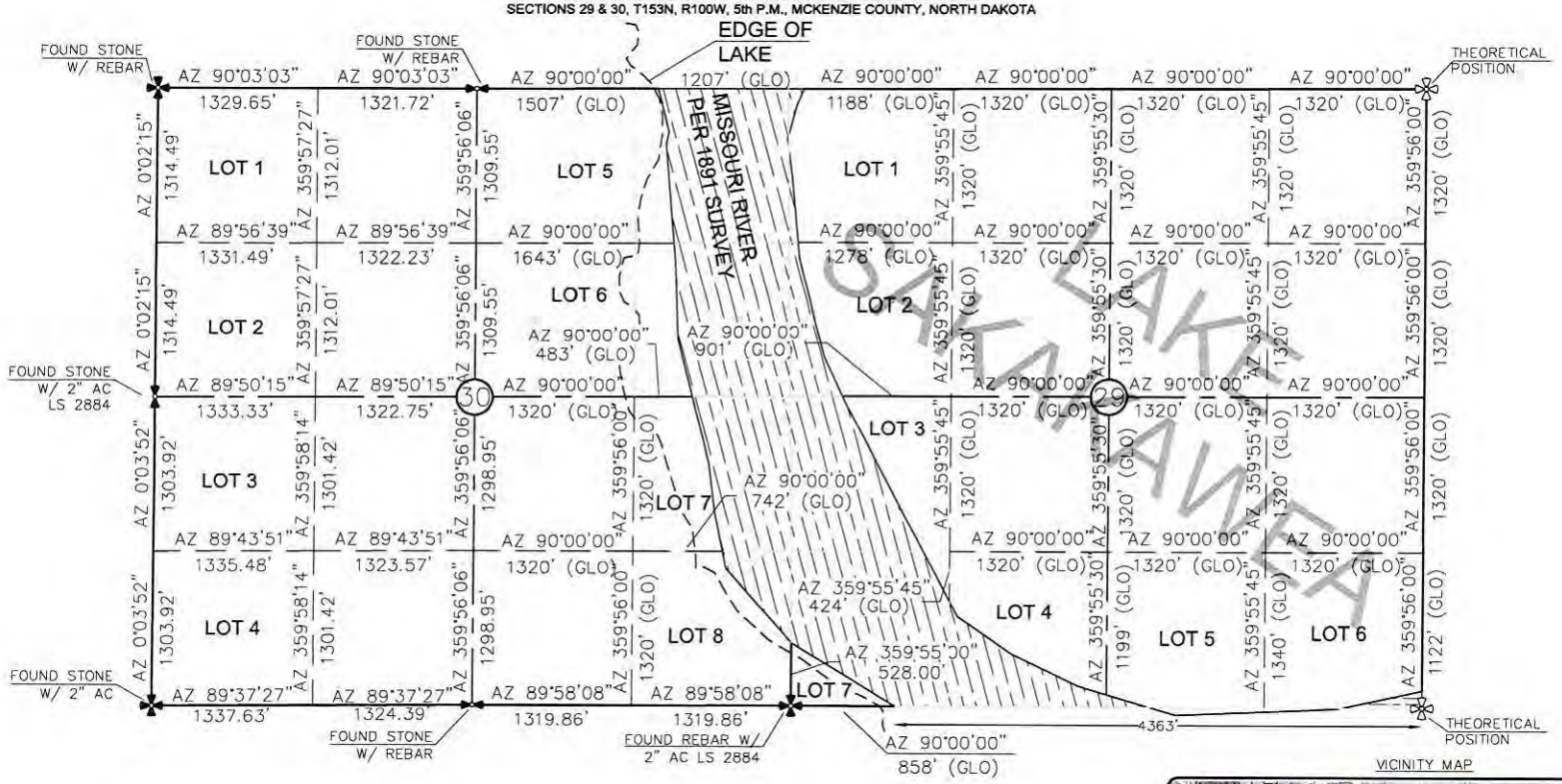
SHEET NO. 1/8

SECTION BREAKDOWN

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"WADE FEDERAL 5300 21-30 13B"
1574 FEET FROM NORTH LINE AND 270 FEET FROM WEST LINE
SECTIONS 29 & 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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- MONUMENT – RECOVERED
- MONUMENT – NOT RECOVERED

0 1000
1" = 1000'

ALL AZIMUTHS ARE BASED ON G.P.S. OBSERVATIONS. THE ORIGINAL SURVEY OF THIS AREA FOR THE GENERAL LAND OFFICE (G.L.O.) WAS 1897. THE CORNERS FOUND ARE AS INDICATED AND ALL OTHERS ARE COMPUTED FROM THOSE CORNERS FOUND AND BASED ON G.L.O. DATA. THE MAPPING ANGLE FOR THIS AREA IS APPROXIMATELY 0°03'.

REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE: 3/19/14
NORTH DAKOTA

VICINITY MAP

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Revision No.	Date	By	Description

OASIS PETROLEUM NORTH AMERICA, LLC
SECTION BREAKDOWN
SECTIONS 29 & 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S14-09-019.01
Checked By: D.D.K. Date: MAR. 2014

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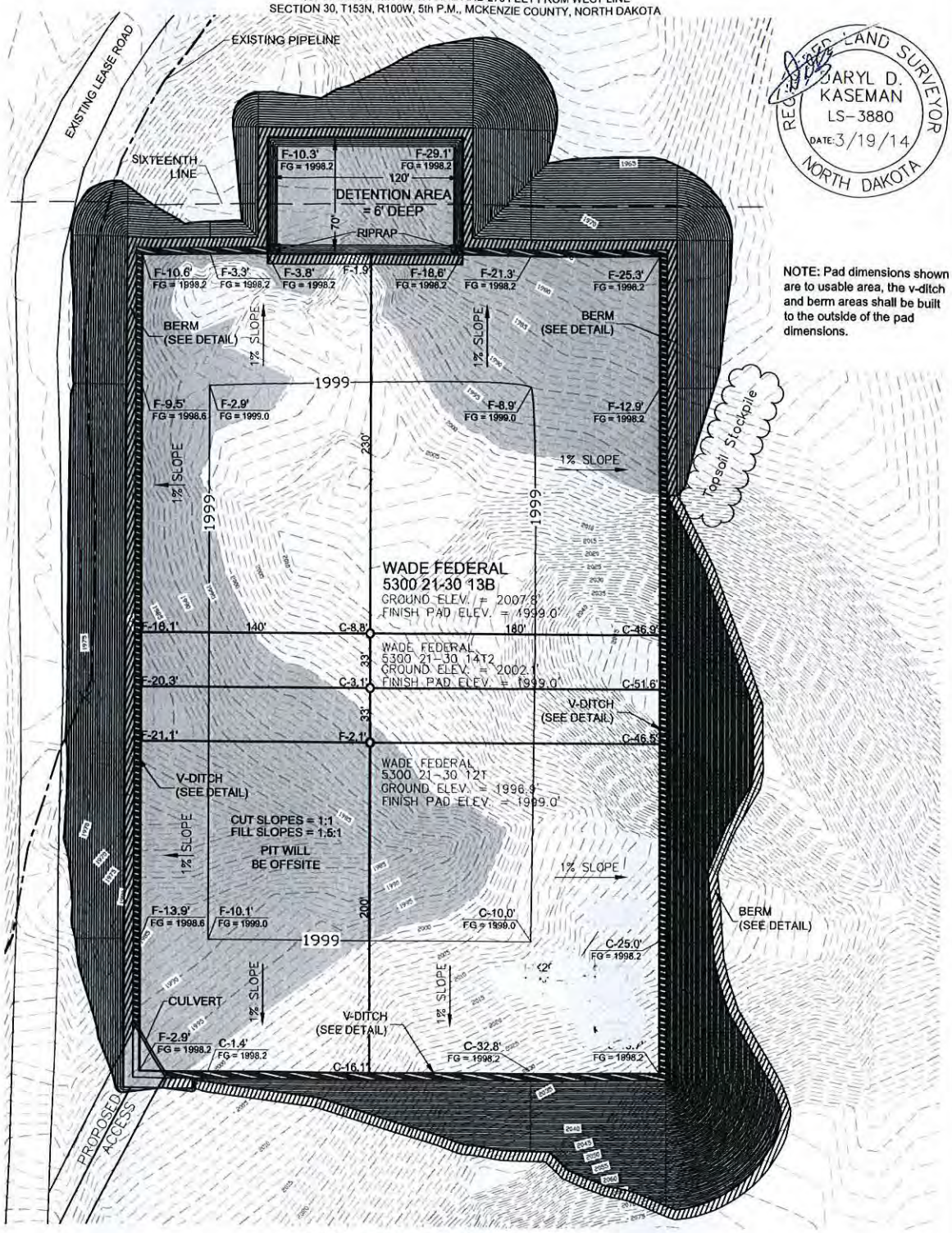
2/8
SHEET NO.

PAD LAYOUT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"WADE FEDERAL 5300 21-30 13B"
1574 FEET FROM NORTH LINE AND 270 FEET FROM WEST LINE
SECTION 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



NOTE: Pad dimensions shown are to usable area, the v-ditch and berm areas shall be built to the outside of the pad dimensions.

REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE: 3/19/14
NORTH DAKOTA

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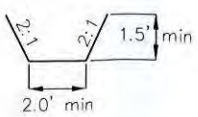
V-DITCH DETAIL



Proposed Contours
Original Contours
BERM
DITCH

0 60
1" = 60'

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NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

3/8
SHEET NO.

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OASIS PETROLEUM NORTH AMERICA, LLC
PAD LAYOUT
SECTION 30, T153N, R100W
MKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.L. Project No.: S14-09-019.01
Checked By: D.D.K. Date: MAR. 2014

Revision No.	Date	By	Description

WELL LOCATION SITE QUANTITIES

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"WADE FEDERAL 5300 21-30 13B"
1574 FEET FROM NORTH LINE AND 270 FEET FROM WEST LINE
SECTION 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

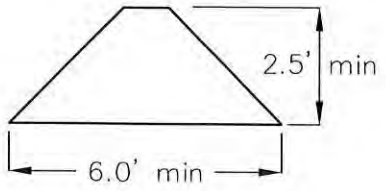
WELL SITE ELEVATION	2007.8
WELL PAD ELEVATION	1999.0

EXCAVATION	65,093
PLUS PIT	0
	65,093
EMBANKMENT	49,762
PLUS SHRINKAGE (25%)	12,441
	62,203
STOCKPILE PIT	0
STOCKPILE TOP SOIL (6")	4,561
BERMS	2,074 LF = 671 CY
DITCHES	1,461 LF = 223 CY
DETENTION AREA	1,377 CY
ADDITIONAL MATERIAL NEEDED	666
DISTURBED AREA FROM PAD	5.65 ACRES

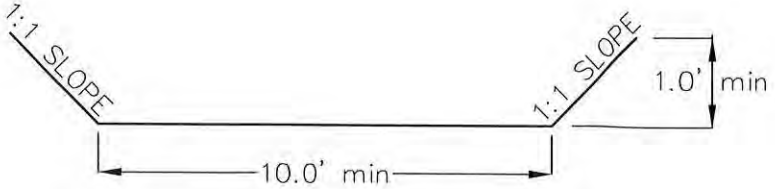
NOTE: ALL QUANTITIES ARE IN CUBIC YARDS (UNLESS NOTED)
CUT END SLOPES AT 1:1
FILL END SLOPES AT 1.5:1

WELL SITE LOCATION
1574' FNL
270' FWL

BERM DETAIL



DITCH DETAIL



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QUANTITIES
SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. | Project No.: S14-09-019.01
Checked By: D.D.K. | Date: MAR. 2014

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CROSS SECTIONS

OASIS PETROLEUM NORTH AMERICA, LLC
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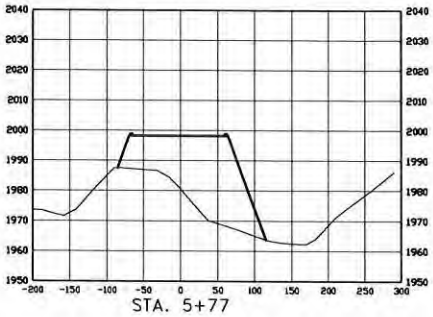

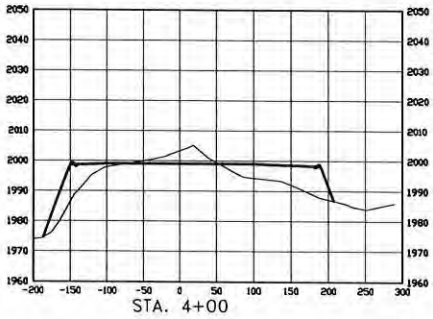

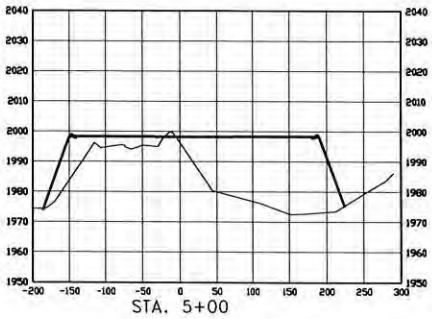

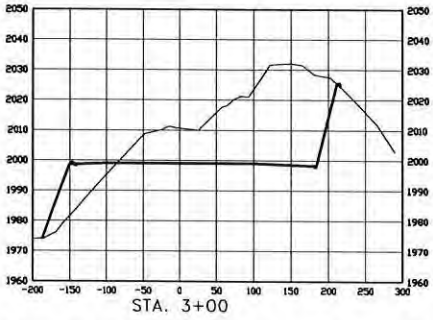

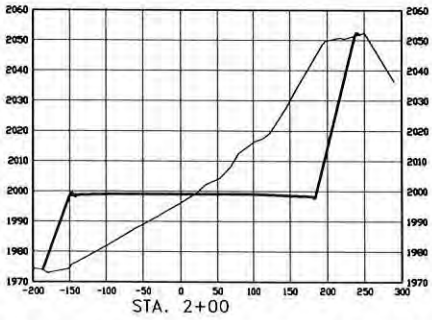

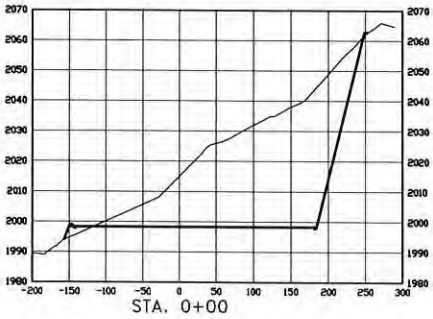

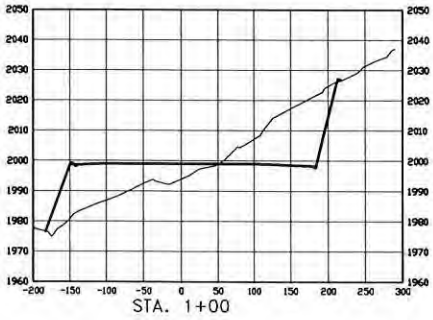

REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE: 3/19/14
NORTH DAKOTA

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SCALE
HORIZ 1"=160'
VERT 1"=40'

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CROSS SECTIONS
SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. | Project No.: S14-09-019.01
Checked By: D.D.K. | Date: MAR. 2014

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ACCESS APPROACH

OASIS PETROLEUM NORTH AMERICA, LLC
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1574 FEET FROM NORTH LINE AND 270 FEET FROM WEST LINE
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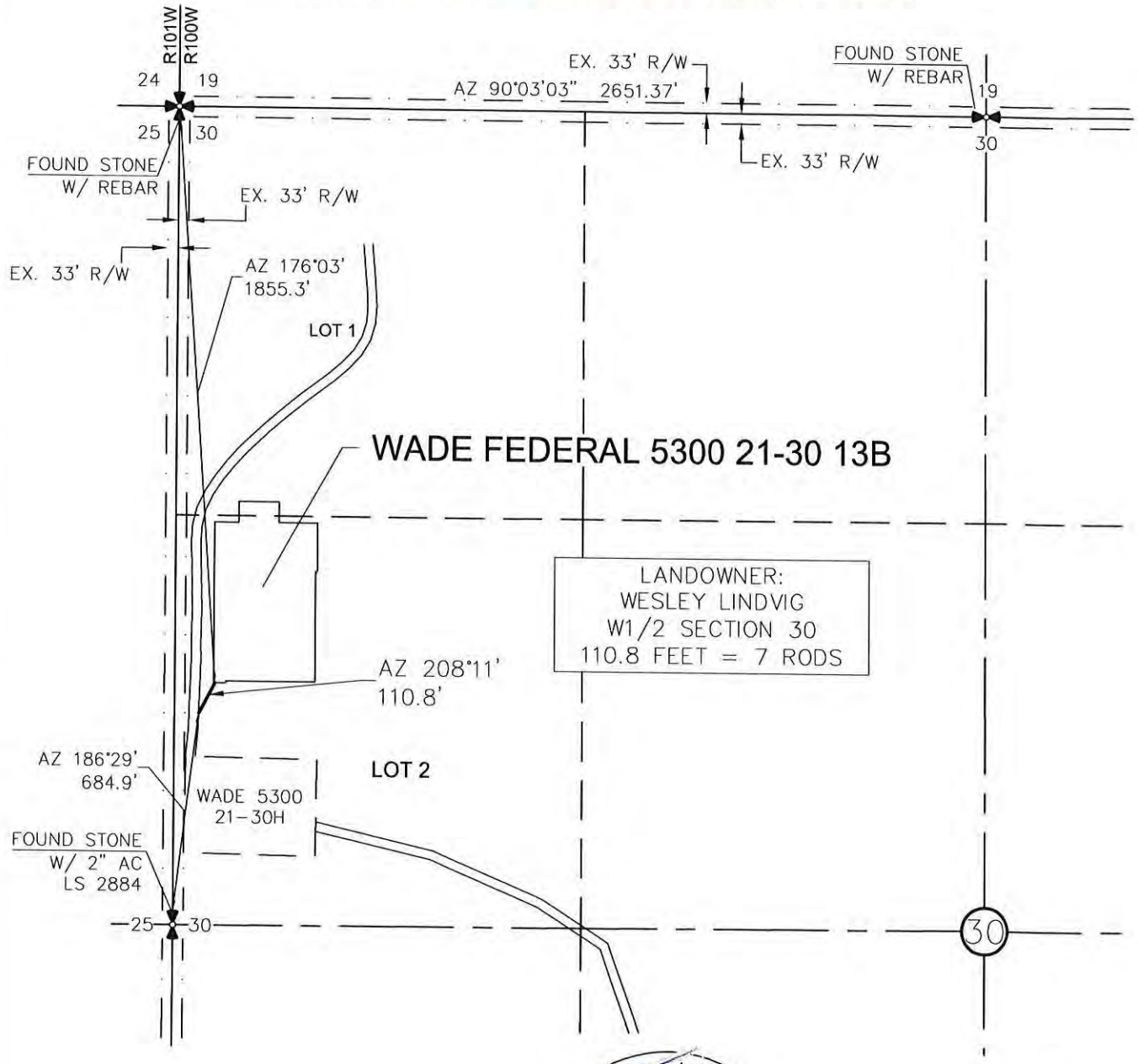

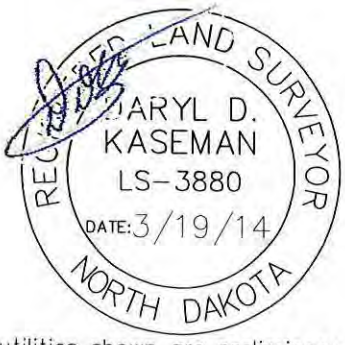

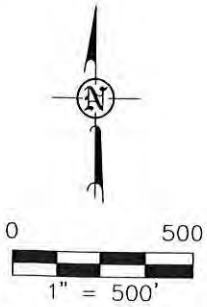

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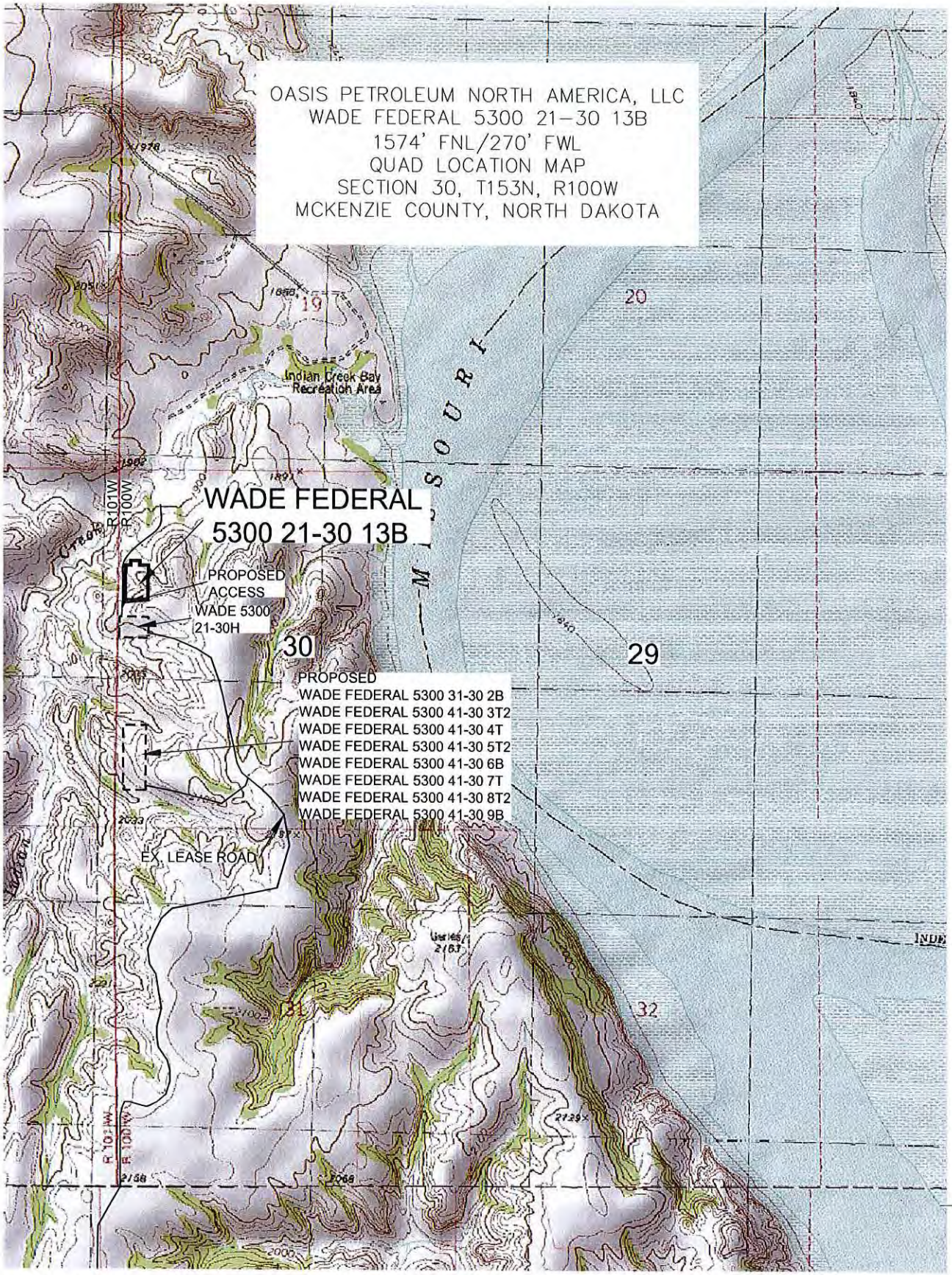
SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H.		Project No.: S14-09-019.01	
Checked By: D.D.K.		Date: MAR. 2014	

Revision No.	Date	By	Description



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425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
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Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
QUAD LOCATION MAP
SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By:	B.H.H.	Project No.:	S14-09-019.01
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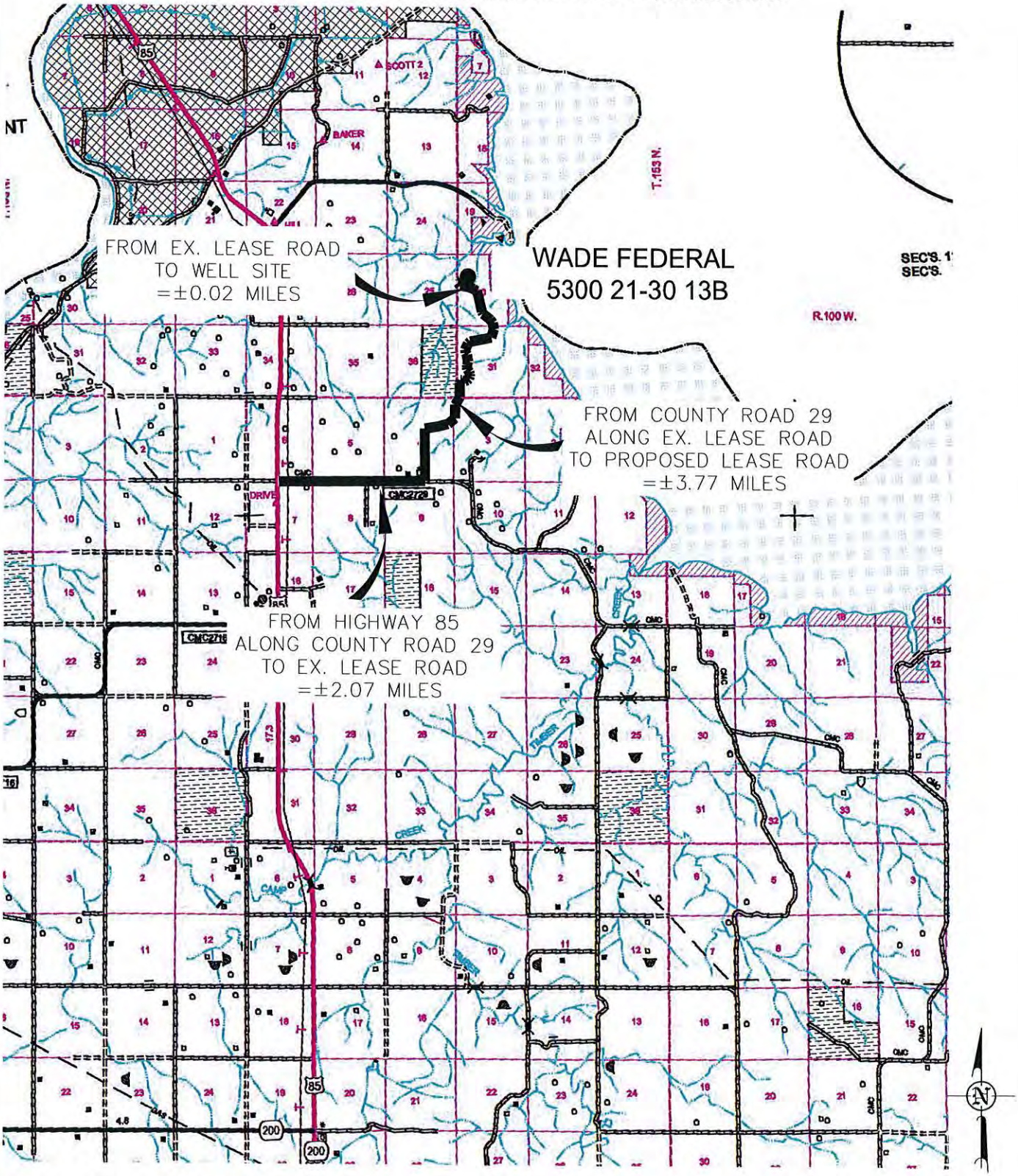
COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"WADE FEDERAL 5300 21-30 13B"
1574 FEET FROM NORTH LINE AND 270 FEET FROM WEST LINE
SECTION 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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SCALE: 1" = 2 MILE

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COUNTY ROAD MAP
SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

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OASIS PETROLEUM

STATEMENT

This statement is being sent in order to comply with NDAC 43-02-03-16 (Application for permit to drill and recomplete) which states (in part that) "confirmation that a legal street address has been requested for the well site, and well facility if separate from the well site, and the proposed road access to the nearest existing public road". On the date noted below a legal street address was requested from the appropriate county office.

McKenzie County
Aaron Chisholm – GIS Specialist for McKenzie County

Wade Federal 5300 21-30 12T – 153N-100W-29/30 – 06/04/2014
Wade Federal 5300 21-30 13B – 153N-100W-29/30 – 06/04/2014
Wade Federal 5300 21-30 14T2 – 153N-100W-29/30 – 06/04/2014

Lauri M. Stanfield
Regulatory Specialist
Oasis Petroleum North America, LLC

From: Burk, Ashley N.
To: Webber, Alice D.
Subject: FW: Wade Federal 5300 21-30 Pad
Date: Friday, July 25, 2014 1:17:57 PM

From: Michael Kukuk [mailto:mkukuk@oasispetroleum.com]
Sent: Friday, July 25, 2014 12:59 PM
To: Burk, Ashley N.
Subject: RE: Wade Federal 5300 21-30 Pad

Hi Ashley,

I was able to confirm that the cuttings will be hauled to the Indian Hills Waste Disposal.

Thank you,
Michael

From: Burk, Ashley N. [mailto:anburk@nd.gov]
Sent: Friday, July 25, 2014 9:46 AM
To: Michael Kukuk
Subject: RE: Wade Federal 5300 21-30 Pad

Michael,

Just realized that since there is not cuttings pit, we need to know what facility you will be hauling the cuttings to for all three wells.

Thanks and have a great Friday!

Ashley

From: Michael Kukuk [mailto:mkukuk@oasispetroleum.com]
Sent: Thursday, July 24, 2014 4:56 PM
To: Burk, Ashley N.
Subject: RE: Wade Federal 5300 21-30 Pad

Hi Ashley,

Thank you for the update on the Wade Federals. We will address this ASAP –

Best,
Michael

From: Burk, Ashley N. [mailto:anburk@nd.gov]
Sent: Thursday, July 24, 2014 2:57 PM
To: Michael Kukuk
Subject: Wade Federal 5300 21-30 Pad

Michael,

It is my understanding that you just called looking to get the Wade Federal well pad issued ASAP, however we have two issues with this well pad.

I talked to Brandi this morning about the 14T2 well. Right now you have the surface casing at 1974', and we need that to be modified so that it is at least 1980'.

For the 13B well, the bottom hole location violates the 200' setback. According to the drilling plan, it is located 10,142' from the wellhead (the other two wells are 10,042'), which would therefore put it at 101' from the east line. I left Brandi a voicemail about this issue a few minutes ago.

The sooner we get these, the sooner we can get your permit.

Feel free to contact me with any questions regarding these issues.

Thanks,

Ashley N. Burk
Permitting Technician
NDIC, Dept. Mineral Resources
Oil and Gas Division
(701) 328-8093
anburk@nd.gov

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